



A Likelihood Ratio Framework for Change Point Detection with Applications to Financial Time Series

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Figure 3.1 and Figure 3.5	I have modified L ^A T _E X code produced by AI to draw a Tikz Picture to produce diagrams.	Google Gemini

Abstract

This report uses a likelihood ratio framework to develop methods for detecting changes in the mean and variance of a time series. We begin by considering a single change point under a univariate signal-plus-noise model and extend this to cover multiple change points and multivariate and high-dimensional data, with parallels drawn to the Bayesian paradigm. We subsequently relax the independence assumption through autoregressive and vector autoregressive models, which capture temporal and cross-sectional dependence respectively. We derive thresholds for the limiting behaviour of the generalised log-likelihood ratio test statistic and align these with established information criteria. The methods are applied to S&P 500 returns and a portfolio of stocks during the period 2018-2022. Multiple detection algorithms consistently identify a change point in February 2020 corresponding to the onset of the COVID-19 pandemic, with INSPECT identifying airline, energy, and financial stocks as the drivers of this change, consistent with the impact of travel restrictions, the oil price crash, and financial uncertainty during the pandemic.

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Notation

$\mathbb{N}$	Natural numbers
$\mathbb{R}$	Real numbers
$\mathbb{Z}$	Integers
c	Threshold
d	Dimension of multivariate time series
h	Sliding window half-width
k	Candidate change point index
m	Number of change points
n	Number of time series observations
p	Number of lags in autoregressive model
r	Parameter dimension
s	Sparsity level
β	PELT penalty
δ	Jump size
ε	Noise
η	Sliding window buffer
μ	Mean
ϕ	Autoregressive coefficients
σ^2	Variance
τ	True change point location(s)
Ω	Parameter space
Δ	Minimum segment length
$\mathbf{X}$	Univariate time series
$\mathbf{X}$	Multivariate time series
$\boldsymbol{\theta}$	Parameter
$\hat{\boldsymbol{\theta}}$	Maximum likelihood estimate
$\boldsymbol{\Sigma}$	Covariance matrix
$B(\cdot)$	Brownian bridge
$D(\cdot, \cdot)$	Kullback-Leibler divergence
$F(t)$	Cost of optimal partition up to time t
$\mathcal{L}_M$	Marginal likelihood function
ℓ	Log-likelihood function
$\mathcal{L}$	Likelihood function
$W(\cdot)$	Standard Brownian motion
$\mathbf{H}(\cdot)$	Hessian matrix
$\mathbf{J}(\cdot)$	Expected Fisher information
$\mathbf{Z}_n(\cdot)$	Partial sum score process
C_k	CUSUM Statistic at candidate index k
LR_k	Generalised log-likelihood ratio test statistic at candidate index k
$\mathcal{W}_k$	Wald statistic at candidate index k
Λ_k	Generalised likelihood ratio test statistic at candidate index k
$\ \cdot\ _*$	Nuclear norm, defined as sum of singular values of input

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Chapter 1

Introduction

Many processes, such as financial returns, weather data and medical monitoring, undergo abrupt structural shifts at which their behaviour changes. Identifying such shifts is crucial: if they are ignored, then analysis and predictions risk modelling based on patterns that no longer hold. These structural breaks are called change points, and represent a change in the underlying data-generating distribution. Whereas an outlier may occur as a result of the stochasticity of a system, change points mark a difference in underlying signal. For example, a single unusual daily return does not indicate the presence of a change in the underlying regime, but an ongoing period of sustained volatility does. Change point analysis dates back to Page in the context of quality control [50], and has since expanded into genomics [64], econometrics [3], and high-dimensional settings [67].

This report discusses changes in the mean and variance of a process in the offline setting, where the full time series is available retrospectively, as opposed to online detection, where changes are detected in real time. We begin by assuming that the observations follow a simple signal-plus-noise model with i.i.d. Gaussian noise, and consider a single change point in the univariate case, before extending to the multivariate and then high-dimensional setting. We discuss the alternative Bayesian framework and its parallels, as well as explicitly deriving two thresholds for the limiting behaviour of the GLLR test statistic, aligning them with established information-theoretic penalties. We extend this further by considering several approaches for detecting multiple changes. We then relax the assumption of independence and allow the data to exhibit temporal dependence between observations, which is a more realistic assumption for many time series applications. We extend this to the VAR case, which captures both temporal and cross-sectional dependence. Each extension relaxes a modelling assumption, broadening the range of data to which the framework can be applied. We use a unified likelihood ratio framework throughout to highlight the parallels between various techniques.

To demonstrate the empirical validity of these models, we apply a variety of the developed techniques to returns data, where the algorithms successfully identify change points corresponding to key events in the COVID-19 pandemic. Multiple different methods find the same change point, and the use of the INSPECT algorithm allows us to identify which sectors drove the change, distinguishing series which were more heavily affected by COVID from those that were not. This emphasises the robustness of the algorithms as they highlight dates aligning with identifiable real-world events.

Code reproducing the figures throughout and used in the analysis in chapter 5 is available at <https://github.com/tarawelchh/Change-Point>.

Chapter 2

Detecting a Change Point

To develop methods for detecting the structural shifts described in the previous section, we formalise the problem of modelling and detecting a change point. We introduce the CUSUM statistic and show that it is a special case of the generalised log-likelihood ratio test statistic, which provides a framework for detecting changes in mean, variance, or both. We discuss when these test statistics indicate a change, then extend to the multivariate and high-dimensional cases, and consider a Bayesian approach to change point detection. We begin by formally defining time series and weak stationarity [28, Ch. 3].

Definition 2.1 (Time Series). *A time series is an ordered sequence of observations indexed by time. We consider only discrete time series, in which each observation is indexed by an integer. For a discrete time series of length n , we write $\mathbf{X} = (X_1, X_2, \dots, X_n)^\top \in \mathbb{R}^n$.*

Definition 2.2 (Weak Stationarity). *A discrete-time stochastic process, $\{X_t\}_{t \in \mathbb{Z}}$ is weakly stationary if:*

- $\mathbb{E}[X_s] = \mathbb{E}[X_t] = \mu \quad \forall s, t \in \mathbb{Z},$
- $\text{Cov}[X_s, X_{s+k}] = \text{Cov}[X_t, X_{t+k}] = \gamma_k \quad \forall s, t, k \in \mathbb{Z}.$

We assume that the time series can be modelled as an underlying signal plus an additive noise term. Letting X_t be the random variable representing the observation at time t , this model can be written:

$$X_t = f(t) + \varepsilon_t. \quad (2.1)$$

The signal, $f(t)$, represents the systematic part of the data, whereas ε_t represents fluctuation around this underlying pattern. We assume that we have independent and identically distributed (i.i.d.) Gaussian noise, such that $\varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$. We can introduce a single change point into our time series at time $\tau \in \{1, \dots, n-1\}$ by specifying the following:

$$f(t) = \begin{cases} f_1(t) & \text{for } t \leq \tau, \\ f_2(t) & \text{for } t > \tau, \end{cases} \quad (2.2)$$

where $f_1(t) \neq f_2(t)$ implies a change in the underlying signal. In the simplest example, we consider $f(t)$ as piecewise constant. Setting $f_1(t) = \mu_1$ and $f_2(t) = \mu_2$, with $\mu_1, \mu_2 \in \mathbb{R}$, results in the change in mean model (example shown in Figure 2.1a). Under the assumption of i.i.d. Gaussian noise, with $f(t)$ piecewise constant, the process $\mathbf{X}$ is characterised by its first two moments [28]. As a result, weak stationarity implies strict stationarity, and so we use the term *stationary* hereafter. Therefore, in the change in mean model, since the variance is assumed to be fixed, a constant mean along each segment implies that our data consists of two distinct stationary segments, separated by the change point at index τ .

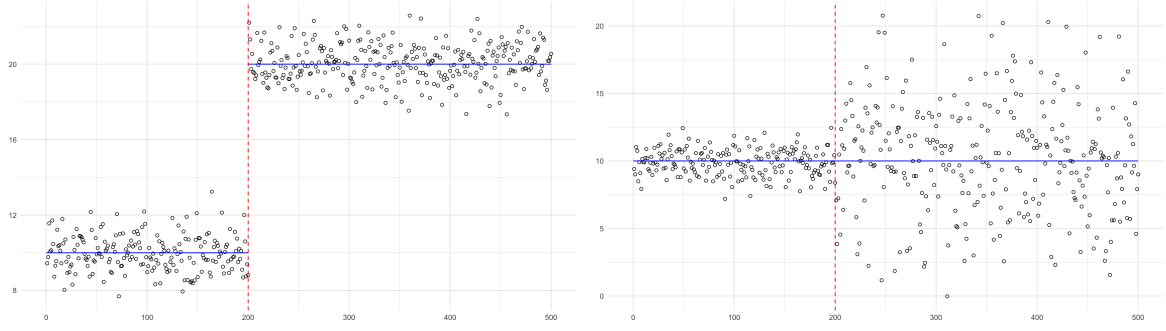
As well as modelling a change in mean, we can also incorporate a change in variance. This may or may not occur simultaneously with the mean change. For example, considering a time series of stock prices, a major economic event is likely to impact the mean and variance simultaneously, as uncertainty causes increased volatility, therefore increasing variance. In order to account for a possible change in variance, we adjust the model in (2.1) to include a second function, $g(t)$:

$$X_t = f(t) + g(t)\zeta_t. \quad (2.3)$$

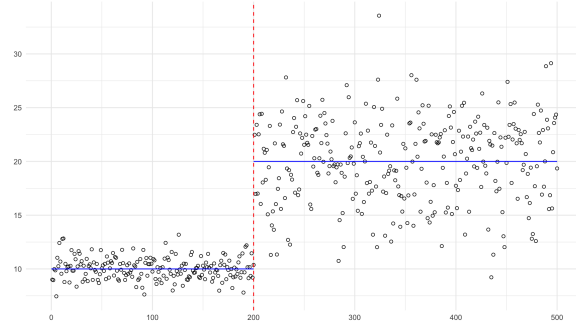
We can specify a simultaneous change in mean and variance at index τ by setting

$$f(t) = \begin{cases} \mu_1 & \text{for } t \leq \tau, \\ \mu_2 & \text{for } t > \tau, \end{cases} \quad g(t) = \begin{cases} \sigma_1 & \text{for } t \leq \tau, \\ \sigma_2 & \text{for } t > \tau, \end{cases} \quad (2.4)$$

with $\zeta_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$. In order for this framework to specify a change, we can take either $\mu_1 \neq \mu_2$, $\sigma_1 \neq \sigma_2$ or both, with an example of each of these cases given in figure 2.1. Note that setting $\sigma_1 = \sigma_2 = 1$ gives our change in mean model specified by (2.1) and (2.2).



(a) Model (2.1) with (2.2), where $f_1(t) = \mu_1 = 10$, $f_2(t) = \mu_2 = 20$, and ε_t i.i.d. Gaussian noise. (b) Model (2.3) with (2.4), where $\mu_1 = \mu_2 = 10$, $\sigma_1 = 1$, $\sigma_2 = 4$, and ζ_t i.i.d. Gaussian noise.



(c) Model (2.3) with (2.4), where $\mu_1 = 10$, $\mu_2 = 20$, $\sigma_1 = 1$, and $\sigma_2 = 4$, and ζ_t i.i.d. Gaussian noise.

Figure 2.1: Examples of data generated under signal-plus-noise model (2.1) with observations X_t (black), signal $f(t)$ (blue solid), and a single change point at $\tau = 200$ (red dashed). Example of a change in mean (a), variance (b), and both mean and variance (c).

Change point analysis can be used to determine whether a change occurred and, if so, at which index. We pose this formally with hypothesis notation, where $\mathcal{H}_0$ represents no change in the signal, and $\mathcal{H}_1$ represents a change at time $t = \tau$. Consider a time series $\mathbf{X} = (X_1, \dots, X_n)^\top$ whose underlying distribution is given by $f(x; \theta)$, where $\theta \in \mathbb{R}^r$ represents a vector of r

parameters. Then the hypothesis test for a single change point is of the form:

$$\begin{aligned} \mathcal{H}_0 : \boldsymbol{\theta} &= \boldsymbol{\theta}_0 \quad \forall 1 \leq t \leq n, \\ \mathcal{H}_1 : \text{There exists a time } \tau \in \{1, \dots, n-1\} \text{ such that } \boldsymbol{\theta} &= \begin{cases} \boldsymbol{\theta}_1 & \text{for } 1 \leq t \leq \tau, \\ \boldsymbol{\theta}_2 & \text{for } \tau < t \leq n, \end{cases} \end{aligned} \quad (2.5)$$

where $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$.

2.1 The CUSUM Statistic

One of the most widely used methods for detecting a change in mean is through use of the CUSUM statistic [50]. This works by treating each index as a candidate for a possible change point, and calculating the empirical mean of the data before and after this point. The CUSUM statistic is the absolute difference of these values multiplied by a variance stabilising constant. The index with the largest CUSUM statistic is then the most likely candidate for a change point, as it indicates the largest difference in means.

Definition 2.3 (CUSUM Statistic). *The CUSUM statistic C_k for a time series $\mathbf{X}$ is given by*

$$C_k(\mathbf{X}) = \frac{1}{\sigma} \sqrt{\frac{k(n-k)}{n}} |\bar{X}_{1:k} - \bar{X}_{k+1:n}|, \quad (2.6)$$

for $k = 1, \dots, n-1$, where $\bar{X}$ is the standard empirical mean:

$$\bar{X}_{i:j} = \frac{1}{j-i+1} \sum_{t=i}^j X_t.$$

To simplify notation, we write C_k , omitting the dependence on $\mathbf{X}$ when it is clear from the context.

A higher value of the CUSUM statistic at index k indicates a greater difference in the mean of the data before and after $t = k$. Therefore we estimate that if a change did occur, then

$$\hat{\tau} = \arg \max_{k=1, \dots, n-1} C_k, \quad (2.7)$$

is an estimator for τ , the true change point. If $C_{\hat{\tau}} > c$ for a predetermined threshold c , then we can reject the null hypothesis and declare a change point at $\hat{\tau}$. We note that the variance normalising constant in (2.6) is required, since variance increases with subsample size. Its value is given by:

$$\frac{1}{\sigma} \sqrt{\frac{k(n-k)}{n}}.$$

This comes from the fact that

$$\text{Var} [\bar{X}_{1:k}] = \frac{\sigma^2}{k}, \quad \text{Var} [\bar{X}_{k+1:n}] = \frac{\sigma^2}{n-k}.$$

Then, by the assumption that the data are independent,

$$\begin{aligned} \text{Var} [\bar{X}_{1:k} - \bar{X}_{k+1:n}] &= \frac{\sigma^2}{k} + \frac{\sigma^2}{n-k}, \\ &= \frac{\sigma^2 n}{k(n-k)}. \end{aligned} \quad (2.8)$$

To standardise the test statistic, we divide our absolute difference in means by the square root of this variance to obtain (2.6). We now demonstrate the use of the CUSUM statistic through an example.

Example 2.4. *The hypotheses for the change in mean problem of the form (2.2) are given by:*

$$\mathcal{H}_0 : f(t) = \mu_0 \quad \forall 1 \leq t \leq n,$$

$$\mathcal{H}_1 : f(t) = \begin{cases} \mu_1 & \text{for } 1 \leq t \leq \tau, \\ \mu_2 & \text{for } \tau < t \leq n. \end{cases}$$

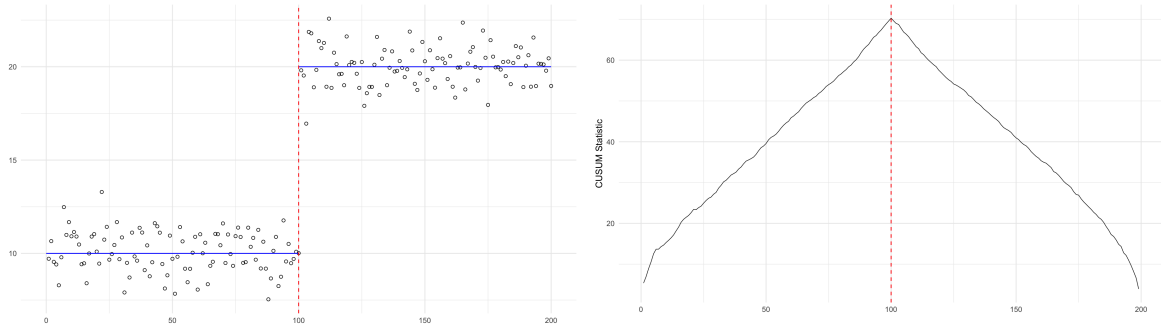
We set $n = 200$ and model the noise by $\varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$. The underlying signal (assumed unknown) is given by setting:

$$\theta_1 = \mu_1 = 10,$$

$$\theta_2 = \mu_2 = 20,$$

$$\tau = 100.$$

The plot of these data is shown in figure 2.2a. Assuming we do not know the underlying signal, we can compute $\hat{\tau}$ as in (2.7). As shown by the red line in figure 2.2b, we have $\hat{\tau} = 100$, and so we correctly identify the true change point.



(a) Simulated data (black) with true means $\mu_1 = 10$, $\mu_2 = 20$ (blue solid) and change point $\tau = 100$ (red dashed). (b) CUSUM statistic (black) for the data in (a) with correctly identified change point at $\tau = 100$ (red dotted).

Figure 2.2: Simulated data with a change in mean (a) and their CUSUM statistic (b) for example 2.4.

2.1.1 Optimisation

Calculating the CUSUM statistic as in (2.6) for each index has complexity $O(n^2)$, and ignores the fact that the statistics are highly correlated with each other, with C_k and C_{k+1} differing by only a single data point. This naive implementation involves calculating the sums $\sum_{t=1}^k X_t$ and $\sum_{t=k+1}^n X_t$ which has complexity $O(n)$. This is repeated for each of the $n - 1$ candidate change points, giving a total complexity of $O(n^2)$. To reduce this complexity, in practice we can pre-calculate a vector of cumulative sums [50], $\mathbf{S} = (S_1, \dots, S_{n-1})$, such that each element S_i is the sum of all data up to index i :

$$S_i = \sum_{t=1}^i X_t.$$

2.2 The Generalised Log-Likelihood Ratio Test

Computing this list involves n additions (complexity $O(n)$). From this, the CUSUM statistic for any value of k can then be computed in $O(1)$ time, since the components of (2.6) are given by:

$$\begin{aligned}\bar{X}_{1:k} &= \frac{1}{k}S_k, \\ \bar{X}_{k+1:n} &= \frac{1}{n-k}(S_n - S_k).\end{aligned}$$

We then substitute these values into (2.6) to get:

$$\begin{aligned}C_k(\mathbf{X}) &= \frac{1}{\sigma} \sqrt{\frac{k(n-k)}{n}} \left| \frac{S_k}{k} - \left(\frac{S_n - S_k}{n-k} \right) \right|, \\ &= \frac{1}{\sigma} \sqrt{\frac{n}{k(n-k)}} \left| S_k - \frac{k}{n}S_n \right|.\end{aligned}\tag{2.9}$$

For each index, this involves two $O(1)$ look-ups from a pre-computed list $\mathbf{S}$, and an $O(1)$ computation. Therefore, the total complexity for calculating the CUSUM statistics for all candidate change points $k = 1, \dots, n-1$ is equal to the complexity of computing $\mathbf{S}$, $O(n)$.

2.1.2 Minimum Segment Length

We see that (2.8) is larger for small values of $\min\{k, n-k\}$. For this reason, it is common to apply a minimum segment length, $\min\{k, n-k\} \geq a$, such that the first and last a points are not considered as candidate change points, to allow the mean and variance to stabilise [22]. For example, though it is valid to calculate C_1 , this implies a segment of length one, whose mean is simply its own value. For a change in variance, a minimum segment length of 2 applies. In both cases, applying a minimum segment length ensures sufficient statistical power.

2.2 The Generalised Log-Likelihood Ratio Test

The CUSUM statistic is useful for locating a single change in mean when the variance is known. We now show that the CUSUM statistic is a special case of the generalised likelihood ratio test. This allows us to formulate analogous tests for other changes, including changes in variance. We begin by defining the likelihood framework, with the following standard definitions.

Definition 2.5 (Likelihood Function). *Let $\mathbf{X}_{1:m} = (X_1, \dots, X_m)$ be a subsequence of our observed time series ($m \leq n$) such that $\boldsymbol{\theta} \in \Omega \subseteq \mathbb{R}^r$ is fixed on this segment, where Ω denotes the parameter space. We define the likelihood function $\mathcal{L}(\boldsymbol{\theta})$ by the joint density of the observed data as a function of $\boldsymbol{\theta}$:*

$$\mathcal{L}(\boldsymbol{\theta}) \equiv \mathcal{L}(\boldsymbol{\theta}; \mathbf{X}_{1:m}) = f(\mathbf{X}_{1:m}; \boldsymbol{\theta}),$$

and the log-likelihood function as its logarithm:

$$\ell(\boldsymbol{\theta}) \equiv \ell(\boldsymbol{\theta}; \mathbf{X}_{1:m}) = \ln \mathcal{L}(\boldsymbol{\theta}) = \ln f(\mathbf{X}_{1:m}; \boldsymbol{\theta}).$$

Definition 2.6 (Maximum Likelihood Estimator). *The maximum likelihood estimator (MLE) $\hat{\boldsymbol{\theta}}$ for $\boldsymbol{\theta}$ given the data $\mathbf{X}$ is the value of $\boldsymbol{\theta}$ that maximises the likelihood over the parameter space Ω .*

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta} \in \Omega} \mathcal{L}(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta} \in \Omega} \ell(\boldsymbol{\theta}).$$

Let $\hat{\boldsymbol{\theta}}_0$ denote the maximum likelihood estimator under $\mathcal{H}_0$, and let $\hat{\boldsymbol{\theta}}_1$ and $\hat{\boldsymbol{\theta}}_2$ denote the maximum likelihood estimators under $\mathcal{H}_1$ before and after the change point respectively, where we assume that the segment parameters are independent across segments. Then we can define the generalised likelihood ratio test as follows.

Definition 2.7 (Generalised Likelihood Ratio Test Statistic). *For a given time series $\mathbf{X}$ of length n , the Generalised Likelihood Ratio (GLR) test for a candidate change point $k \in \{1, \dots, n-1\}$ of $\mathcal{H}_0$ against $\mathcal{H}_1$ as in (2.5) computes*

$$\Lambda_k(\mathbf{X}) \equiv \Lambda_k = \frac{\mathcal{L}(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:n})}{\mathcal{L}(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k})\mathcal{L}(\hat{\boldsymbol{\theta}}_2; \mathbf{X}_{k+1:n})} = \frac{\mathcal{L}(\hat{\boldsymbol{\theta}}_0)}{\mathcal{L}(\hat{\boldsymbol{\theta}}_1, \hat{\boldsymbol{\theta}}_2)}.$$

Taking logarithms and multiplying by -2 results in the Generalised Log-Likelihood Ratio (GLLR) test statistic:

$$\text{LR}_k(\mathbf{X}) \equiv \text{LR}_k = -2 \ln \Lambda_k = 2[\ell(\hat{\boldsymbol{\theta}}_1, \hat{\boldsymbol{\theta}}_2) - \ell(\hat{\boldsymbol{\theta}}_0)], \quad (2.10)$$

$$= 2 \left[\ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) + \ell(\hat{\boldsymbol{\theta}}_2; \mathbf{X}_{k+1:n}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:n}) \right], \quad (2.11)$$

where we have used the independence of the observations to write the joint log-likelihood under $\mathcal{H}_1$ as the sum of the log-likelihoods of each segment.

Intuitively, $\text{LR}_k(\mathbf{X})$ compares the likelihood of the segments as a result of a change point at $t = k$ with the likelihood of the single segment model under $\mathcal{H}_0$ with no change point in the data. Where it is clear from the context, we omit the argument and write LR_k .

Under the assumption of independence of each X_t , the log-likelihood function is additive, and so we can decompose (2.11) as

$$\text{LR}_k(\mathbf{X}) = 2 \left[\ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:k}) + \ell(\hat{\boldsymbol{\theta}}_2; \mathbf{X}_{k+1:n}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{k+1:n}) \right], \quad (2.12)$$

with the first two terms comparing the likelihood of $\hat{\boldsymbol{\theta}}_1$ against $\hat{\boldsymbol{\theta}}_0$, considering only the data points up to k , and the second two comparing $\hat{\boldsymbol{\theta}}_2$ against $\hat{\boldsymbol{\theta}}_0$ for the rest of the data. This naturally leads to the following definition.

Definition 2.8 (Local Log-Likelihood Ratio Test Statistic). *The local log-likelihood ratio test statistic on any subinterval $1 \leq i < j \leq n$ of a time series $\mathbf{X} = (X_1, \dots, X_n)^\top$ is*

$$\text{LR}_{i,j}(\mathbf{X}; \hat{\boldsymbol{\theta}}_0) = 2 \left[\ell(\hat{\boldsymbol{\theta}}_{i,j}; \mathbf{X}_{i,j}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{i,j}) \right], \quad (2.13)$$

where we use $\hat{\boldsymbol{\theta}}_{i,j}$ to denote the MLE for $\boldsymbol{\theta}$ computed using the data segment $\mathbf{X}_{i,j}$, in contrast to the global estimator $\hat{\boldsymbol{\theta}}_0$, which is calculated using the entire dataset $\mathbf{X}_{1:n}$.

Combining the above definition with (2.12), we see that

$$\text{LR}_k(\mathbf{X}) = \text{LR}_{1,k}(\mathbf{X}; \hat{\boldsymbol{\theta}}_0) + \text{LR}_{k+1,n}(\mathbf{X}; \hat{\boldsymbol{\theta}}_0).$$

We now apply these definitions to derive a GLLR test statistic for the change in mean and change in variance problems. Assuming $X_t \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$, we have

$$\begin{aligned} \mathcal{L}(\mu, \sigma^2; \mathbf{X}) &= \prod_{t=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(X_t - \mu)^2}{2\sigma^2} \right], \\ &= (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{t=1}^n (X_t - \mu)^2 \right), \\ \Rightarrow \ell(\mu, \sigma^2) &= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^n (X_t - \mu)^2. \end{aligned} \quad (2.14)$$

2.2 The Generalised Log-Likelihood Ratio Test

The above holds for a segment with fixed mean and variance, so we compute the likelihood before and after the change point separately. To test for a change in mean, we assume σ^2 is known and fix μ at the segment MLE. Using the standard result that $\hat{\mu} = \bar{X}$, we have:

$$\begin{aligned}\ell(\hat{\mu}_0) &= -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^n (X_t - \bar{X}_{1:n})^2, \\ \ell(\hat{\mu}_1) &= -\frac{k}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^k (X_t - \bar{X}_{1:k})^2, \\ \ell(\hat{\mu}_2) &= -\frac{n-k}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=k+1}^n (X_t - \bar{X}_{k+1:n})^2.\end{aligned}$$

Substituting into (2.10), with cancellation, gives

$$\text{LR}_k = \frac{1}{\sigma^2} \left[\sum_{t=1}^n (X_t - \bar{X}_{1:n})^2 - \sum_{t=1}^k (X_t - \bar{X}_{1:k})^2 - \sum_{t=k+1}^n (X_t - \bar{X}_{k+1:n})^2 \right]. \quad (2.15)$$

The following useful lemma is stated without proof [13, Ch. 11].

Lemma 2.9 (Variance Decomposition Identity). *For any sequence of observations $X_1, \dots, X_n$, divided into two segments at index m , the total sum of squared errors (SS_{total}) can be decomposed into a sum of the between segment separation (SS_{between}), and the within segment sum of squared errors (SS_{within}).*

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}},$$

where

$$\begin{aligned}SS_{\text{total}} &= \sum_{t=1}^n (X_t - \bar{X}_{1:n})^2, \\ SS_{\text{between}} &= m (\bar{X}_{1:m} - \bar{X}_{1:n})^2 + (n-m) (\bar{X}_{m+1:n} - \bar{X}_{1:n})^2, \\ &= \frac{m(n-m)}{n} (\bar{X}_{1:m} - \bar{X}_{m+1:n})^2, \\ SS_{\text{within}} &= \sum_{t=1}^m (X_t - \bar{X}_{1:m})^2 + \sum_{t=m+1}^n (X_t - \bar{X}_{m+1:n})^2.\end{aligned}$$

Using lemma 2.9 with $m = k$, we see that (2.15) can be rewritten as:

$$\begin{aligned}\text{LR}_k &= \frac{1}{\sigma^2} [SS_{\text{total}} - SS_{\text{within}}], \\ &= \frac{1}{\sigma^2} SS_{\text{between}}.\end{aligned} \quad (2.16)$$

Hence (2.15) becomes

$$\text{LR}_k = \frac{1}{\sigma^2} \left[k (\bar{X}_{1:k} - \bar{X}_{1:n})^2 + (n-k) (\bar{X}_{k+1:n} - \bar{X}_{1:n})^2 \right], \quad (2.17)$$

$$\begin{aligned}&= \frac{k(n-k)}{n\sigma^2} (\bar{X}_{1:k} - \bar{X}_{k+1:n})^2, \\ &= C_k^2.\end{aligned} \quad (2.18)$$

Therefore, maximising the CUSUM statistic is equivalent to maximising the GLLR test statistic, LR_k .

We can apply the same principle to find LR_k for alternative problems. The statistic for a fixed mean, change in variance problem for i.i.d. normal data is found by applying the known result that $\hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^n (X_t - \mu)^2$ to (2.14) [14]:

$$\begin{aligned} \ell(\hat{\sigma}^2) &= -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{n}{2}, \\ &\propto -\frac{n}{2} \ln \hat{\sigma}^2. \end{aligned} \quad (2.19)$$

We can ignore the constant terms as our aim is to maximise $\ell(\hat{\sigma}^2)$, which is equivalent up to these constants. Then our GLLR test statistic is:

$$\text{LR}_k = \left[n \ln \hat{\sigma}_0^2 - \left(k \ln \hat{\sigma}_1^2 + (n - k) \ln \hat{\sigma}_2^2 \right) \right], \quad (2.20)$$

$$= k \ln \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \right) + (n - k) \ln \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_2^2} \right). \quad (2.21)$$

Intuitively from (2.20), this asks whether the difference between the variance under $\mathcal{H}_0$ and the weighted variances before and after the split (under $\mathcal{H}_1$) is significant. If it is, then this gives a higher value of LR_k , and hence more evidence for a change point at $t = k$. The formula for a change in both mean and variance is almost identical, with the MLE $\hat{\sigma}^2$ for each segment computed based on the empirical segment mean, rather than a known constant μ . The same procedure for finding LR_k can be followed for other types of change, as well as under different assumptions, including the distribution of $\mathbf{X}$. In general, this allows us to find a value $t = \hat{\tau}$ such that:

$$\begin{aligned} \hat{\tau} &= \arg \max_{k \in \{1, \dots, n-1\}} \{ \text{LR}_k \}, \\ \text{LR}_{\hat{\tau}} &= \max_{k \in \{1, \dots, n-1\}} \{ \text{LR}_k \}. \end{aligned} \quad (2.22)$$

If a change point exists within the time series, $\hat{\tau}$ represents the index at which it is most likely to occur, based on the GLLR test. Once we have a value of $\hat{\tau}$ and a statistic, $\text{LR}_{\hat{\tau}}$, we must determine whether or not this represents a change. In order to do this, we set a threshold by choosing a value of c such that a change is declared if $\text{LR}_{\hat{\tau}} > c$.

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In order to choose an appropriate threshold, c , at which we declare a change, we investigate properties of our test statistic $\text{LR}_{\hat{\tau}}$. Doing so enables us to consider the expected variation in the data under the null hypothesis, to which we can compare our test statistic, indicating whether it is a result of expected noise or an indication of a true change.

2.3.1 A Conservative Threshold

We first consider a naive threshold, which overestimates the expected noise of the distribution. To aid this, we introduce the useful theorem.

Theorem 2.10 (Wilks' Theorem [69]). *Let $\mathbf{X} = (X_1, \dots, X_n)$ be an i.i.d. sample taken from $f(x|\boldsymbol{\theta})$, where $\boldsymbol{\theta} \in \mathbb{R}^r$. Suppose that the total number of parameters in the model $\mathcal{H}_1$ is q , and*

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$\mathcal{H}_0$ specifies the values for ν parameters, leaving $q - \nu$ unknown. Then, under some regularity conditions, under $\mathcal{H}_0$, we have that as $n \rightarrow \infty$ the scaled log-likelihood ratio test statistic satisfies:

$$\text{LR} \sim \chi_\nu^2.$$

In the case of a single change point, we have that $q = 2r$, and $\nu = r$. More generally, a total of m change points results in $m + 1$ segments and so $q = (m + 1)r$. Our current procedure given by (2.22) is a test of the hypotheses in (2.5). However, this test violates the regularity conditions of Wilks' theorem, as we are attempting to test for a change at every possible value of k , resulting in an unidentifiable parameter k under $\mathcal{H}_0$. However, Wilks' theorem is applicable for a single, fixed value of $k = k^*$ (for $1 \leq k^* \leq n - 1$), where we can test the hypotheses:

$$\mathcal{H}_0 : \text{No change point exists; } \boldsymbol{\theta} = \boldsymbol{\theta}_0 \quad \forall 1 \leq t \leq n,$$

$$\mathcal{H}_1 : \text{There is a change point at time } k^*; \boldsymbol{\theta} = \begin{cases} \boldsymbol{\theta}_1 & \text{for } 1 \leq t \leq k^*, \\ \boldsymbol{\theta}_2 & \text{for } k^* < t \leq n. \end{cases}$$

Instead of considering our test statistic as $\max_{k \in \{1, \dots, n-1\}} \text{LR}_k$, we consider a single test for a change at k^* . In the case of a change in mean or variance, we have $r = 1$, which results in $\text{LR}_{k^*} \sim \chi_1^2$ by Wilks' theorem, since $\mathcal{H}_1$ specifies two parameters (e.g. μ_1 and μ_2), whereas $\mathcal{H}_0$ specifies one parameter across the data set (e.g. μ_0). If we have a change in both mean and variance, then $r = \nu = 2$, and so $\text{LR}_{k^*} \sim \chi_2^2$. After mitigating the unidentifiable k problem, we have a set of test statistics $\{\text{LR}_1, \dots, \text{LR}_{n-1}\}$, obtained by performing $n - 1$ tests, one at each candidate index.

In order to evaluate the order of a corresponding threshold, we rely on the following definition

Definition 2.11 (Familywise Error Rate). *The Familywise Error Rate (FWER) [43, Ch. 9] is the probability of making at least one Type I error when performing a hypothesis test n times. Letting V be the number of Type I errors, we say the FWER is:*

$$\mathbb{P}[V \geq 1 | \mathcal{H}_0].$$

In this context, the FWER is:

$$\mathbb{P} \left[\max_{k \in \{1, \dots, n-1\}} \text{LR}_k > c | \mathcal{H}_0 \right] = \mathbb{P} \left[\bigcup_{k=1}^{n-1} (\text{LR}_k > c | \mathcal{H}_0) \right], \quad (2.23)$$

since the global maximum test statistic exceeds the threshold if and only if at least one test statistic exceeds the threshold.

We can then bound the FWER as follows:

$$\mathbb{P} \left[\max_{k \in \{1, \dots, n-1\}} \text{LR}_k > c | \mathcal{H}_0 \right] \leq \sum_{k=1}^{n-1} \mathbb{P}[\text{LR}_k > c | \mathcal{H}_0] \quad (2.24)$$

$$= (n - 1) \mathbb{P}[\chi_\nu^2 > c], \quad (2.25)$$

where (2.24) comes from applying Boole's inequality to the right hand side of (2.23), and (2.25) is a result of the Bonferroni correction with $n - 1$ test statistics (see appendix A.1 for formal statements of both of these lemmas).

For a low probability of Type I errors, we assume the FWER takes some small value of $\alpha > 0$. Since Wilks' theorem applies asymptotically, we derive an asymptotic threshold by considering $n \rightarrow \infty$. Rearranging, we have

$$\mathbb{P}[\chi_\nu^2 > c] \leq \frac{\alpha}{n-1},$$

with the right hand side vanishing as $n \rightarrow \infty$, allowing us to use tail bounds of the χ_1^2 distribution, at which $\mathbb{P}[\chi_1^2 > c] \approx \exp(-c/2)$ [40].

$$\begin{aligned} \exp\left(-\frac{c}{2}\right) &\leq \frac{\alpha}{n-1}, \\ c &\geq 2 \ln\left(\frac{n-1}{\alpha}\right) \xrightarrow{n \rightarrow \infty} 2 \ln n. \end{aligned} \quad (2.26)$$

This implies that we should use a threshold that grows like $\ln n$ to ensure that asymptotically, the probability of incorrectly detecting a change point tends to zero. We note that the use of Boole's inequality results in an overestimate of the threshold, since equality would hold in the case of each test statistic being independent. In practice, we know that adjacent test statistics are computed based on $n-1$ shared points, and are therefore highly correlated. Ignoring this correlation results in a conservative threshold.

Interestingly, this is the same as the threshold implied by the Bayes Information Criterion.

Definition 2.12 (Bayes Information Criterion[55]). *The Bayes Information Criterion (BIC) is defined as*

$$\text{BIC} = -2\ell(\hat{\theta}) + q \ln(n),$$

where q is the number of parameters in the model. The BIC is a measure of how well data fits a model, with a lower BIC representing a better fit.

The BIC can be applied in the context of change point detection to compare two models [70]. We apply the BIC in the case of a single change point, assuming we have already calculated our GLLR test statistic $\text{LR}_{\hat{\tau}}$. For $\theta \in \mathbb{R}^r$, the BIC under the null and alternative hypotheses are given by

$$\begin{aligned} \text{BIC}_{\mathcal{H}_0} &= -2\ell(\hat{\theta}_0; \mathbf{X}_{1:n}) + r \ln(n), \\ \text{BIC}_{\mathcal{H}_1} &= -2 \left[\ell(\hat{\theta}_1; \mathbf{X}_{1:\hat{\tau}}) + \ell(\hat{\theta}_2; \mathbf{X}_{\hat{\tau}+1:n}) \right] + (2r+1) \ln(n), \end{aligned}$$

where the penalty $2r+1$ comes from estimating $\hat{\tau}$ and the r -dimensional parameter for each segment. We can then compare the BIC under $\mathcal{H}_0$ with that under $\mathcal{H}_1$. It would be sensible to declare a change if $\text{BIC}_{\mathcal{H}_1} < \text{BIC}_{\mathcal{H}_0}$, as this indicates the alternative hypothesis represents a better-fitting model.

$$\begin{aligned} &\text{BIC}_{\mathcal{H}_1} < \text{BIC}_{\mathcal{H}_0}, \\ \iff &-2 \left[\ell(\hat{\theta}_1; \mathbf{X}_{1:\hat{\tau}}) + \ell(\hat{\theta}_2; \mathbf{X}_{\hat{\tau}+1:n}) \right] + (2r+1) \ln(n) < -2\ell(\hat{\theta}_0; \mathbf{X}_{1:n}) + r \ln(n), \\ \iff &2 \left[\ell(\hat{\theta}_1; \mathbf{X}_{1:\hat{\tau}}) + \ell(\hat{\theta}_2; \mathbf{X}_{\hat{\tau}+1:n}) - \ell(\hat{\theta}_0; \mathbf{X}_{1:n}) \right] > \ln(n)(2r+1-r), \\ \iff &\text{LR}_{\hat{\tau}}(\mathbf{X}) > (r+1) \ln(n). \end{aligned} \quad (2.27)$$

To extend this to multiple change points, we can follow the same logic to get a threshold of order $O(\ln(n))$, differing only by the constant factor.

Whereas the first threshold (2.26) was obtained by modelling the testing problem across all $n - 1$ candidate change points, the BIC relies on the pre-calculated optimal $\hat{\tau}$ which maximises the likelihood of a change point. However, both result in identical thresholds of $2 \ln(n)$ for a single change in mean ($r = 1$). In practice, we set the threshold as $c = K \ln(n)$, with $K > 2$ calculated empirically, which is demonstrated in chapter 5.

While the thresholds established so far are strongly consistent, they can be overly conservative in practice. This may result in smaller, more subtle changes not being detected as they fail to surpass the strict threshold. It turns out that smaller thresholds can be used, which align more closely with the upper bound of the expected noise, while still being strongly consistent.

2.3.2 A Tighter Threshold

Whereas the previous subsection produced a conservative threshold that ignored correlation between adjacent test statistics, in this section we will consider the exact limiting behaviour of the test statistic under the null hypothesis. To do so, we view the test statistic as a process indexed by the candidate change point location. As shown in section 2.1.1, the test statistic for the change in mean problem can be expressed using the partial sums of the data as in (2.9), which can be generalised to the GLLR test statistic [4]. We show that under $\mathcal{H}_0$, these partial sums converge to a squared Brownian bridge and we can use known behaviour of Brownian motion to obtain a more precise threshold. The first step involves transforming the data onto the interval $[0, 1]$ so that we can apply known results.

Definition 2.13 (Continuous Approximation). *We define the normalised time scale $u = t/n$ for $t = 1, \dots, n$. This produces an approximation of the discrete sample $1, \dots, n$ on the continuous interval $[0, 1]$.*

Definition 2.14 (Brownian Motion [28]). *Standard Brownian Motion $W(\cdot)$ is a continuous-time stochastic process mapping each index $t \in [0, 1]$ with the scalar $W(t)$ such that:*

- $W(0) = 0$
- For $0 \leq t_1 < t_2 < \dots < t_k \leq 1$, the changes $[W(t_2) - W(t_1)], \dots, [W(t_k) - W(t_{k-1})]$ are independent multivariate Gaussian with $[W(s) - W(t)] \sim \mathcal{N}(0, s - t)$.
- For any given realisation, $W(t)$ is continuous in t with probability 1.

Definition 2.15 (Brownian Bridge [9]). *A Brownian bridge $B(u)$ on $[0, 1]$ is given by*

$$B(u) = W(u) - uW(1),$$

where $W(\cdot)$ is standard Brownian motion.

We have that $\mathbb{E}[B(u)] = 0$ and $\text{Cov}[B(u), B(v)] = \min\{u, v\} - uv$ [9]. We can now state the following theorem which connects our partial sum process framework to the Brownian bridge.

Theorem 2.16 (Functional Central Limit Theorem [28]). *Let $(\mathbf{Y}_1, \dots, \mathbf{Y}_n)$ be d -dimensional i.i.d. observations with $\mathbb{E}[\mathbf{Y}_t] = 0$ and $\text{Cov}[\mathbf{Y}_t] = \boldsymbol{\Sigma}$. Then for $u \in [0, 1]$,*

$$\frac{1}{\sqrt{n}} \boldsymbol{\Sigma}^{-1/2} \sum_{t=1}^{\lfloor nu \rfloor} \mathbf{Y}_t \rightarrow_d \mathbf{W}(\cdot),$$

where $\mathbf{W}(\cdot)$ is standard d -dimensional Brownian motion and $\rightarrow_d$ denotes convergence in distribution.

Theorem 2.17 (Continuous Mapping Theorem [28]). *Let V_n be a sequence of random elements with $V_n(\cdot) \rightarrow_d V(\cdot)$ and let $g(\cdot)$ be a continuous functional, then*

$$g(V_n(\cdot)) \rightarrow_d g(V(\cdot)).$$

Since the Functional Central Limit Theorem (FCLT) applies to any partial sum of i.i.d. random variables, we express the GLLR test statistic as a partial sum of score functions so that we may apply the FCLT and continuous mapping theorem to obtain the limiting distribution. The remainder of this subsection is based on [31] and [63].

Definition 2.18. *Let $\boldsymbol{\theta} \in \mathbb{R}^r$ and $\ell(\boldsymbol{\theta}; X_t)$ be the contribution of observation X_t to the log-likelihood. We define the first order partial derivative of $\ell(\boldsymbol{\theta}; X_t)$ as*

$$\mathbf{s}(X_t, \boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \ell(\boldsymbol{\theta}; X_t) \in \mathbb{R}^r.$$

The second order partial gives the Hessian matrix:

$$\mathbf{H}(\boldsymbol{\theta}) = \left[\frac{\partial^2 \ell(\boldsymbol{\theta}; \mathbf{X})}{\partial \boldsymbol{\theta}_i \partial \boldsymbol{\theta}_j} \right]_{ij} = \sum_{t=1}^n \frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{s}(X_t, \boldsymbol{\theta}), \quad (2.28)$$

since the total log-likelihood is a sum of the individual log-likelihoods.

Definition 2.19 (Fisher Information[13]). *The expected Fisher information is given by*

$$\mathbf{J}(\boldsymbol{\theta}) = \mathbb{E} \left[-\frac{1}{n} \mathbf{H}(\boldsymbol{\theta}) \right],$$

and we write $\hat{\mathbf{J}} \equiv \mathbf{J}(\hat{\boldsymbol{\theta}}_0)$ for the expected Fisher information at the global MLE. For true parameter $\boldsymbol{\theta}_0$, $\mathbf{J}(\boldsymbol{\theta}_0)$ can be consistently estimated by the average observed Fisher information, $-\mathbf{H}(\hat{\boldsymbol{\theta}}_0)/n$. The weak law of large numbers gives the known result $-\mathbf{H}(\hat{\boldsymbol{\theta}}_0)/n \rightarrow_p \mathbf{J}(\boldsymbol{\theta}_0)$, where $\rightarrow_p$ denotes convergence in probability.

We now introduce a partial sum score process:

$$\mathbf{Z}_n(u, \boldsymbol{\theta}) = \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nu \rfloor} \mathbf{s}(X_t, \boldsymbol{\theta}), \quad (2.29)$$

which is a stochastic process indexed by $u \in [0, 1]$, satisfying the assumptions for theorem 2.16.

Lemma 2.20. *By performing a Taylor expansion of the score function around the global MLE $\hat{\boldsymbol{\theta}}_0$, the GLLR test statistic for a change point at index k can be written as*

$$\text{LR}_k(\mathbf{X}) \approx \frac{1}{k'(1-k')} \mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0)^\top \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0) \mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0), \quad (2.30)$$

where $k' = k/n$ is the fraction of the sample for the candidate change point k , and $\mathbf{J}(\hat{\boldsymbol{\theta}}_0)$ is the expected Fisher information evaluated at the global MLE.

Theorem 2.21. *Under standard maximum likelihood regularity conditions, under $\mathcal{H}_0$, the score process converges weakly in $D^r[0, 1]$:*

$$\mathbf{Z}_n(u, \hat{\boldsymbol{\theta}}_0) \rightarrow_d \mathbf{Z}(u), \quad (2.31)$$

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where $D^r[0, 1]$ is the Skorokhod space of r -dimensional functions that are continuous from the right with limits from the left [9]. $\mathbf{Z}(u) = (Z^{(1)}(u), \dots, Z^{(r)}(u))$ is a Gaussian process with mean zero and covariance structure

$$\mathbb{E} [Z^{(i)}(u_1)Z^{(j)}(u_2)] = J_{i,j}(\min\{u_1, u_2\} - u_1u_2),$$

for $i, j \in \{1, \dots, r\}$. This means that $J_{i,i}^{-1/2}Z^{(i)}(u)$ is a Brownian bridge for each $1 \leq i \leq r$, but the bridges are correlated. Since we have established $\hat{\mathbf{J}}$ is a consistent estimator for $\mathbf{J}$, we can decorrelate them by writing

$$\hat{\mathbf{J}}^{-1/2}\mathbf{Z}_n(u, \hat{\boldsymbol{\theta}}_0) \rightarrow_d \mathbf{J}^{-1/2}\mathbf{Z}(u) = (B_1(u), \dots, B_r(u))^\top, \quad (2.32)$$

where $B_1(u), \dots, B_r(u)$ are r independent, standard Brownian bridges [32].

Note that $\hat{\mathbf{J}}^{-1/2}$ is calculated by eigen-analysis. By the continuous mapping theorem, we can substitute (2.32) into (2.30), as well as using Slutsky's theorem (appendix A.4) to substitute the consistent estimator $\hat{\mathbf{J}}$ for $\mathbf{J}$. We then obtain

$$\begin{aligned} \frac{1}{u(1-u)}\mathbf{Z}(u)^\top \mathbf{J}^{-1}\mathbf{Z}(u) &= \frac{1}{u(1-u)} \left[\mathbf{J}^{1/2}(B_1(u), \dots, B_r(u))^\top \right]^\top \mathbf{J}^{-1} \left[\mathbf{J}^{1/2}(B_1(u), \dots, B_r(u))^\top \right] \\ &= \frac{1}{u(1-u)} (B_1(u), \dots, B_r(u)) \mathbf{J}^{1/2} \mathbf{J}^{-1} \mathbf{J}^{1/2} (B_1(u), \dots, B_r(u))^\top \\ &= \frac{1}{u(1-u)} \sum_{j=1}^r B_j(u)^2, \end{aligned}$$

where we have used that $\mathbf{J}$ is a covariance matrix and therefore symmetric, so $\mathbf{J}^{1/2}$ is also symmetric and that $\mathbf{J}^{1/2} \mathbf{J}^{-1} \mathbf{J}^{1/2} = \mathbf{I}$. We then have that

$$\text{LR}_{[nu]}(\mathbf{X}) \rightarrow_d \frac{1}{u(1-u)} \sum_{j=1}^r B_j(u)^2. \quad (2.33)$$

Therefore, under $\mathcal{H}_0$, the test statistic converges in distribution to the supremum of a sum of squared standard Brownian bridges. By [15] a standard Brownian bridge has maximum fluctuation less than $\sqrt{2 \ln \ln(n)}$ as $n \rightarrow \infty$. As a result, a threshold that grows with n at a rate of $2 \ln \ln n$ can be used for the test statistic. Similarly to the conservative threshold, which had equivalent order to the BIC penalty, this threshold has the same order as the penalty in the Hannan-Quinn information criterion (HQIC) [29], given by $\text{HQIC} = -2\ell(\boldsymbol{\theta}) + 2q \ln \ln n$, where q is the number of parameters. In both cases, the GLLR test statistic matches the order of the penalty of an established information criterion. In practice, we avoid the instability at the boundaries as $u \rightarrow 0$ or $u \rightarrow 1$ by use of a trimming parameter, ϵ . The GLLR test statistic is then given by

$$\sup_{u \in [\epsilon, 1-\epsilon]} \text{LR}_{[nu]}(\mathbf{X}) \rightarrow_d \sup_{u \in [\epsilon, 1-\epsilon]} \frac{1}{u(1-u)} \sum_{j=1}^r B_j(u)^2, \quad (2.34)$$

and we can use exact critical values tabulated by Andrews [4] as our threshold, dependent on the choice of ϵ .

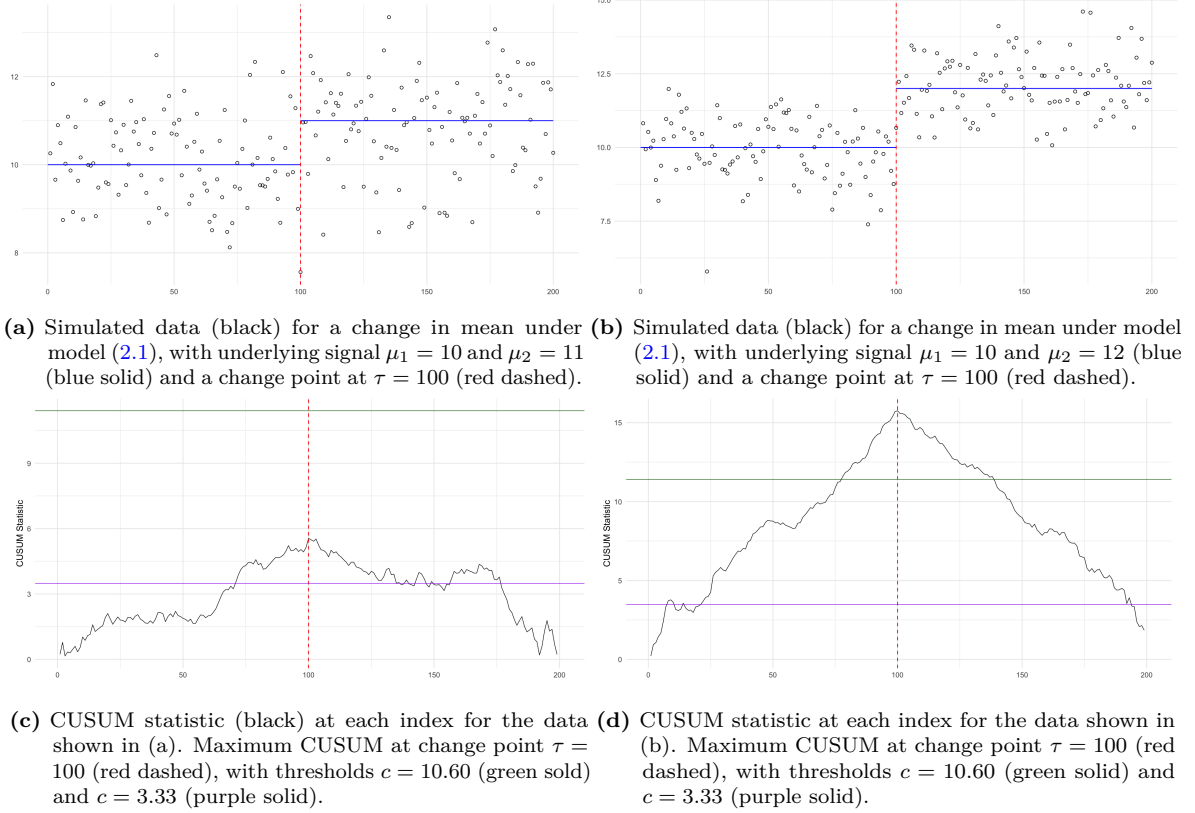


Figure 2.3: Application of the 2 derived thresholds $2 \ln(n)$ (green) and $2 \ln(\ln(n))$ (purple) on change in mean data for fixed n , from example 2.22.

2.3.3 Applying Thresholds

We have derived two theoretical thresholds for the GLLR test statistic, each of which has different practical benefits. For data that deviates from the model assumptions, for example non-normality (tested via a quantile-quantile plot), choosing the conservative threshold protects against false positives. On the other hand, since the conservative threshold may result in missing more subtle changes. The following example demonstrates the latter case.

Example 2.22. We model a change in mean for $n = 200$ with $\tau = 100$ in two cases with $\sigma^2 = 1$. In the first, the mean increases from 10 to 11, in the second it increases from 10 to 12 (figure 2.3). The conservative threshold is $c = 2 \ln n \approx 10.60$, and the tighter threshold is $c = 2 \ln \ln n \approx 3.33$. In the first case (smaller change), the conservative threshold does not detect the change, but the tighter threshold does. However, in the latter case (larger change), both thresholds detect the change. This demonstrates a limitation of the conservative threshold: it has weaker statistical power, and so is more likely to miss smaller changes. As n increases, the difference between the two thresholds diverges, and so this limitation becomes more apparent.

2.3.4 Detectability

Up to this point, we have established that a change is declared if $\text{LR}_{\hat{\tau}} > c$. While this threshold provides a criterion for rejecting the null hypothesis, it does not guarantee that every change point will be detected. The detectability of a change is not only determined by the global sample size n , but also the local properties of the change. Under the assumption that a change occurs at index τ , it is intuitive that the jump size of a change, which we define as $\delta := \theta_2 - \theta_1$, and the smallest segment length, $\Delta := \min\{\tau, n - \tau\}$, will impact its detectability. A short-lived

change is difficult to distinguish from a random anomaly, and a change with a small jump size can be mistaken for expected noise. This can be seen in example 2.22, where the more conservative threshold detects the change from $\mu = 10$ to $\mu = 12$, but fails to detect the change from $\mu = 10$ to $\mu = 11$, despite both cases consisting of $n = 200$ data points, highlighting the effect of jump size on detectability.

By [23], for any distribution in the exponential family, we can rewrite the local likelihood ratio test statistic (2.13) as

$$\text{LR}_{i,j}(\mathbf{X}, \hat{\boldsymbol{\theta}}_0) = (j - i + 1)D(\hat{\boldsymbol{\theta}}_{i,j}, \hat{\boldsymbol{\theta}}_0), \quad (2.35)$$

where D represents the Kullback-Leibler divergence, which can be thought of in this context as a measure of dissimilarity of the distributions under each parameter. That is, the greater the difference between the new local parameter $\hat{\boldsymbol{\theta}}_{i,j}$ and the global parameter $\hat{\boldsymbol{\theta}}_0$, the larger the value of $D(\hat{\boldsymbol{\theta}}_{i,j}, \hat{\boldsymbol{\theta}}_0)$.

Applying this to the generalised log-likelihood ratio test for a candidate change point k , we have:

$$\begin{aligned} \text{LR}_k &= \text{LR}_{1,k}(\mathbf{X}_{1:k}, \hat{\boldsymbol{\theta}}_0) + \text{LR}_{k+1,n}(\mathbf{X}_{k+1:n}, \hat{\boldsymbol{\theta}}_0), \\ &= 2kD(\hat{\boldsymbol{\theta}}_{1:k}, \hat{\boldsymbol{\theta}}_0) + 2(n - k)D(\hat{\boldsymbol{\theta}}_{k+1:n}, \hat{\boldsymbol{\theta}}_0). \end{aligned}$$

The factorisation into these two multiplicative components demonstrates the properties of the change that determine the magnitude of this statistic, and therefore whether it will exceed the threshold. The first of these, $(j - i + 1)$, represents the length of the interval. The longer the interval, the more evidence that accumulates under the new parameter. The other factor represents the intensity, so that a larger change is easier to identify.

The exact details of D will be omitted, but equations (2.17) and (2.21) show examples of this multiplicative relationship, where the length of each segment (k and $n - k$) is multiplied with some comparison of $\hat{\boldsymbol{\theta}}_1$ and $\hat{\boldsymbol{\theta}}_2$ with $\hat{\boldsymbol{\theta}}_0$. The sum of these then produces LR_k , as above. Therefore, for a change of small magnitude to be detected, the duration of the change must be longer in order to increase the statistic $\text{LR}_{i,j}$ for that segment. Similarly, a change which only lasts for a short time period must have a larger jump size to be detected. Considering the signal to noise ratio (SNR), the signal is amplified by larger values of $\boldsymbol{\delta}$ and Δ , whereas under the Brownian bridge model (2.33), the variance weighting $(u(1 - u))^{-1}$ explodes at the edges. As a result, the power of the generalised likelihood ratio test is non-uniform across the domain. Changes that occur near the boundaries, where mean estimates have few data points and so high variance, are more difficult to distinguish from stochastic noise since Δ may be too small to accumulate a strong signal. This can be observed by the characteristic drop in statistical power at the boundaries. This can be mitigated by applying a minimum segment length, as discussed in section 2.1.2.

2.4 Multivariate and High-Dimensional Extensions

2.4.1 Multivariate Data

We can extend the change point detection techniques from the univariate setting and apply them to a vector of observations. This section is largely based on [59], with the model as follows.

Definition 2.23 (Multivariate Signal-Plus-Noise Model). *Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top \in \mathbb{R}^{n \times d}$ be the observation matrix of d -dimensional vector observations $\mathbf{X}_t$. We assume that each $\mathbf{X}_t$ can*

be written as:

$$\mathbf{X}_t = \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t, \quad (2.36)$$

such that $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\varepsilon}_t \stackrel{i.i.d.}{\sim} \mathcal{N}_d(\mathbf{0}, \boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ a positive definite covariance matrix.

We first assume the population covariance matrix, $\boldsymbol{\Sigma}$ is known and fixed, resulting in the multivariate single change in mean problem. To formulate our multivariate test statistic, we use the likelihood function for the multivariate normal distribution:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{X}_1, \dots, \mathbf{X}_n) &= \prod_{t=1}^n f(\mathbf{X}_t | \boldsymbol{\mu}, \boldsymbol{\Sigma}), \\ &= (2\pi)^{-nd/2} |\boldsymbol{\Sigma}|^{-n/2} \exp\left(-\frac{1}{2} \sum_{t=1}^n (\mathbf{X}_t - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_t - \boldsymbol{\mu})\right). \end{aligned} \quad (2.37)$$

For known, constant variance, we then have:

$$\ell(\boldsymbol{\mu} | \boldsymbol{\Sigma}, \mathbf{X}_1, \dots, \mathbf{X}_n) \propto -\frac{1}{2} \sum_{t=1}^n (\mathbf{X}_t - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}_t - \boldsymbol{\mu}).$$

Following almost identical logic to (2.15), we see

$$\text{LR}_k = \frac{k(n-k)}{n} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n})^\top \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n}), \quad (2.38)$$

which is the scaled squared Mahalanobis distance between the two sample means. We note that this is a direct parallel to the univariate case, following a χ_d^2 distribution. In the case of unknown, constant variance, we can replace $\boldsymbol{\Sigma}$ with an estimator $\widehat{\boldsymbol{\Sigma}}$ [59]. Under $\mathcal{H}_0$, we have that the sample covariance matrix, given by

$$\frac{1}{n-1} \mathbf{V} = \frac{1}{n-1} \sum_{t=1}^n (\mathbf{X}_t - \bar{\mathbf{X}})(\mathbf{X}_t - \bar{\mathbf{X}})^\top,$$

is an unbiased estimator for $\boldsymbol{\Sigma}$. On the other hand, under $\mathcal{H}_1$, if we assume a change point occurs at time k , then the sample pooled covariance is a better estimate for $\boldsymbol{\Sigma}$, given by:

$$\mathbf{W}_k = \frac{1}{n-2} \left[\sum_{t=1}^k (\mathbf{X}_t - \bar{\mathbf{X}}_{1:k})(\mathbf{X}_t - \bar{\mathbf{X}}_{1:k})^\top + \sum_{t=k+1}^n (\bar{\mathbf{X}}_t - \bar{\mathbf{X}}_{k+1:n})(\bar{\mathbf{X}}_t - \bar{\mathbf{X}}_{k+1:n})^\top \right].$$

[59] shows that maximising the following test statistic, known as the Hotelling's T^2 statistic, obtained by substituting $\widehat{\boldsymbol{\Sigma}} = \mathbf{W}_k$ in (2.38) is equivalent to maximising the generalised log-likelihood ratio test statistic:

$$T_k^2 = \frac{k(n-k)}{n} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n})^\top \mathbf{W}_k^{-1} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n}). \quad (2.39)$$

However, calculating a covariance matrix $\mathbf{W}_k$ at each candidate change point is computationally very slow. This is mitigated by introducing a new statistic, D_k , which instead uses $\widehat{\boldsymbol{\Sigma}} = \mathbf{V}$:

$$D_k = \frac{k(n-k)}{n} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n})^\top \mathbf{V}^{-1} (\bar{\mathbf{X}}_{1:k} - \bar{\mathbf{X}}_{k+1:n}).$$

By [59], it holds that

$$D_k = \frac{T_k^2}{n - 2 + T_k^2},$$

and so maximising D_k is equivalent to maximising T_k^2 by monotonicity. The $\mathbf{W}_k$ estimators for $\boldsymbol{\Sigma}$ are replaced by the global sum of squares matrix $\mathbf{V}$, which only needs to be calculated once. Moreover, this also implies only calculating a single matrix inverse, as opposed to one for each candidate change as in (2.39). Under the null hypothesis, it can be shown that $D_k \sim \beta(\frac{1}{2}d, \frac{1}{2}(n - d - 1))$, so that thresholds can be calculated exactly.

We now consider the case of a change in variance. Using appendix A.5, the likelihood for a change in variance evaluated at the MLE $\widehat{\boldsymbol{\Sigma}}$ (2.37) becomes:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\mu}, \widehat{\boldsymbol{\Sigma}} | \mathbf{X}_1, \dots, \mathbf{X}_n) &= (2\pi)^{-nd/2} |\widehat{\boldsymbol{\Sigma}}|^{-n/2} \exp\left(-\frac{1}{2}nd\right), \\ &\propto |\widehat{\boldsymbol{\Sigma}}|^{-n/2}, \\ \ell(\boldsymbol{\mu}, \widehat{\boldsymbol{\Sigma}} | \mathbf{X}_1, \dots, \mathbf{X}_n) &\propto -\frac{n}{2} \ln |\widehat{\boldsymbol{\Sigma}}|. \end{aligned}$$

This results in the GLLR test statistic

$$\text{LR}_k = n \ln |\widehat{\boldsymbol{\Sigma}}_{1:n}| - k \ln |\widehat{\boldsymbol{\Sigma}}_{1:k}| - (n - k) \ln |\widehat{\boldsymbol{\Sigma}}_{k+1:n}|,$$

where

$$\widehat{\boldsymbol{\Sigma}}_{i:j} = \frac{1}{j - i + 1} \sum_{t=i}^j (\mathbf{X}_t - \overline{\mathbf{X}}_{i:j})(\mathbf{X}_t - \overline{\mathbf{X}}_{i:j})^\top.$$

This is again analogous to the univariate case, (2.20). Applying a degrees of freedom correction for calculating the unknown segment means results in the following formula for the change in both mean and variance case [30]:

$$\text{LR}_k = (n - 1) \ln |\widehat{\boldsymbol{\Sigma}}_{1:n}| - (k - 1) \ln |\widehat{\boldsymbol{\Sigma}}_{1:k}| - (n - k - 1) \ln |\widehat{\boldsymbol{\Sigma}}_{k+1:n}|.$$

In all cases outlined in this section, the multivariate GLLR test statistic aggregates signal across multiple dimensions, such that it may detect subtle shifts which may have been missed in the case of univariate independent tests. This idea of building strength is a result of the covariance matrix standardising the signal to noise ratio across dimensions by accounting for cross-dimensional dependence.

2.4.2 High-Dimensional Data

In the previous section, we derived multivariate test statistics involving the inverse of the covariance matrix, which can only be calculated if the matrix is non-singular. In the case that we have more dimensions than observations, $d > n$, the covariance matrix is singular, since it is a $d \times d$ matrix of rank at most $n < d$. In some cases, we can apply shrinkage, such as Ridge regularisation, which adds a diagonal matrix to the sample covariance to produce a positive definite covariance matrix [42]. However, if we have that $d \gg n$, this is often ineffective as the accumulated noise can dominate the test statistic [67]. To overcome this we assume sparsity, that is, assuming that a change occurs in only a subset of the d dimensions. This is applicable in contexts such as a small subset of genes undergoing a mutation. By applying dimension reduction through projection, we isolate the subset of rows undergoing a structural change, and filter out the noise of dimensions without a change point in order to maximise the signal to noise ratio.

2.4.3 INSPECT

As we showed, the CUSUM statistic is a special case of the GLLR test statistic in one dimension. The INSPECT algorithm [67] performs dimension reduction on a CUSUM matrix, producing a one-dimensional sequence of CUSUM statistics. Then, we compare the maximum absolute statistic against a threshold, and take the corresponding index to be our estimator for the change point. The aim of the algorithm is to identify the optimal sparse vector on which to project the matrix in order to maximise the signal to noise ratio in this univariate CUSUM sequence. This subsection is based on [67].

Definition 2.24 (CUSUM Matrix). *Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top \in \mathbb{R}^{n \times d}$ be a vector time series of length $n \geq 2$, with each $\mathbf{X}_t \in \mathbb{R}^d$. Then the CUSUM matrix is given by $\mathcal{T}(\mathbf{X})$, where*

$$[\mathcal{T}(\mathbf{X})]_{j,t} = \sqrt{\frac{t(n-t)}{n}} \left(\frac{1}{n-t} \sum_{r=t+1}^n \mathbf{X}_{j,r} - \frac{1}{t} \sum_{r=1}^t \mathbf{X}_{j,r} \right),$$

for $j = 1, \dots, d$ and $t = 1, \dots, n-1$.

We assume our time series is modelled as specified in (2.36), with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_d$ and σ^2 known. Writing in matrix form, we have:

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{W}, \quad (2.40)$$

where $\mathbf{W}$ is a $d \times n$ random matrix with independent $\mathcal{N}_d(0, \sigma^2 \mathbf{I}_d)$ columns. Since CUSUM is a linear operator, we can apply it to each term of (2.40). Defining $\mathbf{T} := \mathcal{T}(\mathbf{X})$, $\mathbf{A} := \mathcal{T}(\boldsymbol{\mu})$, and $\mathbf{E} := \mathcal{T}(\mathbf{W})$, we obtain:

$$\mathbf{T} = \mathbf{A} + \mathbf{E},$$

with $\mathbf{T}, \mathbf{A}, \mathbf{E} \in \mathbb{R}^{d \times (n-1)}$. Then $T_{j,t}^2$ is equivalent to the GLLR test statistic for testing a change in mean at index t of row j of $\mathbf{X}$, by (2.18). The INSPECT algorithm uses the raw CUSUM matrix $\mathbf{T}$ without squaring to preserve the direction of the shift, which is required in constructing the optimal direction for projection. We assume the change is sparse, noting that the INSPECT algorithm does not require knowledge of the level of sparsity or which rows undergo a change, and it can also be used for non-sparse changes. The aim is to aggregate the rows of $\mathbf{X}$ in such a way that the signal of the rows undergoing a change is amplified, and the noise of the other rows is minimised.

Under the alternative hypothesis that there is a change in mean at index τ in a sparse subset of rows of $\mathbf{X}$, we denote the difference in means by $\boldsymbol{\delta}$. That is, $\boldsymbol{\delta} := \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1$. If $\boldsymbol{\delta}$ were known, then the SNR would be maximised by the unit length projection in the direction of $\boldsymbol{\delta}$:

$$\frac{\boldsymbol{\delta}^\top}{\|\boldsymbol{\delta}\|_2} \mathbf{A}.$$

In practice, both $\boldsymbol{\delta}$ and $\mathbf{A}$ are unknown. The elements of $\mathbf{A}$ can be written explicitly as:

$$\begin{aligned} \mathbf{A} &= \begin{cases} \sqrt{\frac{t}{n(n-t)}}(n-\tau)\boldsymbol{\delta} & \text{for } 1 \leq t \leq \tau, \\ \sqrt{\frac{n-t}{nt}}\tau\boldsymbol{\delta} & \text{for } \tau+1 \leq t \leq n-1, \end{cases} \\ &= \boldsymbol{\delta} \boldsymbol{\gamma}^\top \end{aligned}$$

where

$$\boldsymbol{\gamma}^\top = \frac{1}{\sqrt{n}} \left(\sqrt{\frac{1}{n-1}}(n-\tau), \sqrt{\frac{2}{n-2}}(n-\tau), \dots, \sqrt{\tau(n-\tau)}, \sqrt{\frac{n-\tau-1}{\tau+1}}\tau, \dots, \sqrt{\frac{1}{n-1}}\tau \right).$$

Lemma 2.25 (Singular Value Decomposition [26]). *For a rank q matrix $\mathbf{Q}$, the compact singular value decomposition (SVD) of $\mathbf{Q}$ is*

$$\mathbf{Q} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \sum_{i=1}^q \sigma_i \mathbf{u}_i \mathbf{v}_i^\top,$$

with $\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_q)$. The leading left singular vector $\mathbf{u}_1$ satisfies:

$$\mathbf{u}_1 = \arg \max_{\|\mathbf{u}\|_2=1} \|\mathbf{u}^\top \mathbf{Q}\|_2.$$

Since $\mathbf{A}$ can be represented by the product of two vectors, it has rank 1. Therefore applying lemma 2.25 with $q = 1$ to $\mathbf{A}$ results in:

$$\mathbf{A} = \frac{\boldsymbol{\delta}}{\|\boldsymbol{\delta}\|_2} \sigma_1 \frac{\boldsymbol{\gamma}^\top}{\|\boldsymbol{\gamma}\|_2},$$

where $\mathbf{A}$ has one non-zero singular value, $\sigma_1 = \|\boldsymbol{\delta}\|_2 \|\boldsymbol{\gamma}\|_2$. In other words, the leading left singular vector of $\mathbf{A}$ is guaranteed to be parallel to $\boldsymbol{\delta}$. In practice, we observe only $\mathbf{T}$, not $\mathbf{A}$. By our assumptions on $\mathbf{W}$, the leading left singular vector of $\mathbf{T}$ should approximate that of $\mathbf{A}$. If we know the sparsity level of the change (s), such that $\|\boldsymbol{\delta}\|_0 \leq s$, then by the condition in lemma 2.25 the optimal vector would be:

$$\hat{\mathbf{u}} = \arg \max_{\tilde{\mathbf{u}} \in \mathbb{S}^{d-1}(s)} \|\tilde{\mathbf{u}}^\top \mathbf{T}\|_2 = \arg \max_{\tilde{\mathbf{u}} \in \mathbb{S}^{d-1}(s), \mathbf{w} \in \mathbb{S}^{n-2}} (\tilde{\mathbf{u}}^\top \mathbf{T} \mathbf{w}),$$

where $\mathbb{S}^{d-1}(s)$ is the set of unit vectors of length d with exactly s non-zero entries and $\mathbb{S}^{n-2}$ is the set of unit vectors of length $n - 1$. This is non-convex and hard to implement. Considering the dimensions of $\tilde{\mathbf{u}}^\top \mathbf{T} \mathbf{w}$, we see that this is a scalar and is therefore equal to its trace. By the cyclic property of the trace, we have that $\text{tr}(\tilde{\mathbf{u}}^\top \mathbf{T} \mathbf{w}) = \text{tr}(\mathbf{w} \tilde{\mathbf{u}}^\top \mathbf{T})$ and so we can formulate the objective function in matrix notation. Defining $\mathbf{M} := \tilde{\mathbf{u}} \mathbf{w}^\top$ for $\tilde{\mathbf{u}} \in \mathbb{S}^{d-1}$, $\mathbf{w} \in \mathbb{S}^{n-2}$, and

$$\mathcal{M} := \{\mathbf{M} \in \mathbb{R}^{d \times (n-1)} \mid \text{rank}(\mathbf{M}) = 1, \|\mathbf{M}\|_* = 1, \mathbf{M} \text{ has at most } s \text{ nonzero rows}\},$$

we then have

$$\hat{\mathbf{M}} = \arg \max_{\mathbf{M} \in \mathcal{M}} \langle \mathbf{T}, \mathbf{M} \rangle,$$

from which $\hat{\mathbf{u}}$ can be extracted as the leading left singular vector of $\hat{\mathbf{M}}$. Note that although this objective function is linear, the problem is non-convex as $\mathcal{M}$ is a non-convex set. Wang and Samworth rewrite this as the convex optimisation problem:

$$\hat{\mathbf{M}} = \arg \max_{\mathbf{M} \in \mathcal{S}} \{\langle \mathbf{T}, \mathbf{M} \rangle - \lambda \|\mathbf{M}\|_1\}, \quad (2.41)$$

where $\lambda > 0$ is a tuning parameter, and $\mathcal{S} = \{\mathbf{M} \in \mathbb{R}^{d \times (n-1)} : \|\mathbf{M}\|_* \leq 1\}$. The non-convex condition $\text{rank}(\mathbf{M}) = 1$ can be relaxed to the constraint $\|\mathbf{M}\|_* \leq 1$. Similarly, the non-convex L_0 constraint that requires $\mathbf{M}$ to have at most s non-zero rows can be relaxed by adding the convex L_1 penalty. Both of these relaxations represent the exact convex envelopes of the respective non-convex functions [20, 60]. As well as being non-convex, the L_0 constraint requires prior knowledge about which rows of $\mathbf{M}$ are non-zero. Without this, even with s known, the

combinatorial problem of choosing these rows is NP-hard. The use of the L_1 norm bypasses this, with the optimal subset of non-zero rows determined by tuning λ .

The matrix $\widehat{\mathbf{M}}$ is obtained with a method called Alternating Direction Method of Multipliers (ADMM). It works by creating two new matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ and alternating between optimising $\mathbf{M}_1$ with respect to the nuclear norm constraint while holding $\mathbf{M}_2$ fixed, and optimising $\mathbf{M}_2$ with respect to the L_1 constraint while fixing $\mathbf{M}_1$. It updates a Lagrange multiplier at each step that penalises the difference $\mathbf{M}_1 - \mathbf{M}_2$, repeating these steps until this difference converges to zero, resulting in the matrix $\widehat{\mathbf{M}}$ which is of low rank and sparse, by optimisation of $\mathbf{M}_1$ and $\mathbf{M}_2$ respectively. We now consider the detectability of the signal which, as in section 2.3.4,

Algorithm 1 The INSPECT Algorithm

Require: $\mathbf{X} \in \mathbb{R}^{d \times n}$, $\lambda > 0$, $c > 0$

- 1: $\mathbf{T} \leftarrow \mathcal{T}(\mathbf{X})$ ▷ Perform CUSUM transform
 - 2: $\widehat{\mathbf{M}} \leftarrow \text{ADMM}(\mathbf{T}, \lambda)$ ▷ Solves convex relaxation of (2.41)
 - 3: $\widehat{\mathbf{u}} \leftarrow \arg \max_{\|\tilde{\mathbf{u}}\|_2=1} \|\tilde{\mathbf{u}}^\top \widehat{\mathbf{M}}\|_2$ ▷ Sparse SVD to find estimator for optimal projection vector
 - 4: $\widehat{\tau} \leftarrow \arg \max_{1 \leq k \leq n-1} |\widehat{\mathbf{u}}^\top \mathbf{T}_k|$ ▷ Find time that maximises CUSUM statistic of projected data
 - 5: $\mathbf{T}_{max} \leftarrow |\widehat{\mathbf{u}}^\top \mathbf{T}_{\widehat{\tau}}|$
 - 6: **if** $\mathbf{T}_{max} > c$ **then** ▷ Compare with threshold value
 - 7: **Output:** $\widehat{\tau}$
 - 8: **end if**
-

depends on the segment lengths and the difference in means between segments. We let the signal strength be represented by $\vartheta = \|\boldsymbol{\delta}\|_2 n^{-1} \min\{\tau, n - \tau\}$. Then Theorem 3 of [67] concludes that if we have:

$$\frac{\vartheta}{\sigma} \geq C \sqrt{\frac{s \ln(d \ln n)}{n}},$$

then the estimated change point location $\widehat{\tau}$ converges to the true change point τ at a minimax optimal rate up to a factor of $\ln \ln n$. Intuitively, this represents the idea that the local signal to noise ratio of a change must exceed the statistical penalty of searching for a change across d dimensions. Recall from section 2.3.2 that the CUSUM threshold in the univariate case scales with $(\ln(\ln n))^{1/2}$, by the fact that $C_k = (\text{LR}_k)^{1/2}$. By the assumption of independence of the d dimensions, the equivalent high-dimensional threshold scales like $(\ln(d \ln n))^{1/2}$, where the $\ln d$ comes from penalising across the search space.

Typically, regularisation requires empirical cross-validation for selection of the tuning parameter λ , whereas [67] provides an optimal $\lambda = 2\sigma(\ln(d \ln n))^{1/2}$ to achieve the above minimax rate. The penalty scales with the standard deviation σ , and incorporates the high-dimensional threshold $(\ln(d \ln n))^{1/2}$ in order to shrink noise that falls below this threshold to zero, characterising it as expected deviation about the mean, as opposed to an underlying signal change.

2.5 A Bayesian Approach

The change point detection methods studied so far have been based on maximum likelihood estimation, a common frequentist framework that identifies a point estimate for a change point. In contrast, Bayesian approaches to change point analysis provide a probabilistic measure of how much evidence exists to support the existence of a change point, quantifying uncertainty about estimates. Whereas the frequentist approach uses a rigid null hypothesis stating that

there is no change, Bayesian methods specify a prior and condition on the data to build a posterior probability mass function which indicates probability of change points across the interval. Instead of searching for a fixed, unknown change point τ , the change point is treated as a random variable with a probability distribution [57]. More specifically, we say the data follows a piecewise model, represented by the i.i.d. signal-plus-noise model (2.1), with a change point at τ modelled by:

$$X_t \sim \begin{cases} f(\cdot|\boldsymbol{\theta}_1) & \text{for } t = 1, \dots, \tau, \\ g(\cdot|\boldsymbol{\theta}_2) & \text{for } t = \tau + 1, \dots, n, \end{cases}$$

where we assume that the parametric families of $f(\cdot|\boldsymbol{\theta}_1)$ and $g(\cdot|\boldsymbol{\theta}_2)$ are known but their parameters are unknown [7]. We assume τ is unknown and can take values in $\{1, \dots, n\}$, where we use $\tau = n$ to represent the case of no change point, for which the second segment is empty.

If we know the location of the change point, and the parameters before and after the change, then the probability of the data given this information is the likelihood:

$$\mathcal{L}(\mathbf{X}|\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = \prod_{t=1}^{\tau} f(X_t|\boldsymbol{\theta}_1) \prod_{t=\tau+1}^n g(X_t|\boldsymbol{\theta}_2).$$

We assume a prior on τ , $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$, assuming independence of the parameters so that $p(\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = p(\tau)p(\boldsymbol{\theta}_1)p(\boldsymbol{\theta}_2)$. Then by Bayes' theorem we have the joint posterior:

$$p(\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2|\mathbf{X}) = \frac{\mathcal{L}(\mathbf{X}|\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)p(\tau)p(\boldsymbol{\theta}_1)p(\boldsymbol{\theta}_2)}{p(\mathbf{X})},$$

where

$$p(\mathbf{X}) = \sum_{\tau=1}^n \int \int \ell(\mathbf{X}|\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)p(\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)d\boldsymbol{\theta}_1d\boldsymbol{\theta}_2,$$

which acts as a normalising constant, hence the numerator is sufficient for describing the distribution shape.

This joint posterior is not informative for detecting change points, as it does not allow us to easily make inferences about τ since it also depends on the distribution parameters $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$. We can isolate τ by marginalisation with respect to $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$:

$$\begin{aligned} p(\tau|\mathbf{X}) &= \iint p(\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2|\mathbf{X})d\boldsymbol{\theta}_1d\boldsymbol{\theta}_2, \\ &= \frac{1}{p(\mathbf{X})} \iint \mathcal{L}(\mathbf{X}|\tau, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2)p(\tau)p(\boldsymbol{\theta}_1)p(\boldsymbol{\theta}_2)d\boldsymbol{\theta}_1d\boldsymbol{\theta}_2, \\ &= \frac{p(\tau)}{p(\mathbf{X})} \int f(\mathbf{X}_{1:\tau}|\tau, \boldsymbol{\theta}_1)p(\boldsymbol{\theta}_1)d\boldsymbol{\theta}_1 \int g(\mathbf{X}_{\tau+1:n}|\tau, \boldsymbol{\theta}_2)p(\boldsymbol{\theta}_2)d\boldsymbol{\theta}_2, \end{aligned} \quad (2.42)$$

where we can split the integral by independence of the parameters τ , $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ and conditional independence of the observations given these parameters. These terms represent marginal likelihoods, defined as follows.

Definition 2.26 (Marginal Likelihood). *The marginal likelihood, also known as Bayes evidence, is a measure of model fit. For a model with prior $p(\boldsymbol{\theta})$ on parameter $\boldsymbol{\theta} \in \Omega$ and likelihood $\mathcal{L}(\boldsymbol{\theta}) = f(\mathbf{X}|\boldsymbol{\theta})$, the marginal likelihood is defined as:*

$$\mathcal{L}_M(\mathbf{X}) = \int_{\Omega} \mathcal{L}(\mathbf{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}.$$

It represents the probability of observing the data $\mathbf{X}$, averaging the likelihood across all plausible $\boldsymbol{\theta} \in \Omega$ weighted by the prior.

Therefore, (2.42) can be rewritten in terms of marginal likelihoods on each subinterval:

$$p(\tau|\mathbf{X}) = \frac{p(\tau)\mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau)\mathcal{L}_M(\mathbf{X}_{\tau+1:n}|\tau)}{p(\mathbf{X})}. \quad (2.43)$$

This resulting marginal posterior distribution represents our predictions about the location of the change point given the data, irrespective of the distribution parameters. Under the assumptions that the X_t are i.i.d. and that the prior has fixed parameters and is conjugate to the likelihood function, we can solve this integral analytically. Then we can compute the posterior probability of a change point at each value of $\tau = 1, 2, \dots, n$. The highest probabilities represent the most likely change points, and uncertainty can be represented by flat peaks across multiple values of τ .

We apply this formula to the change in mean problem, in which we assume variance is known and that our data has the following distribution:

$$X_t \sim \begin{cases} \mathcal{N}(\mu_1, 1) & \text{for } t = 1, \dots, \tau, \\ \mathcal{N}(\mu_2, 1) & \text{for } t = \tau + 1, \dots, n. \end{cases} \quad (2.44)$$

We must now specify priors for μ_1, μ_2 and τ . We assume that:

$$\mu_1, \mu_2 \sim \mathcal{N}(m_0, v_0).$$

By specifying the same prior for both cases, we ensure any change in the posterior is driven solely by the data. Since we have i.i.d. data and conjugate priors with fixed parameters for μ_1 and μ_2 , the joint posterior $p(\tau|\mathbf{X})$ is analytically tractable.

By normal-normal conjugacy (details in appendix A.6) we have that

$$\begin{aligned} \mu_1|\mathbf{X}_{1:\tau}, \tau &\sim \mathcal{N}(m_1, v_1), \\ \mu_2|\mathbf{X}_{\tau+1:n}, \tau &\sim \mathcal{N}(m_2, v_2), \end{aligned}$$

where

$$\begin{aligned} m_1 &= \frac{v_1}{v_0}m_0 + \tau v_1 \bar{X}_{1:\tau}, & v_1 &= \left(\frac{1}{v_0} + \tau\right)^{-1}, \\ m_2 &= \frac{v_2}{v_0}m_0 + (n - \tau)v_2 \bar{X}_{\tau+1:n}, & v_2 &= \left(\frac{1}{v_0} + n - \tau\right)^{-1}. \end{aligned}$$

We are also required to specify a prior for τ , representing prior beliefs about where the change point may be. Assuming that all values of $t = 1, \dots, n$ are equally likely to be the change point results in the discrete uniform prior $p(\tau) = n^{-1}$. Note that this weights the model towards a change occurring, since the prior probability of no change is $1/n$, which may not be realistic in practice. Since this prior is independent of τ we can use (2.43) to write the posterior as:

$$p(\tau|\mathbf{X}) \propto \mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau)\mathcal{L}_M(\mathbf{X}_{\tau+1:n}|\tau). \quad (2.45)$$

Substituting into the normal probability density function, we obtain

$$\mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau) = (2\pi)^{-\tau/2} \sqrt{\frac{v_1}{v_0}} \exp \left[-\frac{1}{2} \left(\sum_{t=1}^{\tau} (X_t - m_1)^2 + \frac{1}{v_0} (m_1 - m_0)^2 \right) \right],$$

2.5 A Bayesian Approach

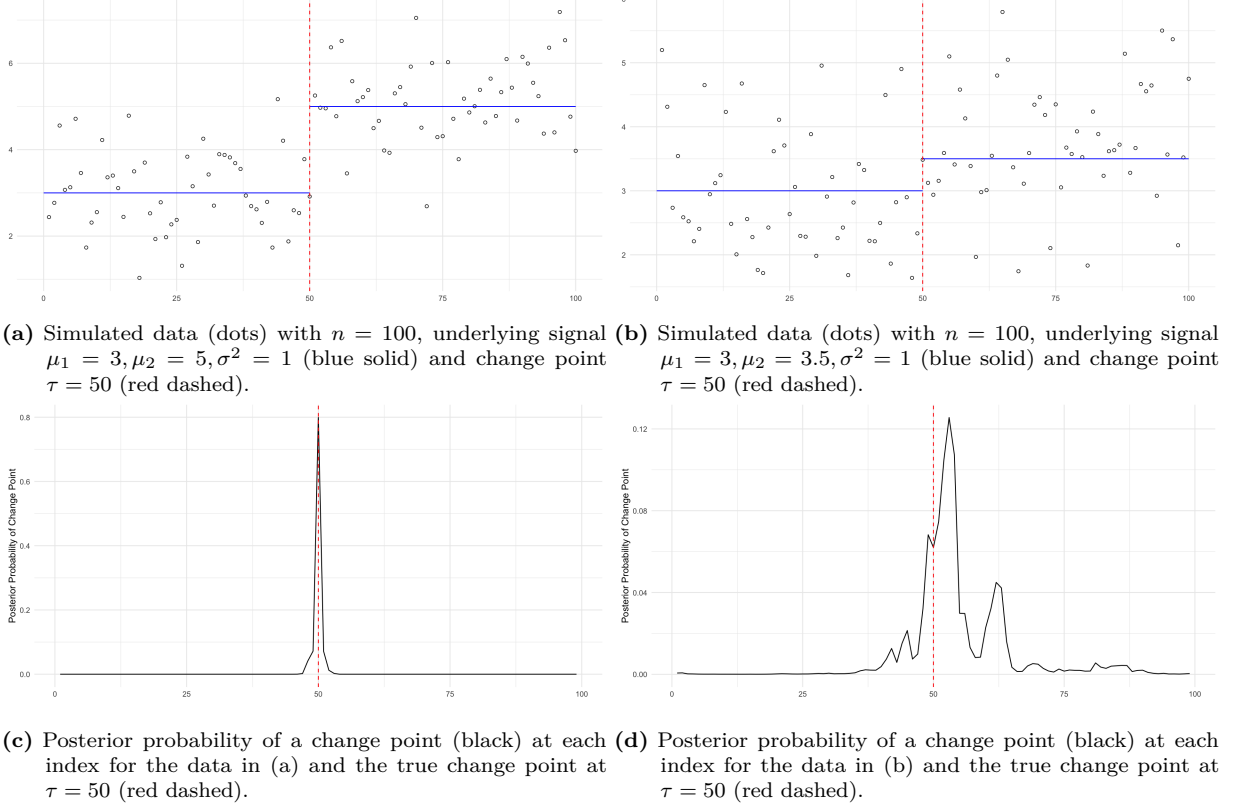


Figure 2.4: Simulation of data in example 2.27 for a larger change in mean (a) and a smaller change (b), with $\tau = 50$ (red dashed), showing the contrasting posterior distributions (c) and (d).

where the term $(X_t - m_1)^2$ represents the residuals of the data against the posterior mean (m_1), the term $(m_1 - m_0)^2/v_0$ checks how well the posterior mean fits the initial prior beliefs, and the factor $\sqrt{v_1/v_0}$ weights the likelihood by the ratio of posterior to prior variance. Applying the same logic to the post-change point data, and substituting into (2.45) gives

$$p(\tau|\mathbf{X}) \propto \sqrt{v_1 v_2} \exp \left[-\frac{1}{2} \left(\sum_{t=1}^{\tau} (X_t - m_1)^2 + \sum_{t=\tau+1}^n (X_t - m_2)^2 + \frac{(m_1 - m_0)^2 + (m_2 - m_0)^2}{v_0} \right) \right].$$

Example 2.27. If the true model has $\mu_1 = 3, \mu_2 = 5, n = 100, \tau = 50, \sigma^2 = 1$ (figure 2.4a) and we set an uninformative prior of $m_0 = 0, v_0 = 10$, then the distribution has a single sharp peak, identifying the true change point $\tau = 50$ as having the highest probability (≈ 0.8) (figure 2.4c). In contrast, an otherwise identical model with a smaller jump of $\mu_1 = 3$ to $\mu_2 = 3.5$ (figure 2.4b) gives a lower (≈ 0.12), more spread peak representing uncertainty (figure 2.4d).

The integrals in (2.42) are tractable under the assumptions that the prior is conjugate and non-hierarchical (prior parameters are constants, not estimated with hyperparameters), and that the data $\mathbf{X}$ is i.i.d. [25]. Under these assumptions, the computational complexity is $O(n)$, since we calculate the posterior at each candidate change point τ . In general, these conditions do not hold and the integral is analytically intractable, and so typically Markov Chain Monte Carlo (MCMC) simulation is used to approximate the posterior distribution.

Once we have obtained the posterior distribution $p(\tau|\mathbf{X})$, either analytically or via simulation, we can make inferences about whether a change point is likely to have occurred and which values of τ are most probable. We can then use this to estimate the distributions of the parameters θ_1 and θ_2 , quantifying our uncertainty about these estimations.

Chapter 3

Detecting Multiple Change Points

The signal-plus-noise model (2.3) naturally extends to multiple change points. We assume that we have i.i.d. Gaussian noise, as before, and that there exist m change points ($m \in \mathbb{N}$), with:

$$\tau_0 = 0 < \tau_1 < \cdots < \tau_m < \tau_{m+1} = n,$$

where we assign τ_0 and τ_{m+1} as the indices 0 and n , respectively, for notational convenience. Then we can rewrite the general change point model

$$X_t = f(t) + g(t)\zeta_t,$$

for multiple changes by setting

$$\begin{aligned} f(t) &= \mu_i \text{ for } \tau_{i-1} < t \leq \tau_i, \\ g(t) &= \sigma_i \text{ for } \tau_{i-1} < t \leq \tau_i, \\ i &\in \{1, \dots, m+1\}, \end{aligned} \tag{3.1}$$

where changes in mean and variance can occur simultaneously or independently, with a valid change point occurring for either $\mu_i \neq \mu_{i+1}$, $\sigma_i \neq \sigma_{i+1}$ or both, for each $i \in \{1, \dots, m+1\}$. As in (2.3), we have $\zeta_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$.

The GLLR test statistic can be used in the case of multiple change points. However, whereas in the single change point case we can perform a global test directly on the entire dataset, the introduction of multiple change points adds significant complexity. The main heuristic methods for detecting multiple change points can be divided into three structural classifications: top-down, bottom-up, and sliding window. While these algorithmic methods work by detecting change points one at a time, there also exist exact methods which evaluate the entire dataset simultaneously. Exact methods are guaranteed to find the optimal set of change points for the data, but are more theoretical and can be slower. The heuristic methods often give more intuition for the problem, and solution path methods are available which find change points in order of significance, thus giving a hierarchy that optimal methods do not provide. This can be beneficial if, for example, we want to rank the significance of the change points.

In some cases, we may have a pre-specified number of change points we wish to detect. This can be used as a stopping criterion in our algorithms. Alternatively, we may be estimating not only the location of change points, but also the number. In these cases, the number of change points detected is implicitly determined by our choice of threshold. In what follows, to maintain generality, we assume that the number of change points is unknown.

3.1 Top-Down Methods

Top-down methods, believed to have originated with [56], consider all of the data in the first step, treating the full time series as a single segment, and iteratively partition it into sub-segments at

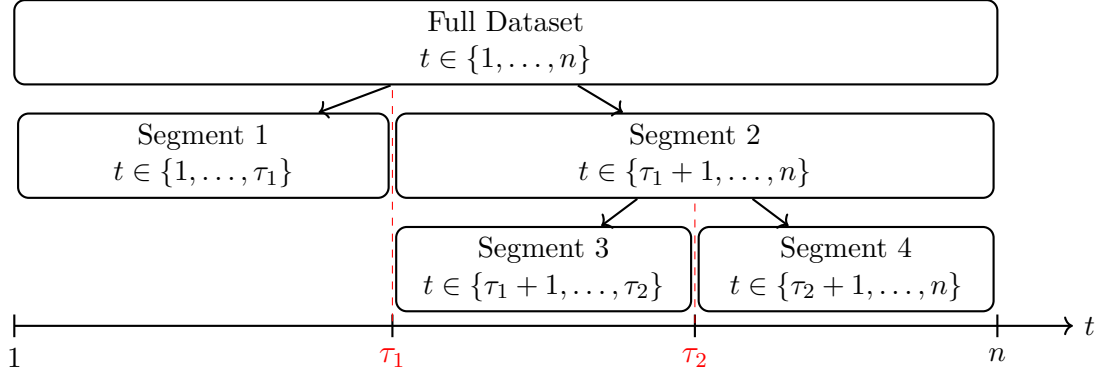


Figure 3.1: Recursive splitting in a top-down algorithm for change points at τ_1 and τ_2 (red). The full data set is first split at τ_1 , and then the right segment is split at τ_2 .

points where the maximum test statistic is greater than the threshold. This splitting continues recursively until there are no more change points detected, with the resulting tree of segments illustrated in figure 3.1 for two change points.

3.1.1 Binary Segmentation

Binary segmentation is an example of a top-down method. Its first step is to scan the entire time series to detect the first point at which to split. It is a greedy algorithm, choosing $\hat{\tau}$ as the index k which maximises LR_k at each step. If the test statistic at index $\hat{\tau}$ is greater than the threshold, it splits the data into two segments before and after this index. It repeats these steps for each segment, until there is no segment for which $\max_k \text{LR}_k > c$.

Algorithm 2 Binary Segmentation

Require: $\mathbf{X} = (X_1, \dots, X_n)$, $c > 0$

```

1:  $\mathcal{T} \leftarrow \emptyset$ 
2: function BINSEG( $a, b$ ) ▷ Input indices to search for change points in between
3:   if  $b - a < 2$  then return
4:   end if
5:    $\hat{\tau} \leftarrow \arg \max_{a \leq k < b} \text{LR}_k(\mathbf{X}_{a:b})$  ▷ Find  $\hat{\tau}$  and  $\text{LR}_{\hat{\tau}}$  by maximising GLLR test statistic
6:   if  $\text{LR}_{\hat{\tau}} > c$  then
7:      $\mathcal{T} \leftarrow \mathcal{T} \cup \{\hat{\tau}\}$ 
8:     BINSEG( $a, \hat{\tau}$ ) ▷ Recursively call to find change points before and after  $\hat{\tau}$ 
9:     BINSEG( $\hat{\tau} + 1, b$ )
10:  end if
11: end function
12: BINSEG(1,  $n$ )
13: Output:  $\mathcal{T}$ 

```

This algorithm performs $O(n)$ operations per layer, since it calculates the test statistic for each of the $n - 1$ candidate change points. Note that this assumes implementation of the pre-computed sum optimisation as specified in section 2.1.1. The expected tree depth is $\ln n$, giving an overall complexity of $O(n \ln n)$. Note that this is based on the assumption that splits will be fairly balanced, and complexity would increase closer to $O(n^2)$ in the case of multiple change points close together.

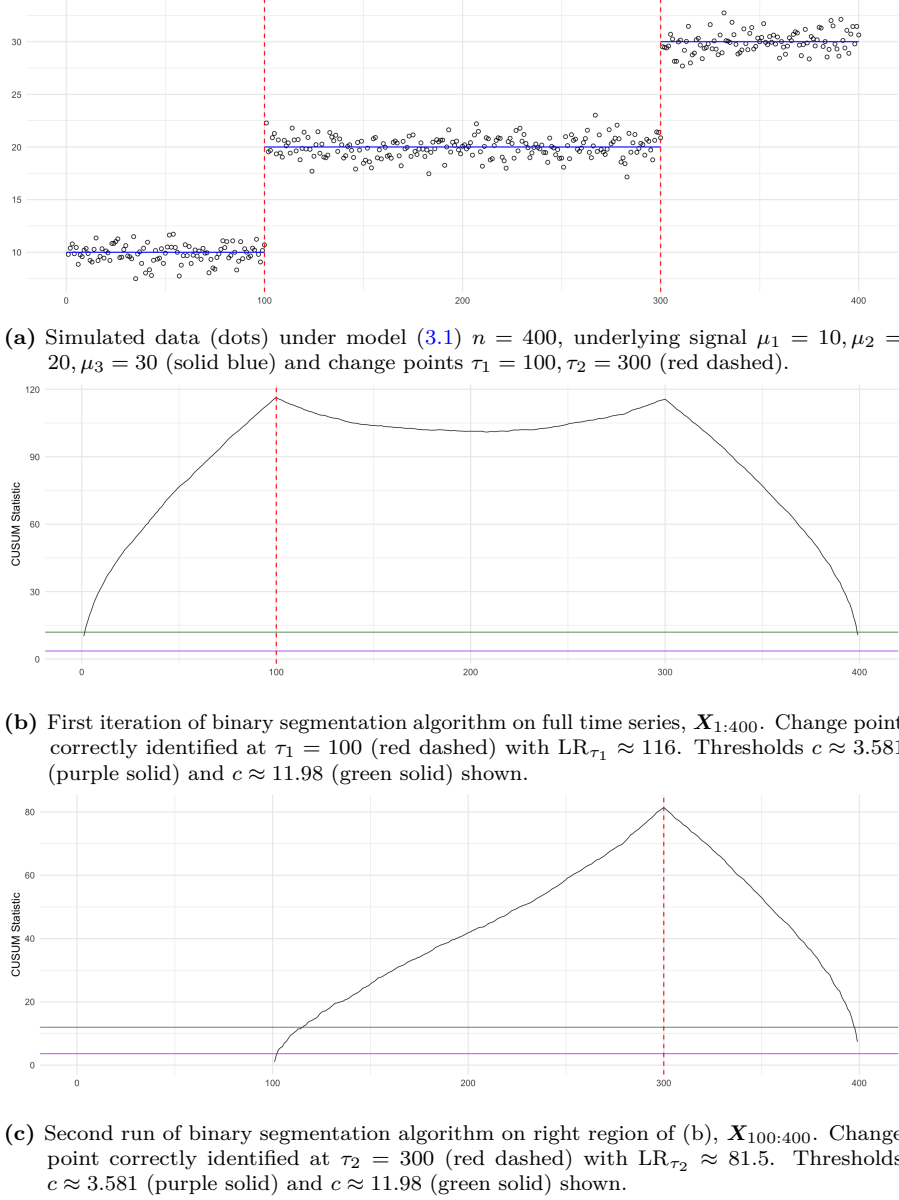


Figure 3.2: Binary segmentation simulation for example 3.1, identifying true change points $\tau_1 = 100$, $\tau_2 = 300$ (red dashed).

The first step is analogous to fitting the data (potentially containing multiple change points) to a model for a single change-point [24]. For this reason, the first step can suffer from early-termination, where the algorithm terminates after the first iteration, declaring no change points. This happens because the global variance estimate is artificially inflated due to the ill-fitting model. This reduces the GLLR test statistic which then may not surpass the threshold. A common cause of this is masking, where changes in opposing directions cancel out in the global view.

Example 3.1. Setting $n = 400$, we have potential thresholds $2 \ln(\ln(n)) \approx 3.581$ and $2 \ln n \approx 11.98$. As in figure 3.2, the two changes are correctly detected at $\tau_1 = 100$ and $\tau_2 = 300$ under both thresholds, and there are no false positives detected in either case.

3.1.2 Wild Binary Segmentation

Wild binary segmentation [24] works by sampling pairs (u_1, u_2) of integers uniformly on $[1, n]$, and creating a sub-interval $[\min(u_1, u_2), \max(u_1, u_2)] \subseteq [1, n]$ for each pair. The GLLR test statistic is then calculated locally for each random sub-interval at each candidate change point, given that the interval length is greater than the minimum segment length. If the global maximum of these across all intervals is greater than the threshold, a change point is declared at the corresponding index. Then, the data is split into two segments as in standard binary segmentation, and the process is repeated with the same intervals, using only those that fit entirely within the segment. At each step, instead of calculating a single statistic for each segment, it calculates multiple statistics, one for each interval. This is effective as by random chance there is likely to be an interval containing only one change, which mitigates the issue of early termination.

Performing wild binary segmentation with K random sub-intervals increases the complexity by a factor of K compared with regular binary segmentation, so that the expected complexity is $O(Kn \ln n)$. Despite being slower, wild binary segmentation outperforms binary segmentation, particularly for brief changes, or when changes of opposite signs occur, as demonstrated in the following example.

Example 3.2. Taking $\tau_1 = 50, \tau_2 = 55, \mu_1 = \mu_3 = 20, \mu_2 = 22$, results in figure 3.3a and we see that regular binary segmentation fails (figure 3.3b). It finds no change point, since at the first step $\max_{k \in \{1, \dots, n-1\}} \text{LR}_k$ does not surpass even the smaller threshold, $c = 2 \ln \ln(150)$. This is a result of the small values of $\delta = 2$ and $\Delta = 5$. The GLLR test statistic for each k is shown in figure 3.3b. On the other hand, wild binary segmentation correctly identifies the change points $\tau_1 = 50$ and $\tau_2 = 55$. These changes are detected in multiple of the random subintervals, with the sub-interval with the largest test statistic for each change point shown in figure 3.3c and figure 3.3d.

3.2 Bottom-Up Methods

The previous section discussed the consequences of attempting to fit a single change model to a time series containing multiple changes. As an alternative, we introduce an agglomerative set of approaches called bottom-up methods [36]. The general idea is to start by assuming a model where a change point occurs at every observation, that is, viewing each observation as its own unique segment. The algorithm then proceeds by sequentially merging adjacent segments, selecting which pair to merge based on minimising a cost function. This continues recursively until the cost of the next potential merge exceeds our threshold, c , as discussed in section 2.3. Therefore, as with top-down methods, the number of change points, m , is implicitly defined by selection of the threshold value.

We consider the naive bottom-up algorithm [36]. A naive implementation considers every possible adjacent pair at each step and so has complexity $O(n^2)$ as we expect to perform n merges. However, optimising to store the test statistics in a min-heap structure results in an improved $O(n \ln n)$ complexity, since only the test statistics adjacent to the merge point must be updated.

In the previous section, we used the GLLR test statistic to compare the likelihood of no change point to the likelihood of a change point at index k . The bottom-up algorithm we introduce performs an almost identical comparison in the opposite direction. While top-down methods

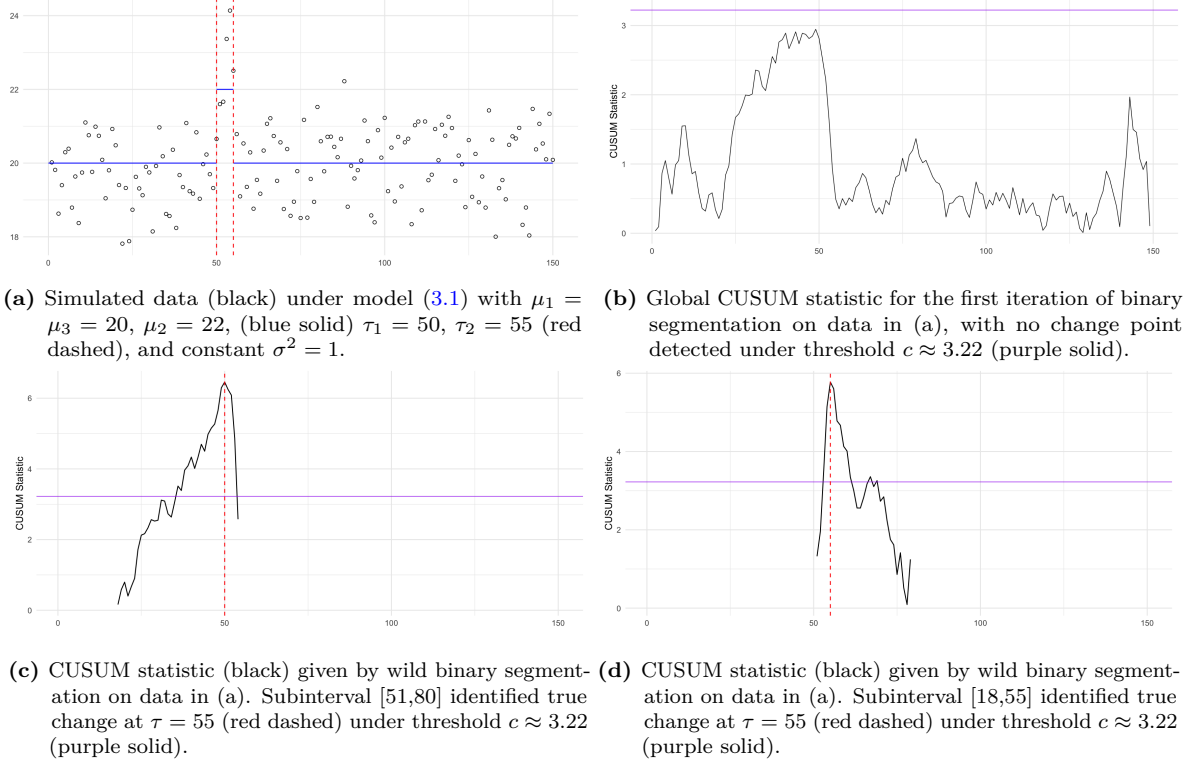


Figure 3.3: Simulated data for example 3.2 (a) with standard binary segmentation detecting no change point (b), and wild binary segmentation identifying the true change points $\tau = 50$ (c) and $\tau = 55$ (d).

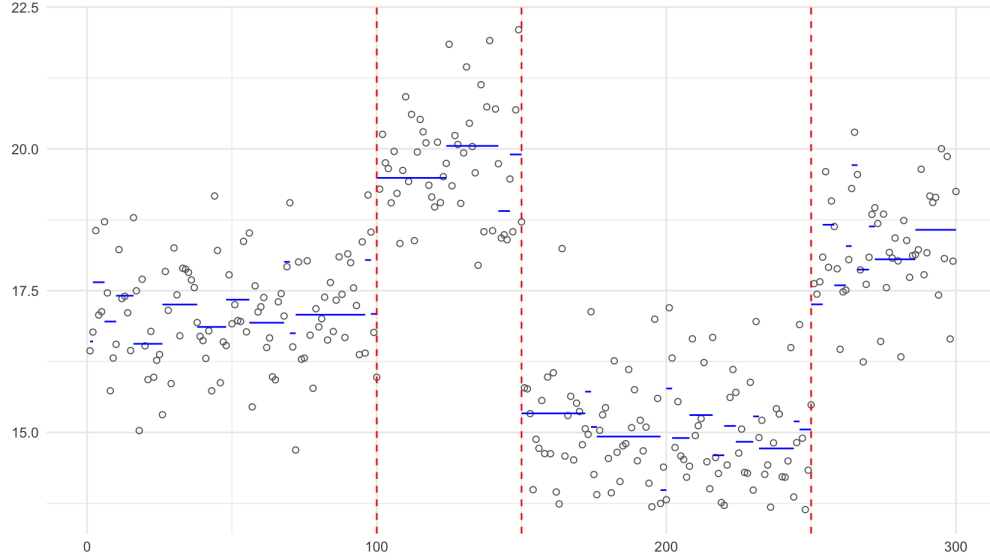
aim to maximise LR_k as they decide whether or not to split the data at index k , bottom-up methods aim to minimise LR_k , since they decide whether or not to merge.

To initialise the bottom-up algorithm, we partition the set of time indices $\{1, \dots, n\}$ into P disjoint segments $S_1, \dots, S_P$, such that $\bigcup_{i=1}^P S_i = \{1, \dots, n\}$ and $S_i \cap S_j = \emptyset$ for $i \neq j$. We define the sub-vector $\mathbf{X}_{S_i}$ of the time series as $\mathbf{X}_{S_i} = \{X_t\}_{t \in S_i}$. For the change in mean problem, we can set $P = n$ so that each initial segment is length one, whereas in the change in variance problem, each starting segment must be a minimum length of two, since variance of a single point is undefined.

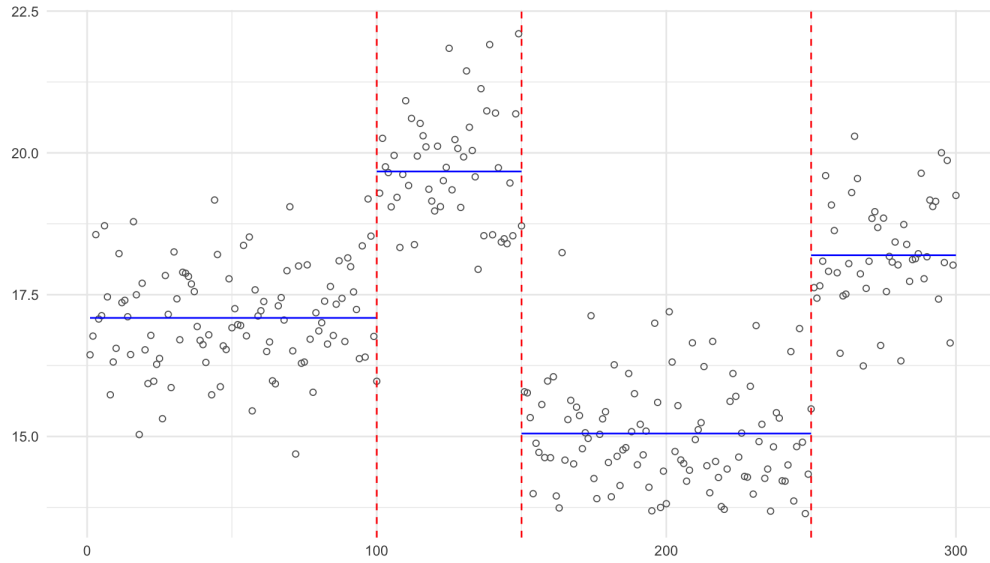
Note that the reference parameter here, $\hat{\theta}_{A \cup B}$, is explicitly recalculated locally for each pair, unlike the typical reference parameter $\hat{\theta}_0$ which is a global parameter that uses the entire series.

Example 3.3. Setting $n = 300$ and $m = 3$ so that there are three change points at $\tau_1 = 100$, $\tau_2 = 150$, $\tau_3 = 250$ with segment means $\mu_1 = 17$, $\mu_2 = 20$, $\mu_3 = 15$ and $\mu_4 = 18$, the bottom-up algorithm correctly identifies the change points as shown in figure 3.4b with an intermediate step at iteration 110 shown in figure 3.4a. A threshold of $2 \ln(n)$ is used to prevent over-segmentation.

Unlike binary segmentation, which suffers from a global masking effect, bottom-up considers local test statistics, making it highly effective in detecting shorter changes. A disadvantage of this is its tendency to over-segment the data, mistaking noise for an underlying change. To prevent this, we could initialise with larger segments or specify a minimum segment length.



(a)



(b)

Figure 3.4: Iterations of the bottom-up algorithm applied to example 3.3 with true change points at $\tau_1 = 100$, $\tau_2 = 150$, $\tau_3 = 250$ (red dashed) and disjoint segments (blue solid) representing mean of merged observations. (a) shows iteration 110, (b) shows final iteration (148) with change points only at true values.

Algorithm 3 Naive Bottom-Up

Require: $\mathbf{X} = (X_1, \dots, X_n), \mathbf{S} = (S_1, \dots, S_P), c > 0$

```

1: while TRUE do
2:    $\text{LR}_{\min} \leftarrow \infty$ 
3:   for  $i \in \{1, \dots, P-1\}$  do
4:      $A \leftarrow S_i$ 
5:      $B \leftarrow S_{i+1}$  ▷ Takes each pair of neighbouring segments
6:      $\text{LR} \leftarrow 2 \left[ \ell(\hat{\boldsymbol{\theta}}_A; \mathbf{X}_A) + \ell(\hat{\boldsymbol{\theta}}_B; \mathbf{X}_B) - \ell(\hat{\boldsymbol{\theta}}_{A \cup B}; \mathbf{X}_{A \cup B}) \right]$ 
7:     if  $\text{LR} < \text{LR}_{\min}$  then
8:        $\text{LR}_{\min} \leftarrow \text{LR}$ 
9:        $\text{merge} \leftarrow i$  ▷ Stores index at which to merge
10:    end if
11:  end for
12:  if  $\text{LR}_{\min} > c$  then ▷ Stopping criterion, all "similar" segments merged
13:    return  $\mathbf{S}$ 
14:  end if
15:   $\mathbf{S} \leftarrow (S_1, \dots, S_{\text{merge}-1}, (S_{\text{merge}} \cup S_{\text{merge}+1}), S_{\text{merge}+2}, \dots, S_P)$  ▷ Performs merge
16:   $P \leftarrow P-1$ 
17: end while

```

3.3 Sliding Window Methods

[33, 19] proposed the use of moving sum (MOSUM) statistics for the multiple change point problem (3.1). Intuitively, the MOSUM statistic can be interpreted as a local CUSUM statistic, performed in a fixed window of width $2h$, where the candidate change point k is the midpoint of the window. A direct comparison is given below:

$$C_k(\mathbf{X}) = \frac{1}{\sigma} \sqrt{\frac{k(n-k)}{n}} \left| \bar{X}_{1:k} - \bar{X}_{k+1:n} \right|,$$

$$M_k(\mathbf{X}) = \frac{1}{\sigma} \sqrt{\frac{h}{2}} \left| \bar{X}_{k-h+1:k} - \bar{X}_{k+1:k+h} \right|.$$

Since the window width is fixed, the variance normalising factor is independent of k . The local approach of MOSUM is generally more successful in detecting multiple changes than CUSUM as CUSUM is a global measure, and so suffers from the ill-fitting model problem like binary segmentation, where the statistic is contaminated with data containing multiple change points. In contrast, for a minimum segment length greater than $2h$, each MOSUM statistic will use observations spanning at most one change. Though the sliding window method is generally used as MOSUM for change in mean, the same approach of fixing a window size and comparing left and right segments can be applied to the GLLR test, and so the method is not restricted to change in mean problems. We use M_k to represent the generalisation of the MOSUM statistic.

The algorithm is outlined below, where we note that all MLEs are calculated locally, with $\hat{\boldsymbol{\theta}}_L$ denoting the MLE based on the first h points in the window, $\hat{\boldsymbol{\theta}}_R$ denoting the MLE based on the remaining h points, and $\hat{\boldsymbol{\theta}}_W$ denoting the MLE computed using all $2h$ points in the window. Note that the optimisation from section 2.1.1 applies here so that updates to the parameters can be performed in $O(1)$ time. This results in $O(n)$ overall complexity, since only a single pass of the data is required. Note that this assumes $j := |\mathcal{C}| \ll n$, since the pruning step (line 12

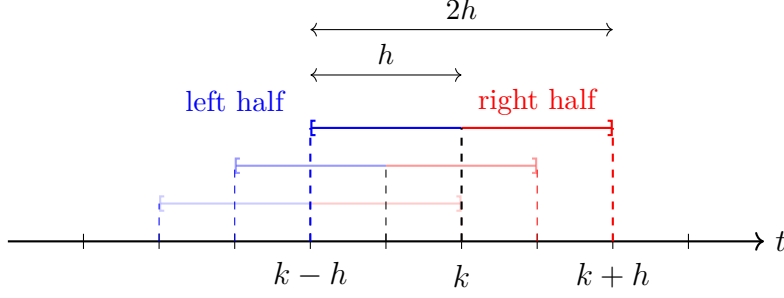


Figure 3.5: Illustration of the sliding window algorithm showing 3 iterations with the left half (blue) and right half (red) of each window separated by the candidate change point at its centre. The current iteration is shown at candidate change point k , with previous iterations in faded red and blue.

of algorithm 4) has complexity $O(j \log j)$, and so choosing a threshold too low would result in more candidate change points in $\mathcal{C}$ and a slower runtime.

Algorithm 4 Sliding Window

Require: $\mathbf{X} = (X_1, \dots, X_n), c > 0, \eta \geq 1, 1 \leq h \leq n/2$

- 1: $\mathcal{T} \leftarrow \emptyset$ ▷ Stores change points
 - 2: $\mathcal{C} \leftarrow \emptyset$ ▷ Stores potential change points before local maximisation
 - 3: **for** $k \in \{h+1, h+2, \dots, n-h\}$ **do**
 - 4: $M_k \leftarrow 2 \left[\ell(\hat{\theta}_L; \mathbf{X}_{k-h+1:k}) + \ell(\hat{\theta}_R; \mathbf{X}_{k+1:k+h}) - \ell(\hat{\theta}_W; \mathbf{X}_{k-h+1:k+h}) \right]$
 - 5: **if** $M_k > c$ **then**
 - 6: $\mathcal{C} \leftarrow \mathcal{C} \cup \{k\}$
 - 7: **end if**
 - 8: **end for**
 - 9: **while** $\mathcal{C} \neq \emptyset$ **do**
 - 10: $k^* \leftarrow \arg \max_{k \in \mathcal{C}} M_k$
 - 11: $\mathcal{T} \leftarrow \mathcal{T} \cup \{k^*\}$ ▷ Find which candidate has the highest test statistic
 - 12: $\mathcal{C} \leftarrow \mathcal{C} \setminus \{k \in \mathcal{C} : |k - k^*| \leq \eta\}$ ▷ Remove k^* and other candidates within distance η
 - 13: **end while**
 - 14: **Output:** $\mathcal{T}$
-

While using the standard CUSUM can result in imbalanced segments, the sliding window has a fixed width, and tests for the change point at its midpoint, which maximises statistical power. On the other hand, the main limitation of this algorithm is that its accuracy highly depends on the choice of parameter h . Choosing a small value of h means that the data will be sensitive to brief changes, but results in a small sample size ($2h$) for each test statistic and so is sensitive to noise, similar to bottom-up methods. Increasing h improves statistical power, but risks including multiple true changes in a single window which dilutes the test statistic and may mean no changes are detected, a parallel with the binary segmentation case. This trade-off is addressed by the Multiscale Sliding Window approach [48] which considers multiple values of h , enabling detection of changes of varying scales. It calculates test statistics $M_{k,h}$ for each combination of k and h , resulting in a 2D heat map of $M_{k,h}$, to which a scale-dependent threshold c_h is applied. A true change point will surpass the threshold for multiple values of h , with the smallest of these giving the most accurate estimator for the change point location.

3.4 Exact Methods

The methods discussed so far for the multiple change point problem are greedy or local, and so not guaranteed to find the global optimal partition of the observations into segments. Exact methods detect all change points jointly and guarantee that these result in an optimal partitioning of the data. For a time series of length n , there are 2^{n-1} possible partitions, so a naive search of every combination is computationally intractable. Exact algorithms utilise dynamic programming techniques to intelligently search the space of valid combinations of change points to identify the globally optimal partition. One such method is optimal partitioning, the computational complexity of which is improved by PELT. The algorithms for each of these are outlined in the subsequent sections, based on [34] and [37].

3.4.1 Optimal Partitioning

Exact algorithms use dynamic programming to find the optimal partition of a dataset by maximising with respect to a global objective function. By [34] this works for any additive function goodness-of-fit measure such that the total score of a partition is equal to the sum of scores of its individual segments. We use the log-likelihood function (2.11) which satisfies this property since the log-likelihood is additive under our assumption of independent observations. In other words, we have that:

$$\ell(\hat{\boldsymbol{\theta}}_1, \hat{\boldsymbol{\theta}}_2, \dots, \hat{\boldsymbol{\theta}}_{m+1}; \mathbf{X}_{1:n}) = \sum_{i=0}^m \ell(\hat{\boldsymbol{\theta}}_{i+1}; \mathbf{X}_{\tau_i+1:\tau_{i+1}}).$$

This additive property guarantees that the problem exhibits optimal substructure, such that the optimality of a segment is independent of future segments, and so dynamic programming results in a global optimum. However, since the maximised likelihood function has the property that adding parameters never decreases the likelihood, optimising the log-likelihood function without a penalty would result in overfitting. This is clear since if Ω_i is the parameter space for a model with i change points, then $\Omega_m \subset \Omega_{m+1}$. Hence:

$$\sup_{\boldsymbol{\theta} \in \Omega_{m+1}} \ell(\boldsymbol{\theta}; \mathbf{X}) \geq \sup_{\boldsymbol{\theta} \in \Omega_m} \ell(\boldsymbol{\theta}; \mathbf{X}),$$

where the left and right sides of the inequality represent the maximised log-likelihood in the case of $m+1$ and m change points respectively. For example, adding a single change at k on the interval between a and b , we see

$$\ell(\hat{\boldsymbol{\theta}}_{a:k}; \mathbf{X}_{a:k}) + \ell(\hat{\boldsymbol{\theta}}_{k+1:b}; \mathbf{X}_{k+1:b}) \geq \ell(\hat{\boldsymbol{\theta}}_{a:b}; \mathbf{X}_{a:b}). \quad (3.2)$$

In this case, the algorithm would declare a change point at each time index $k = 1, \dots, n-1$, since the optimal partition will be achieved when each observation represents a change point. To prevent this, we add an L_0 penalty, β , so that the extra degrees of freedom represented by a more complex model are penalised. The problem of maximising fit is equivalent to that of minimising cost, where we define the cost as twice the negative maximised log-likelihood function,

$$C(\mathbf{X}_{a:b}) = -2\ell(\hat{\boldsymbol{\theta}}_{a:b}; \mathbf{X}_{a:b}). \quad (3.3)$$

Then we aim to optimise:

$$\begin{aligned} & \min_{m \in \mathbb{N}} \left[\min_{\tau_1, \dots, \tau_m} \sum_{i=0}^m -2\ell(\hat{\boldsymbol{\theta}}_{\tau_i+1:\tau_{i+1}}; \mathbf{X}_{\tau_i+1:\tau_{i+1}}) + \beta m \right], \\ & = \min_{m \in \mathbb{N}} \left[\min_{\tau_1, \dots, \tau_m} \sum_{i=0}^m C(\mathbf{X}_{\tau_i+1:\tau_{i+1}}) + \beta m \right], \end{aligned}$$

where $\tau_1, \dots, \tau_m$ represent the m change points as in (3.1) and β is the L_0 penalty. We write $F(t)$ for the optimal cost of segmenting $\mathbf{X}_{1:t}$. The below algorithm derives the optimal partition of all n data points using the optimal partitions of the preceding data points $1, \dots, n-1$.

Algorithm 5 Optimal Partitioning

Require: $\mathbf{X} = (X_1, \dots, X_n), \beta$

- 1: $F(0) \leftarrow -\beta$
 - 2: $\mathcal{T}(0) \leftarrow \text{NULL}$
 - 3: **for** $t = 1, \dots, n$ **do**
 - 4: $F(t) \leftarrow \min_{k=0,1,\dots,t-1} [F(k) + C(\mathbf{X}_{k+1:t}) + \beta]$
 - 5: $k^* \leftarrow \arg \min_{\tau=0,1,\dots,t-1} [F(k) + C(\mathbf{X}_{k+1:t}) + \beta]$
 - 6: $\mathcal{T}(t) \leftarrow (\mathcal{T}(k^*), k^*)$
 - 7: **end for**
 - 8: **Output:** $\mathcal{T}(n)$
-

This is a deterministic $O(n^2)$ algorithm which finds the set of change points that minimise the negative log-likelihood function [34]. Note that the initialisation of $F(0) = -\beta$ ensures that no penalty is added to the model without change points, since in that case we would have:

$$\begin{aligned} F(n) &= F(0) - 2\ell(\hat{\boldsymbol{\theta}}_{1:n}; \mathbf{X}_{1:n}) + \beta, \\ &= -2\ell(\hat{\boldsymbol{\theta}}_{1:n}; \mathbf{X}_{1:n}). \end{aligned}$$

3.4.2 PELT

Though optimal partitioning guarantees a globally optimal segmentation, the $O(n^2)$ complexity is a significant drawback. It involves computing $F(t)$ a total of n times, and within each iteration it considers t possible splits. The Pruned Exact Linear Time (PELT) algorithm [37] aims to reduce the computational complexity by performing a “pruning” step which reduces the number of possible splits considered.

Multiplying (3.2) by a factor of -2 , and setting $a = t+1, b = T$, we see

$$\begin{aligned} -2\ell(\hat{\boldsymbol{\theta}}_{t+1:k}; \mathbf{X}_{t+1:k}) + -2\ell(\hat{\boldsymbol{\theta}}_{k+1:T}; \mathbf{X}_{k+1:T}) &\leq -2\ell(\hat{\boldsymbol{\theta}}_{t+1:T}; \mathbf{X}_{t+1:T}), \\ \iff C(\mathbf{X}_{t+1:k}) + C(\mathbf{X}_{k+1:T}) &\leq C(\mathbf{X}_{t+1:T}). \end{aligned} \tag{3.4}$$

Theorem 3.4. *With the cost function (3.3), if we have*

$$F(t) + C(\mathbf{X}_{t+1:k}) \geq F(k), \tag{3.5}$$

then t will never be the last optimal change point at time $T > k$.

Proof. Note that the left hand side of (3.5) represents the cost of the optimal split up to k conditioned on there being a change point at time t . This value is compared to the right

hand side, $F(k)$, which is the unconstrained optimal split up to time k . If the true change point preceding k is not at t , then there is a cheaper (or equal) split of the data up to point k . Therefore any future split of the data up to time T inherits this cheaper path.

The cost of an optimal partitioning up to time T , given that a change point occurs at t is

$$F(t) + C(\mathbf{X}_{t+1:T}) + \beta.$$

Then (3.4) implies that

$$F(t) + C(\mathbf{X}_{t+1:T}) + \beta \geq F(t) + C(\mathbf{X}_{t+1:k}) + C(\mathbf{X}_{k+1:T}) + \beta.$$

Applying the condition in (3.5) to the right hand side gives

$$F(t) + C(\mathbf{X}_{t+1:k}) + C(\mathbf{X}_{k+1:T}) + \beta \geq F(k) + C(\mathbf{X}_{k+1:T}) + \beta.$$

Combining these two inequalities and cancelling the β terms results in

$$F(t) + C(\mathbf{X}_{t+1:T}) \geq F(k) + C(\mathbf{X}_{k+1:T}), \quad (3.6)$$

and so for any $T > k$, the last optimal change point will never be t , since the cost of the partition with its last change point at t is greater or equal to that of the last change point being at k . $\square$

Note that in the general PELT algorithm there is an additional constant K in the pruning condition (3.5). However, when the cost function is set to the negative log-likelihood, as in this case, $K = 0$ and so it is omitted here. The PELT algorithm utilises a set $\mathcal{R}_t$ to indicate indices to include in the next step, reducing the number of required computations at each iteration. [37] shows that under certain conditions, including that the number of change points

Algorithm 6 PELT Algorithm

Require: $\mathbf{X} = (X_1, \dots, X_n)$, $\beta > 0$

```

1:  $F(0) \leftarrow -\beta$ 
2:  $\mathcal{R}_1 \leftarrow \{0\}$ 
3:  $\mathcal{T}(0) \leftarrow \text{NULL}$ 
4: for  $t = 1, \dots, n$  do
5:    $F(t) \leftarrow \min_{k \in \mathcal{R}_t} [F(k) + C(\mathbf{X}_{k+1:t}) + \beta]$ 
6:    $k^* \leftarrow \arg \min_{k \in \mathcal{R}_t} [F(k) + C(\mathbf{X}_{k+1:t}) + \beta]$ 
7:    $\mathcal{T}(t) \leftarrow (\mathcal{T}(k^*), k^*)$ 
8:    $\mathcal{R}_{t+1} \leftarrow \{k \in \mathcal{R}_t \cup \{t\} \mid F(k) + C(\mathbf{X}_{k+1:t}) \leq F(t)\}$ 
9: end for
10: Output:  $\mathcal{T}(n)$ 
```

increases linearly with n , the algorithm has $O(n)$ complexity. This demonstrates a significant improvement to the $O(n^2)$ complexity of the optimal partitioning algorithm in the previous section.

3.4.3 Bayesian Application

The algorithms described in this chapter can be adapted for application in a Bayesian context. In general, the maximised likelihood in frequentist terms can be substituted for the marginal

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likelihood in Bayesian, since the Bayesian parallel to evaluating the likelihood at the MLE is integrating the likelihood over the prior $p(\boldsymbol{\theta})$. Similarly, the likelihood-ratio test has parallels with the Bayes factor, where both are used for model comparison. By making these substitutions, both the heuristic and exact algorithms described can be applied in a Bayesian framework [21, 18].

Definition 3.5 (Bayes factor). *Given a test of the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta}_0 \in \Omega_0$ against $\mathcal{H}_1 : \boldsymbol{\theta}_1 \in \Omega_1$, the Bayes factor in favour of $\mathcal{H}_0$ against $\mathcal{H}_1$ is defined as the ratio of marginal likelihoods:*

$$B_{01} = \frac{\mathcal{L}_M(\mathbf{X}|\mathcal{H}_0)}{\mathcal{L}_M(\mathbf{X}|\mathcal{H}_1)}.$$

In general, as mentioned in section 2.5, it is not always possible to calculate the marginal likelihood analytically. In these situations, simulations are used. In particular, to compare models of multiple change points in the case of an unknown number of changes, reversible jump MCMC [27] allows switching between different dimensions by “birth” and “death” of parameters.

3.4.4 Multivariate Case

The methods discussed also extend to the multivariate case under the assumption of simultaneous breaks across dimensions. For example, the PELT algorithm can be used with a multivariate cost function. However, as the dimensionality d increases, or if true change points are sparse, then the curse of dimensionality means that the signal may not be distinguishable from the noise. This limitation motivates the need for dimension reduction as in the INSPECT algorithm, after which univariate techniques can be applied.

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The heuristic algorithms are intuitive, allowing context to drive the choice between methods. For example, in a scenario where false positives are acceptable and the priority is detecting every potential change, bottom-up methods outperform binary segmentation as they are more sensitive to subtle changes, mitigating the masking effect seen in binary segmentation. If the expected distance between change points is known, the sliding window algorithm may perform best. The solution path method in the top-down and bottom-up techniques creates a hierarchy of change points based on the signal strength by the order in which changes are detected. For example, in binary segmentation the first detected change is likely to be the most significant. Contrastingly, exact methods return all change points simultaneously with no indication of significance. While the greedy nature of heuristic methods causes irreversible partitions, the global optimality of the optimal partitioning algorithm comes at a cost of computational complexity. However, the introduction of the PELT algorithm means that in cases where m grows linearly with n , the exact methods can perform in $O(n)$ time, whilst still guaranteeing global optimality. Exact methods are more robust in a wider range of problems, including the multivariate and Bayesian cases discussed above.

In all cases, a suitable threshold must be selected, typically of order $\ln \ln n$ or $\ln n$ as discussed in section 2.3, since all methods discussed have been formulated based on the GLLR test statistic, though they are agnostic to the chosen metric. Understanding the limitations of the techniques can influence threshold choice, for example applying the tighter threshold ($O(\ln \ln n)$) to binary segmentation to counterbalance its greedy approach.

Chapter 4

Autoregressive Models

The change point detection methods covered so far have relied upon the assumption that elements of the time series are independent of one another, following the signal-plus-noise model (2.1). In many applications, this assumption does not hold. It is natural, for example, that on a particular day, the closing price of a stock or the maximum temperature in a given region is related to the measurement from the day before. This time-dependence violates the independence assumption of (2.1), and motivates the introduction of an alternative model, known as an autoregressive model. This chapter is largely based on [16].

4.1 Univariate Autoregressive Model

4.1.1 Properties of Autoregressive Models

Definition 4.1 (Autoregressive Process). *A time series, $\mathbf{X} = (X_1, \dots, X_n)$, is said to be an autoregressive process of order p , denoted $AR(p)$, if an observation of the time series depends linearly on (at most) the p preceding observations, such that:*

$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t, \quad (4.1)$$

where $\varepsilon_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$, $\phi_i \in \mathbb{R}$ for $i = 0, \dots, p$, and the characteristic polynomial $\Phi(z)$ satisfies the causality condition. While we assume that p is known, we allow $\phi_i = 0$ for $i \leq p$, such that p is an upper bound on the ‘true’ order of the model.

We demonstrate the intuition of the causality condition by introducing the simplest autoregressive model, $AR(1)$, with $\phi_0 = 0$, resulting in the model:

$$X_t = \phi_1 X_{t-1} + \varepsilon_t. \quad (4.2)$$

Say we have some known starting value X_0 . Then

$$\begin{aligned} X_1 &= \phi_1 X_0 + \varepsilon_1, \\ X_2 &= \phi_1 X_1 + \varepsilon_2, \\ &= \phi_1^2 X_0 + \phi_1 \varepsilon_1 + \varepsilon_2. \end{aligned}$$

Continuing recursively up to X_t , we see that:

$$X_t = \phi_1^t X_0 + \phi_1^{t-1} \varepsilon_1 + \phi_1^{t-2} \varepsilon_2 + \dots + \phi_1 \varepsilon_{t-1} + \varepsilon_t,$$

such that X_t is a function of the initial value X_0 and historical values of ε . If we have that $|\phi_1| < 1$, then the weights ϕ_1^i will decay exponentially towards zero over time, so that the effects of historical values of ε and the initial condition X_0 will eventually die out. This constraint is generalised for the $AR(p)$ case by the following definition.

Definition 4.2 (Causality Condition [11]). *The characteristic polynomial of an $AR(p)$ model is given by $\Phi : \mathbb{C} \rightarrow \mathbb{C}$, where*

$$\Phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p.$$

We say that an $AR(p)$ model with characteristic polynomial $\Phi(z)$ is causal if it satisfies

$$\Phi(z) \neq 0 \quad \forall \quad |z| \leq 1.$$

This condition is derived from the following general definition.

Definition 4.3 (Causality [11]). *An $AR(p)$ process, $\mathbf{X}$, is causal if there exist constants $\{\psi_j\}$ with $\sum_{j=0}^{\infty} |\psi_j| < \infty$ and*

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \quad \forall t.$$

Since we have defined ε_t as i.i.d. normal, X_t is a linear sum of Gaussian random variables and therefore a Gaussian process. This formulation is useful since, whereas the X_t are correlated with each other, the ε_t are i.i.d. To derive asymptotic results in the following sections, the time series must satisfy several conditions, including being stationary and strongly mixing. We first note that causality implies weak stationarity, a proof for which is given in appendix B.1 by checking each condition of (2.2), noting that weak and strict stationarity coincide in the case of a Gaussian process. Conceptually, strongly mixing means that the effects of past observations on future observations become negligible over time [10]. The assumptions of causality combined with Gaussian noise are sufficient for each of these [49].

In the next section we define the change point model for an $AR(p)$ process, and a GLLR test statistic as we had for the signal-plus-noise model. Whereas before the problem was formulated based on the type of change, in this chapter we consider a change in the lag coefficients, ϕ_i for $1 \leq i \leq p$ and the intercept ϕ_0 . For a stationary $AR(p)$ process, the unconditional mean, variance and autocovariance are functions of these coefficients, and so testing for a structural change simultaneously tests for a change in these properties, since by [28]:

$$\mathbb{E}[X_t] = \frac{\phi_0}{1 - \sum_{i=1}^p \phi_i}, \quad (4.3)$$

$$\text{Var}[X_t] = \sum_{i=1}^p \phi_i \gamma_i + \sigma^2, \quad (4.4)$$

where we use the convention $\gamma_i = \text{Cov}[X_t, X_{t+i}]$ to represent the autocovariance.

4.1.2 Change Point Model

We now define the change point model for an $AR(p)$ process, following [16]. By definition 4.1 we assume that the process is causal, and therefore stationary and strongly mixing, as above. While a change point implies stationarity is violated, these properties are assumed to hold on each segment. We model distinct stationary $AR(p)$ processes between change points.

We define our parameter $\boldsymbol{\theta}$ as:

$$\boldsymbol{\theta}_t = (\phi_{0,t}, \phi_{1,t}, \dots, \phi_{p,t})^\top \in \mathbb{R}^{(p+1)},$$

with $\phi_{0,t}$ representing the intercept term at each index t , and $\phi_{i,t}$ representing the multiplier of X_{t-i} used in calculating X_t for $i = 1, \dots, p$. We restrict our model to $t = p + 1, \dots, n$, so that

X_{t-p} is defined, and refer to $t = 1, \dots, p$ as the burn-in period, assuming that $X_1, \dots, X_p$ are fixed constants. We test the following hypotheses for a single change point at time τ :

$$\begin{aligned} \mathcal{H}_0 : \boldsymbol{\theta}_{p+1} &= \boldsymbol{\theta}_{p+2} = \dots = \boldsymbol{\theta}_n, \\ \mathcal{H}_1 : \text{There exists an index } \tau < n \text{ such that: } \boldsymbol{\theta}_{p+1} &= \dots = \boldsymbol{\theta}_\tau \neq \boldsymbol{\theta}_{\tau+1} = \dots = \boldsymbol{\theta}_n. \end{aligned} \quad (4.5)$$

More specifically, under the alternative hypothesis we have

$$X_t = \begin{cases} \alpha_0 + \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + \varepsilon_t & \text{for } p+1 \leq t \leq \tau, \\ \beta_0 + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_p X_{t-p} + \varepsilon_t & \text{for } \tau+1 \leq t \leq n, \end{cases} \quad (4.6)$$

where

$$\begin{aligned} \boldsymbol{\alpha} &:= (\alpha_0, \alpha_1, \dots, \alpha_p)^\top = \boldsymbol{\theta}_{p+1} = \dots = \boldsymbol{\theta}_\tau, \\ \boldsymbol{\beta} &:= (\beta_0, \beta_1, \dots, \beta_p)^\top = \boldsymbol{\theta}_{\tau+1} = \dots = \boldsymbol{\theta}_n, \end{aligned}$$

with at least one $\alpha_i \neq \beta_i$, and allowing our autoregression coefficients α_i, β_i to be zero for any $i \leq p$, as in definition 4.1. As such, (4.5) models a change in values of the coefficients ϕ_i and also a change in the order of the model.

4.2 Testing for a Change Point

Under the assumption of Gaussian noise, with σ^2 known, our log-likelihood function for the model in (4.1) under the null hypothesis is given by

$$\ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}, \sigma^2) = -\frac{(n-p)}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left[\sum_{t=p+1}^n \left(X_t - \hat{\phi}_0 - \sum_{j=1}^p \hat{\phi}_j X_{t-j} \right)^2 \right].$$

We again use the Generalised Log-Likelihood Ratio test statistic to test the hypotheses in (4.5):

$$\begin{aligned} \text{LR}_k &= 2 \left[\ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) + \ell(\hat{\boldsymbol{\theta}}_2; \mathbf{X}_{k+1:n}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:n}) \right] \\ \text{LR}_k &= \frac{1}{\sigma^2} \left[- \sum_{t=p+1}^k \left(X_t - \hat{\alpha}_0 - \sum_{j=1}^p \hat{\alpha}_j X_{t-j} \right)^2 - \sum_{t=k+1}^n \left(X_t - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_j X_{t-j} \right)^2 \right. \\ &\quad \left. + \sum_{t=p+1}^n \left(X_t - \hat{\phi}_0 - \sum_{j=1}^p \hat{\phi}_j X_{t-j} \right)^2 \right] \end{aligned} \quad (4.7)$$

Here, each term represents the SSR evaluated at the MLE. To evaluate the MLE at each candidate change point k for $p+1 \leq k \leq n-1$, we formulate the problem using matrix notation.

Definition 4.4 (Design Matrix). *The design matrix M_k for $p+1 \leq k \leq n$ is given by*

$$M_k = \begin{bmatrix} 1 & X_p & X_{p-1} & \cdots & X_2 & X_1 \\ 1 & X_{p+1} & X_p & \cdots & X_3 & X_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_{k-1} & X_{k-2} & \cdots & X_{k-p+1} & X_{k-p} \end{bmatrix} \in \mathbb{R}^{(k-p) \times (p+1)}$$

where the i^{th} row represents the p lagged observations used to calculate X_{p+i} , for $1 \leq i \leq k-p$.

4.2 Testing for a Change Point

In what follows, we require that $\mathbf{M}_k^\top \mathbf{M}_k$ is strictly positive definite and invertible. To satisfy these conditions, we assume that $\mathbf{M}_k$ has full column rank, which requires $k-p \geq p+1 \implies k \geq 2p+1$, so we restrict our continuous interval to $[u_1, u_2]$ with $0 < u_1 < u_2 < 1$, which simultaneously implements the trimming and avoids boundary instability of the $(u(1-u))^{-1}$ factor.

With observations $\mathbf{X}_n = \mathbf{X}_{p+1:n} = (X_{p+1}, \dots, X_n)^\top$, lag coefficients $\boldsymbol{\phi} = (\phi_0, \phi_1, \dots, \phi_p)^\top$, and unknown i.i.d. noise $\boldsymbol{\varepsilon}_n = (\varepsilon_{p+1}, \dots, \varepsilon_n)$ we have that under $\mathcal{H}_0$:

$$\mathbf{X}_n = \mathbf{M}_n \boldsymbol{\phi} + \boldsymbol{\varepsilon}_n. \quad (4.8)$$

Following the standard OLS procedure, we minimise the sum of squared residuals to find the MLE for $\boldsymbol{\phi}$. First, we rearrange (4.8) as $\boldsymbol{\varepsilon}_n = \mathbf{X}_n - \mathbf{M}_n \boldsymbol{\phi}$ and consider the sum of squared errors:

$$\begin{aligned} \|\boldsymbol{\varepsilon}_n\|^2 &= \boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n = (\mathbf{X}_n - \mathbf{M}_n \boldsymbol{\phi})^\top (\mathbf{X}_n - \mathbf{M}_n \boldsymbol{\phi}), \\ &= \mathbf{X}_n^\top \mathbf{X}_n - \mathbf{X}_n^\top \mathbf{M}_n \boldsymbol{\phi} - (\mathbf{M}_n \boldsymbol{\phi})^\top \mathbf{X}_n + \boldsymbol{\phi}^\top \mathbf{M}_n^\top \mathbf{M}_n \boldsymbol{\phi}. \end{aligned}$$

Considering the dimensions of the second term, we see that it is a constant:

$$\underbrace{\mathbf{X}_n^\top}_{1 \times (n-p)} \cdot \underbrace{\mathbf{M}_n}_{(n-p) \times (p+1)} \cdot \underbrace{\boldsymbol{\phi}}_{(p+1) \times 1} \in \mathbb{R}.$$

We note that the third term is the transpose of the second term, $(\mathbf{M}_n \boldsymbol{\phi})^\top \mathbf{X}_n = (\mathbf{X}_n^\top \mathbf{M}_n \boldsymbol{\phi})^\top$, and is therefore also equal to the same constant. This yields:

$$\|\boldsymbol{\varepsilon}_n\|^2 = \mathbf{X}_n^\top \mathbf{X}_n - 2(\mathbf{M}_n \boldsymbol{\phi})^\top \mathbf{X}_n + (\mathbf{M}_n \boldsymbol{\phi})^\top \mathbf{M}_n \boldsymbol{\phi}. \quad (4.9)$$

Differentiating to find the minimiser, we obtain

$$\frac{\partial}{\partial \boldsymbol{\phi}} \|\boldsymbol{\varepsilon}_n\|^2 = -2\mathbf{M}_n^\top \mathbf{X}_n + 2\mathbf{M}_n^\top \mathbf{M}_n \boldsymbol{\phi}, \quad (4.10)$$

$$\begin{aligned} 0 &= -2\mathbf{M}_n^\top \mathbf{X}_n + 2\mathbf{M}_n^\top \mathbf{M}_n \hat{\boldsymbol{\phi}}, \\ \implies \mathbf{M}_n^\top \mathbf{X}_n &= \mathbf{M}_n^\top \mathbf{M}_n \hat{\boldsymbol{\phi}}, \\ \implies \hat{\boldsymbol{\phi}} &= (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \mathbf{X}_n. \end{aligned} \quad (4.11)$$

Taking the derivative of (4.10), we see that $\partial^2 / \partial \boldsymbol{\phi}^2 (\|\boldsymbol{\varepsilon}_n\|^2) = 2\mathbf{M}_n^\top \mathbf{M}_n$, and since $\mathbf{M}_n^\top \mathbf{M}_n$ is positive definite, $\hat{\boldsymbol{\phi}}$ represents a minimum. We then rewrite (4.9) using the identity in (4.11):

$$\begin{aligned} \|\hat{\boldsymbol{\varepsilon}}_n\|^2 &= \mathbf{X}_n^\top \mathbf{X}_n - 2(\mathbf{M}_n \hat{\boldsymbol{\phi}})^\top \mathbf{X}_n + \hat{\boldsymbol{\phi}}^\top \mathbf{M}_n^\top \mathbf{M}_n \hat{\boldsymbol{\phi}}, \\ &= \mathbf{X}_n^\top \mathbf{X}_n - 2\hat{\boldsymbol{\phi}}^\top \mathbf{M}_n^\top \mathbf{X}_n + \hat{\boldsymbol{\phi}}^\top \mathbf{M}_n^\top \mathbf{X}_n, \\ &= \mathbf{X}_n^\top \mathbf{X}_n - \hat{\boldsymbol{\phi}}^\top \mathbf{M}_n^\top \mathbf{X}_n. \end{aligned}$$

Finally, substituting in our estimate $\hat{\boldsymbol{\phi}}$ and using the fact that $\mathbf{M}_n^\top \mathbf{M}_n$ is symmetric:

$$\begin{aligned} \|\hat{\boldsymbol{\varepsilon}}_n\|^2 &= \mathbf{X}_n^\top \mathbf{X}_n - \left[(\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \mathbf{X}_n \right]^\top \mathbf{M}_n^\top \mathbf{X}_n, \\ &= \mathbf{X}_n^\top \mathbf{X}_n - (\mathbf{M}_n^\top \mathbf{X}_n)^\top (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} (\mathbf{M}_n^\top \mathbf{X}_n). \end{aligned} \quad (4.12)$$

This can be interpreted as the uncentered total sum of squares, $\mathbf{X}_n^\top \mathbf{X}_n$ minus the uncentered total explained sum of squares, $(\mathbf{M}_n^\top \mathbf{X}_n)^\top (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} (\mathbf{M}_n^\top \mathbf{X}_n)$.

More specifically, we can expand $\mathbf{M}_n^\top \mathbf{X}_n$ to see:

$$\mathbf{M}_n^\top \mathbf{X}_n = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ X_p & X_{p+1} & \cdots & X_{n-1} \\ \vdots & \vdots & \vdots & \vdots \\ X_1 & X_2 & \cdots & X_{n-p} \end{bmatrix} \begin{pmatrix} X_{p+1} \\ X_{p+2} \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} \sum_{i=p+1}^n X_i \\ \sum_{i=p+1}^n X_i X_{i-1} \\ \vdots \\ \sum_{i=p+1}^n X_i X_{i-p} \end{pmatrix},$$

which represents the uncentered sample cross-covariance vector between X_t and the preceding p lags. $\mathbf{M}_n^\top \mathbf{M}_n$ is the uncentered sample autocovariance matrix of the p lagged predictors.

We can apply this to write matrix equations under $\mathcal{H}_1$. For a candidate change point k with $p+1 \leq k \leq n-1$, we partition the data such that $\mathbf{X}_k = \mathbf{X}_{p+1:k} = (X_{p+1}, \dots, X_k)^\top$ and $\widetilde{\mathbf{X}}_k = \mathbf{X}_{k+1:n} = (X_{k+1}, \dots, X_n)^\top$, with vectors $\boldsymbol{\varepsilon}_t$ and $\widetilde{\boldsymbol{\varepsilon}}_t$ defined analogously. This results in:

$$\mathbf{X}_n = \begin{pmatrix} \mathbf{X}_k \\ \widetilde{\mathbf{X}}_k \end{pmatrix} \in \mathbb{R}^{(n-p)}, \quad \boldsymbol{\varepsilon}_n = \begin{pmatrix} \boldsymbol{\varepsilon}_k \\ \widetilde{\boldsymbol{\varepsilon}}_k \end{pmatrix} \in \mathbb{R}^{(n-p)}. \quad (4.13)$$

Similarly, we set $\mathbf{M}_k$ as the first $k-p$ rows of $\mathbf{M}_n$, and $\widetilde{\mathbf{M}}_k$ as the remaining $n-k$ rows, so that

$$\mathbf{M}_n = \begin{bmatrix} \mathbf{M}_k \\ \widetilde{\mathbf{M}}_k \end{bmatrix}. \quad (4.14)$$

Since the three summations in (4.7) are equivalent to the residual sum of squares of the associated segments, which we estimate using the respective values of $\|\widehat{\boldsymbol{\varepsilon}}\|^2$, then

$$\text{LR}_k = Q_0 - (Q_1 + Q_2), \text{ where } Q_0 = \|\widehat{\boldsymbol{\varepsilon}}_n\|^2, \quad Q_1 = \|\widehat{\boldsymbol{\varepsilon}}_k\|^2, \quad Q_2 = \|\widehat{\boldsymbol{\varepsilon}}_k\|^2,$$

where we can use (4.12) for the full expansion of Q_0 , and we can set $\sigma^2 = 1$ without loss of generality. Following identical steps, Q_1 and Q_2 are obtained by replacing $(\mathbf{M}_n, \mathbf{X}_n, \boldsymbol{\varepsilon}_n)$ with $(\mathbf{M}_k, \mathbf{X}_k, \boldsymbol{\varepsilon}_k)$ and $(\widetilde{\mathbf{M}}_k, \widetilde{\mathbf{X}}_k, \widetilde{\boldsymbol{\varepsilon}}_k)$ respectively. As in previous sections, we declare a change if $\max_{k=p+1, \dots, n-1} \text{LR}_k > c$, for some threshold c . The estimated change point is given by $\widehat{\tau} = \arg \max_{k=p+1, \dots, n-1} \text{LR}_k$.

4.2.1 Inference about the Statistic

Now that we have an expression for the test statistic, we examine its properties and derive its asymptotic distribution. The $\mathbf{X}_n$ and $\mathbf{X}_k$ vectors in the test statistic are auto-correlated, so we rewrite (4.12) in terms of the respective $\boldsymbol{\varepsilon}$ vectors, which we have assumed are i.i.d. and normally distributed. We show this for Q_0 , however the logic can be applied almost identically for Q_1 and Q_2 .

$$\begin{aligned} Q_0 &= \mathbf{X}_n^\top \mathbf{X}_n - \mathbf{X}_n^\top \mathbf{M}_n (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \mathbf{X}_n, \\ &= \mathbf{X}_n^\top \underbrace{\left[\mathbf{I}_{n-p} - \mathbf{M}_n (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \right]}_{\mathbf{A}} \mathbf{X}_n, \end{aligned}$$

where we have factored out $\mathbf{X}_n^\top$ on the left and $\mathbf{X}_n$ on the right. If we label our new matrix $\mathbf{A}$, as above, with $\mathbf{A} \in \mathbb{R}^{(n-p) \times (n-p)}$ and multiply both sides of (4.8) by $\mathbf{A}$, we see:

$$\begin{aligned} \mathbf{A} \mathbf{X}_n &= \mathbf{A} (\mathbf{M}_n \boldsymbol{\phi} + \boldsymbol{\varepsilon}_n), \\ &= \left[\mathbf{I}_{n-p} - \mathbf{M}_n (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \right] \mathbf{M}_n \boldsymbol{\phi} + \mathbf{A} \boldsymbol{\varepsilon}_n, \\ &= \left[\mathbf{M}_n - \mathbf{M}_n (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} \mathbf{M}_n^\top \mathbf{M}_n \right] \boldsymbol{\phi} + \mathbf{A} \boldsymbol{\varepsilon}_n, \\ &= (\mathbf{M}_n - \mathbf{M}_n) \boldsymbol{\phi} + \mathbf{A} \boldsymbol{\varepsilon}_n, \\ &= \mathbf{A} \boldsymbol{\varepsilon}_n. \end{aligned}$$

4.2 Testing for a Change Point

We can therefore interchange $\mathbf{X}_n$ and $\boldsymbol{\varepsilon}_n$ in (4.12). Rewriting in terms of ε gives:

$$Q_0 = \boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n - \boldsymbol{\varepsilon}_n^\top \mathbf{M}_n (\mathbf{M}_n^\top \mathbf{M}_n)^{-1} (\mathbf{M}_n^\top \boldsymbol{\varepsilon}_n).$$

To simplify the test statistic we can relate the terms in each equation. By (4.13) and (4.14), we see that:

$$\boldsymbol{\varepsilon}_n^\top \mathbf{M}_n = \begin{pmatrix} \boldsymbol{\varepsilon}_k \\ \tilde{\boldsymbol{\varepsilon}}_k \end{pmatrix}^\top \begin{bmatrix} \mathbf{M}_k \\ \widetilde{\mathbf{M}}_k \end{bmatrix} = \boldsymbol{\varepsilon}_k^\top \mathbf{M}_k + \tilde{\boldsymbol{\varepsilon}}_k^\top \widetilde{\mathbf{M}}_k.$$

Similarly,

$$\mathbf{M}_n^\top \mathbf{M}_n = \begin{bmatrix} \mathbf{M}_k \\ \widetilde{\mathbf{M}}_k \end{bmatrix}^\top \begin{bmatrix} \mathbf{M}_k \\ \widetilde{\mathbf{M}}_k \end{bmatrix} = \mathbf{M}_k^\top \mathbf{M}_k + \widetilde{\mathbf{M}}_k^\top \widetilde{\mathbf{M}}_k.$$

Therefore, defining:

$$\begin{aligned} \mathbf{S}_n &:= (\boldsymbol{\varepsilon}_n^\top \mathbf{M}_n)^\top, & \mathbf{P}_n &:= (\mathbf{M}_n^\top \mathbf{M}_n)^{-1}, \\ \mathbf{S}_k &:= (\boldsymbol{\varepsilon}_k^\top \mathbf{M}_k)^\top, & \mathbf{P}_k &:= (\mathbf{M}_k^\top \mathbf{M}_k)^{-1}, \\ \tilde{\mathbf{S}}_k &:= (\tilde{\boldsymbol{\varepsilon}}_k^\top \widetilde{\mathbf{M}}_k)^\top, & \tilde{\mathbf{P}}_k &:= (\widetilde{\mathbf{M}}_k^\top \widetilde{\mathbf{M}}_k)^{-1}, \\ &= \mathbf{S}_n - \mathbf{S}_k, & &= (\mathbf{P}_n^{-1} - \mathbf{P}_k^{-1})^{-1}, \end{aligned}$$

we can then rewrite the GLLR test statistic as:

$$\begin{aligned} \text{LR}_k &= Q_0 - Q_1 - Q_2, \\ &= \|\boldsymbol{\varepsilon}_n\|^2 - \|\boldsymbol{\varepsilon}_k\|^2 - \|\tilde{\boldsymbol{\varepsilon}}_k\|^2 - \mathbf{S}_n^\top \mathbf{P}_n \mathbf{S}_n + \mathbf{S}_k^\top \mathbf{P}_k \mathbf{S}_k + \tilde{\mathbf{S}}_k^\top \tilde{\mathbf{P}}_k \tilde{\mathbf{S}}_k, \\ &= -\mathbf{S}_n^\top \mathbf{P}_n \mathbf{S}_n + \mathbf{S}_k^\top \mathbf{P}_k \mathbf{S}_k + (\mathbf{S}_n - \mathbf{S}_k)^\top (\mathbf{P}_n^{-1} - \mathbf{P}_k^{-1})^{-1} (\mathbf{S}_n - \mathbf{S}_k), \end{aligned} \quad (4.15)$$

where we have used that (4.13) implies $\|\boldsymbol{\varepsilon}_n\|^2 = \|\boldsymbol{\varepsilon}_k\|^2 + \|\tilde{\boldsymbol{\varepsilon}}_k\|^2$. The test statistic (4.15) is for a candidate change point at time k . To test each possible candidate, we need $\{\text{LR}_k\}_{k=p+1, \dots, n}$. We rescale our time axis as in definition 2.13 so that for $u \in [0, 1]$, we define a process $R_n(u) = \text{LR}_{\lfloor nu \rfloor}$. As shown by (4.15), the behaviour of the GLLR test statistic, and therefore that of $R(u)$, is determined by $\mathbf{S}$ and $\mathbf{P}$.

4.2.2 Asymptotic Behaviour of $\mathbf{S}$

In what follows, we require the following theorem, which acts to extend the ideas from theorem 2.16, relaxing the i.i.d. assumption.

Theorem 4.5 ([39]). *Let $\{\boldsymbol{\xi}_n, n \geq 1\}$ be a weakly stationary series of random vectors in $\mathbb{R}^d$, centred at expectations and having $(2 + \delta)^{\text{th}}$ moments with $0 < \delta \leq 1$, uniformly bounded by 1. Suppose that $\{\boldsymbol{\xi}_n, n \geq 1\}$ is strongly mixing with mixing rate $\rho(n) \ll n^{-(1+\epsilon)(1+2/\delta)}$ for $\epsilon > 0$. If we write*

$$\boldsymbol{\xi}_n = (\boldsymbol{\xi}_{n1}, \dots, \boldsymbol{\xi}_{nd}),$$

then the two series in

$$\gamma_{ij} = \mathbb{E}[\mathbb{E}[\xi_{1i} \xi_{1j}]] + \sum_{k \geq 2} \mathbb{E}[\xi_{1i} \xi_{kj}] + \sum_{k \geq 2} \mathbb{E}[\xi_{ki} \xi_{1j}]$$

converge absolutely. Writing

$$\Gamma = (\gamma_{ij})_{1 \leq i, j \leq d},$$

we can redefine the sequence $\{\xi_n, n \geq 1\}$ on a new probability space with standard Brownian motion $\mathbf{W}(t)$ such that

$$\sum_{n \leq t} \xi_n - \Gamma^{1/2} \mathbf{W}(t) \ll t^{1/2-\lambda} \text{ almost surely,}$$

for some $\lambda > 0$ depending on ϵ, δ and d only.

We now consider the behaviour of $\mathbf{S}$ and show it satisfies the conditions for this theorem. In general, for $p+1 \leq k \leq n$,

$$\mathbf{S}_k = \begin{pmatrix} \sum_{i=p+1}^k \varepsilon_i \\ \sum_{i=p+1}^k X_{i-1} \varepsilon_i \\ \vdots \\ \sum_{i=p+1}^k X_{i-p} \varepsilon_i \end{pmatrix} = \underbrace{\begin{pmatrix} \varepsilon_{p+1} \\ X_p \varepsilon_{p+1} \\ \vdots \\ X_1 \varepsilon_{p+1} \end{pmatrix}}_{\xi_{p+1}} + \underbrace{\begin{pmatrix} \varepsilon_{p+2} \\ X_{p+1} \varepsilon_{p+2} \\ \vdots \\ X_2 \varepsilon_{p+2} \end{pmatrix}}_{\xi_{p+2}} + \cdots + \underbrace{\begin{pmatrix} \varepsilon_k \\ X_{k-1} \varepsilon_k \\ \vdots \\ X_{k-p} \varepsilon_k \end{pmatrix}}_{\xi_k} \in \mathbb{R}^{(p+1)}.$$

Considering the properties of each element of the sum, ξ_i for $i = p+1, \dots, k$, we can demonstrate that (4.5) applies. By causality, X_t is a strictly stationary (section B.1), strongly mixing [49] process. Since the ξ_i are a product of X_t and i.i.d. ε_t , they inherit these properties. Since X_t and ε_t are Gaussian, their products have finite moments of all orders, so the $(2+\delta)^{th}$ moment condition holds for any $\delta \in (0, 1]$. Therefore, to use theorem 4.5 we must show that each ξ_i is conditionally centred. In general, causality implies that ε_i is independent of past observations X_t , for $t < i$. Here, the entries of ξ_i consist of products $X_t \varepsilon_i$, where $t < i$, and so all products are independent. As a result, we have $\mathbb{E}[X_t \varepsilon_i] = \mathbb{E}[X_t] \mathbb{E}[\varepsilon_i]$. For a given ξ_i , we can use this independence to write:

$$\mathbb{E}[\xi_i] = \begin{pmatrix} \mathbb{E}[\varepsilon_i] \\ \mathbb{E}[X_{i-1} \varepsilon_i] \\ \vdots \\ \mathbb{E}[X_{i-p} \varepsilon_i] \end{pmatrix} = \begin{pmatrix} \mathbb{E}[\varepsilon_i] \\ \mathbb{E}[X_{i-1}] \mathbb{E}[\varepsilon_i] \\ \vdots \\ \mathbb{E}[X_{i-p}] \mathbb{E}[\varepsilon_i] \end{pmatrix} = \mathbb{E}[\varepsilon_i] \begin{pmatrix} 1 \\ \mathbb{E}[X_{i-1}] \\ \vdots \\ \mathbb{E}[X_{i-p}] \end{pmatrix} = \mathbf{0},$$

since $\mathbb{E}[\varepsilon_i] = 0 \forall i$. This implies that the sequence $\{\xi_i\}$ is centred, and therefore the partial sum $\mathbf{S}_k$ is centred. We also have that:

$$\mathbb{E}[\xi_i | X_1, \dots, X_{i-1}] = \mathbb{E}[\varepsilon_i | X_1, \dots, X_{i-1}] \begin{pmatrix} 1 \\ X_{i-1} \\ \vdots \\ X_{i-p} \end{pmatrix} = \mathbb{E}[\varepsilon_i] \begin{pmatrix} 1 \\ X_{i-1} \\ \vdots \\ X_{i-p} \end{pmatrix} = \mathbf{0},$$

where we have used independence and the fact that at time $t = i$, the data $X_{i-p}, \dots, X_{i-1}$ is known and can be treated as constant. This implies that ξ_i are conditionally centred. Similarly, for the variance we have:

$$\begin{aligned} \text{Var}[\xi_i] &= \mathbb{E}[(\xi_i - \mathbb{E}[\xi_i])(\xi_i - \mathbb{E}[\xi_i])^\top], \\ &= \mathbb{E}[\xi_i \xi_i^\top], \\ &= \mathbb{E}[\varepsilon_i^2] \begin{bmatrix} 1 & \mathbb{E}[X_{i-1}] & \cdots & \mathbb{E}[X_{i-p}] \\ \mathbb{E}[X_{i-1}] & \mathbb{E}[X_{i-1}^2] & \cdots & \mathbb{E}[X_{i-1} X_{i-p}] \\ \mathbb{E}[X_{i-2}] & \mathbb{E}[X_{i-1} X_{i-2}] & \cdots & \mathbb{E}[X_{i-2} X_{i-p}] \\ \vdots & \vdots & \vdots & \vdots \\ \mathbb{E}[X_{i-p}] & \mathbb{E}[X_{i-p} X_{i-1}] & \cdots & \mathbb{E}[X_{i-p}^2] \end{bmatrix}, \end{aligned}$$

4.2 Testing for a Change Point

where we have used independence to factor out $\mathbb{E}[\varepsilon_i^2]$. Using the fact that $\text{Var}[\varepsilon_i] = \mathbb{E}[\varepsilon_i^2] = 1$, and by stationarity of $\mathbf{X}$, we then have:

$$\text{Var}[\xi_i] = [\Gamma_{yz}]_{y,z=1,\dots,p+1},$$

where

$$\Gamma_{yz} = \begin{cases} 1 & \text{if } y = z = 1, \\ \mu = \mathbb{E}[X_j] & \text{if } y > 1, z = 1 \text{ or } y = 1, z > 1, \\ \mathbb{E}[X_y X_z] & \text{if } y > 1, z > 1. \end{cases} \quad (4.16)$$

Therefore, each $\mathbf{S}_k$ is the partial sum of $k - p$ uncorrelated terms from a $(p + 1)$ -dimensional weakly stationary strongly mixing sequence with covariance matrix $\mathbf{\Gamma}_{p+1} = [\Gamma_{yz}]_{y,z=1,\dots,p+1}$. It satisfies the conditions for us to use theorem 4.5, noting that X_t being causal with i.i.d. Gaussian noise is sufficient for the mixing rate specified [49]. Applying theorem 4.5 and rescaling to $[0, 1]$ using $u = t/n$, we can redefine $\{\xi_n\}$ on a new probability space with a $(p + 1)$ -dimensional standard Brownian motion $\mathbf{W}(u)$ such that for $u \in [0, 1]$,

$$\frac{\mathbf{S}_{[nu]}}{\sqrt{n}} - \mathbf{\Gamma}_{p+1}^{1/2} \mathbf{W}(u) = O(n^{-\lambda}) \text{ as } n \rightarrow \infty \text{ almost surely,} \quad (4.17)$$

for some $\lambda > 0$. This implies that

$$\frac{\mathbf{S}_{[nu]}}{\sqrt{n}} \rightarrow_d (\mathbf{\Gamma}_{p+1})^{1/2} \mathbf{W}(u), \quad (4.18)$$

in $D_{p+1}[0, 1]$, where $\mathbf{W}(\cdot)$ is $(p + 1)$ -dimensional Brownian motion with covariance matrix $\mathbf{I}_{p+1}$.

4.2.3 Asymptotic Behaviour of $\mathbf{P}$

We now consider the asymptotic behaviour of $\mathbf{P}$, which we will combine with (4.18) to derive the asymptotic behaviour of our test statistic (4.15). In general, for $p + 1 \leq k \leq n$, $\mathbf{P}_k = (\mathbf{M}_k^\top \mathbf{M}_k)^{-1}$ represents the inverse of the sample covariance matrix. By the law of large numbers,

$$\frac{1}{k - p} \mathbf{P}_k^{-1} \xrightarrow{p} \mathbf{\Gamma}_{p+1},$$

and since $\frac{k-p}{k} \rightarrow 1$ as $k \rightarrow \infty$, we can equivalently write:

$$\frac{1}{k} \mathbf{P}_k^{-1} \xrightarrow{p} \mathbf{\Gamma}_{p+1}. \quad (4.19)$$

Element-wise we have:

$$[(\mathbf{P}_k)^{-1}]_{ij} = \begin{cases} k - p & \text{if } i = 1, j = 1 \\ \sum_{s=1}^{k-p} X_{p+1-i+s} & \text{if } i > 1, j = 1 \\ \sum_{s=1}^{k-p} X_{p+1-j+s} & \text{if } i = 1, j > 1 \\ \sum_{s=1}^{k-p} X_{p+1-i+s} X_{p+1-j+s} & \text{if } i > 1, j > 1 \end{cases} \quad (4.20)$$

For $i > 1, j > 1$, we wish to apply the FCLT theorem 2.16. We have that

$$\mathbb{E}[X_{p+1-i+s} X_{p+1-j+s}] = \mathbb{E}[X_i X_j],$$

by stationarity of $\mathbf{X}$. We define a new sequence

$$Y_s = \{X_{p+1-i+s}X_{p+1-j+s} - \mathbb{E}[X_iX_j]\},$$

such that by construction we have $\mathbb{E}[Y_s] = 0$. Again, $\{Y_s\}$ is stationary and strongly mixing, with both properties inherited from $\mathbf{X}$. Then applying theorem 2.16 gives:

$$\sum_{s=1}^{k-p} (Y_s) - V(k-p) = O(k^{1/2-\lambda}) \text{ as } k \rightarrow \infty \text{ almost surely,} \quad (4.21)$$

for some $\lambda > 0$, where $V(k-p)$ is univariate Brownian motion with variance:

$$\text{Var}[X_iX_j] + 2 \sum_{s=1}^{\infty} \text{Cov}[X_{-i}X_{-j}, X_{s-i}X_{s-j}].$$

Note that there exists a sequence $\{Y_s\}$ for each entry of $(\mathbf{P}_k)^{-1}$ with $i > 1, j > 1$, and therefore we can apply (4.21) to all of these entries. For $i = 1, j > 1$, or $i > 1, j = 1$, we can construct a stationary, strongly mixing sequence with mean zero simply by subtracting the mean $\mathbb{E}[X_s]$. We then have

$$\sum_{s=1}^{k-p} (X_s - \mathbb{E}[X_s]) - \tilde{V}(k) = O(k^{1/2-\lambda}) \text{ as } k \rightarrow \infty \text{ almost surely,}$$

for some $\lambda > 0$ and univariate Brownian motion $\tilde{V}$. For $i = j = 1$, the entry is simply $k-p$, which is deterministic and so no stochastic approximation is required. Therefore theorem 2.16 can be applied to all components of $(\mathbf{P}_k)^{-1}$, by (4.20).

Rescaling (4.19) to the continuous interval and setting $k = \lfloor nu \rfloor$, it follows that as $n \rightarrow \infty$,

$$n\mathbf{P}_{\lfloor nu \rfloor} \rightarrow_p u^{-1}\mathbf{\Gamma}_{p+1}^{-1}, \quad (4.22)$$

uniformly on $[u_1, u_2]$, with $0 < u_1 < u_2 < 1$.

4.2.4 Limiting Distribution of the Test Statistic

For brevity, in what follows we write $\mathbf{\Gamma} \equiv \mathbf{\Gamma}_{p+1}$. For $p+1 \leq k \leq n$ such that $u = k/n$, combining (4.18) and (4.22) by the continuous mapping theorem yields

$$\begin{aligned} \mathbf{S}_k^\top \mathbf{P}_k \mathbf{S}_k &= \left(\frac{\mathbf{S}_k}{\sqrt{n}} \right)^\top n \mathbf{P}_k \frac{\mathbf{S}_k}{\sqrt{n}} \rightarrow_d (\mathbf{\Gamma}^{1/2} \mathbf{W}(u))^\top \cdot u^{-1} \mathbf{\Gamma}^{-1} \cdot \mathbf{\Gamma}^{1/2} \mathbf{W}(u), \\ &= \frac{1}{u} \mathbf{W}(u)^\top (\mathbf{\Gamma}^{1/2})^\top \mathbf{\Gamma}^{-1} \mathbf{\Gamma}^{1/2} \mathbf{W}(u), \\ &= \frac{\|\mathbf{W}(u)\|^2}{u}. \end{aligned} \quad (4.23)$$

where we have used the fact that $\mathbf{\Gamma}$ is symmetric and positive definite, so $\mathbf{\Gamma}^{1/2}$ is also symmetric, therefore $(\mathbf{\Gamma}^{1/2})^\top = \mathbf{\Gamma}^{1/2}$. Applying (4.23) to each term of (4.15), the first two terms are:

$$-\mathbf{S}_n^\top \mathbf{P}_n \mathbf{S}_n + \mathbf{S}_k^\top \mathbf{P}_k \mathbf{S}_k \rightarrow_d -\frac{\|\mathbf{W}(1)\|^2}{1} + \frac{\|\mathbf{W}(u)\|^2}{u}. \quad (4.24)$$

For the third term, we have by (4.18) and the continuous mapping theorem that

$$\frac{\mathbf{S}_n - \mathbf{S}_k}{\sqrt{n}} \rightarrow_d (\mathbf{\Gamma}_{p+1})^{1/2} \mathbf{W}(1) - (\mathbf{\Gamma}_{p+1})^{1/2} \mathbf{W}(u) = (\mathbf{\Gamma}_{p+1})^{1/2} [\mathbf{W}(1) - \mathbf{W}(u)],$$

4.3 Vector Autoregressive Model

and by (4.22):

$$\begin{aligned} (n\mathbf{P}_n)^{-1} - (n\mathbf{P}_k)^{-1} &\rightarrow_p \mathbf{\Gamma}_{p+1} - u\mathbf{\Gamma}_{p+1} = (1-u)\mathbf{\Gamma}_{p+1}, \\ \implies n(\mathbf{P}_n^{-1} - \mathbf{P}_k^{-1})^{-1} &\rightarrow_p ((1-u)\mathbf{\Gamma}_{p+1})^{-1}. \end{aligned}$$

where the second equation comes from taking the inverse of both sides. As a result, we have

$$\begin{aligned} (\mathbf{S}_n - \mathbf{S}_k)^\top (\mathbf{P}_n^{-1} - \mathbf{P}_k^{-1})^{-1} (\mathbf{S}_n - \mathbf{S}_k) &= \left(\frac{\mathbf{S}_n - \mathbf{S}_k}{\sqrt{n}} \right)^\top n(\mathbf{P}_n^{-1} - \mathbf{P}_k^{-1})^{-1} \frac{\mathbf{S}_n - \mathbf{S}_k}{\sqrt{n}}, \\ &\rightarrow_d \left[\mathbf{\Gamma}^{1/2}(\mathbf{W}(1) - \mathbf{W}(u)) \right]^\top \cdot (1-u)^{-1} \mathbf{\Gamma}^{-1} \cdot \mathbf{\Gamma}^{1/2}(\mathbf{W}(1) - \mathbf{W}(u)), \\ &= (1-u)^{-1} (\mathbf{W}(1) - \mathbf{W}(u))^\top \mathbf{\Gamma}^{1/2} \mathbf{\Gamma}^{-1} \mathbf{\Gamma}^{1/2} (\mathbf{W}(1) - \mathbf{W}(u)), \\ &= \frac{1}{1-u} (\mathbf{W}(1) - \mathbf{W}(u))^\top (\mathbf{W}(1) - \mathbf{W}(u)). \end{aligned} \quad (4.25)$$

Substituting (4.24) and (4.25) in (4.15) gives

$$\begin{aligned} \text{LR}_{[nu]} &\rightarrow_d -\|\mathbf{W}(1)\|^2 + \frac{\|\mathbf{W}(u)\|^2}{u} + \frac{1}{1-u} (\mathbf{W}(1) - \mathbf{W}(u))^\top (\mathbf{W}(1) - \mathbf{W}(u)), \\ &= \frac{\|\mathbf{W}(u) - u\mathbf{W}(1)\|^2}{u(1-u)}, \end{aligned} \quad (4.26)$$

in $D[u_1, u_2]$, where $0 < u_1 < u_2 < 1$ and $\mathbf{W}(u)$ is standard $(p+1)$ -dimensional Brownian motion. Therefore $\text{LR}_{[nu]}$ converges in distribution to the Euclidean norm of a standard Brownian bridge, divided by the variance normalising constant $u(1-u)$. Then to test for a change at time k , we consider the likelihood ratio test statistic:

$$\max_{k \in [nu_1], [nu_2]} \text{LR}_k \rightarrow_d \sup_{u \in [u_1, u_2]} \frac{\|\mathbf{W}(u) - u\mathbf{W}(1)\|^2}{u(1-u)}. \quad (4.27)$$

Davis [16] showed that the case of unknown, estimated σ^2 results in the same limit. The case where σ^2 is allowed to change follows a similar procedure and is covered in proposition 3.2 of this paper.

4.3 Vector Autoregressive Model

So far, we have considered the test for structural changes in univariate AR(p) models. In many applications, there is likely to be not only time dependence, but also cross-sectional dependence. For example, in a financial context, an economic event is likely to cause change points simultaneously across multiple series. This means that each element of our time series is a vector, and the model accounts for dependence both spatially (between rows) and temporally (between columns). This section is largely based on [5].

Definition 4.6. Let $\mathbf{X} = \mathbf{X}_{p+1}, \dots, \mathbf{X}_n$ be a vector time series, with $\mathbf{X}_t = (X_{t,1}, \dots, X_{t,d})^\top \in \mathbb{R}^d$. A time series is said to be a vector autoregressive process of order p (VAR(p)) if a vector of observations of the time series depends linearly on (at most) the p preceding vectors, such that, for $p+1 \leq t \leq n$

$$\mathbf{X}_t = \mathbf{a}_0 + \sum_{j=1}^p \mathbf{A}_j \mathbf{X}_{t-j} + \boldsymbol{\varepsilon}_t, \quad (4.28)$$

where each $\mathbf{A}_j \in \mathbb{R}^{d \times d}$, $\mathbf{a}_0, \boldsymbol{\varepsilon}_t \in \mathbb{R}^d$, and $\boldsymbol{\varepsilon}_t \stackrel{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ a positive definite covariance matrix [46].

Whereas in the univariate case we could assume σ^2 known as it acts as a scaling factor, the off-diagonal elements of $\mathbf{\Sigma}$ represent relationships between series, such that no general structure can be assumed. For this reason, we only consider the case of $\mathbf{\Sigma}$ unknown and estimated by $\widehat{\mathbf{\Sigma}}$, which is calculated based on OLS residuals under $\mathcal{H}_1$ given k . Parallel to causality in the univariate case, we introduce the notion of stability.

Definition 4.7 (Stability Condition [46]). *A VAR(p) process as defined above is said to be stable if*

$$\det(\mathbf{I}_d - \mathbf{A}_1 z - \dots - \mathbf{A}_p z^p) \neq 0 \text{ for } |z| \leq 1. \quad (4.29)$$

We introduce a change point in this model at time $t = \tau$ by writing [5]:

$$\mathbf{X}_t = \mathbf{a}_0 + \sum_{j=1}^p \mathbf{A}_j \mathbf{X}_{t-j} + \mathbf{1}_{t>\tau} \left(\mathbf{b}_0 + \sum_{j=1}^p \mathbf{B}_j \mathbf{X}_{t-j} \right) + \boldsymbol{\varepsilon}_t, \quad (4.30)$$

where $\mathbf{b}_0 \in \mathbb{R}^d$ and each $\mathbf{B}_j \in \mathbb{R}^{d \times d}$, and we assume the stability condition holds. For ease of notation, we can rewrite this by setting $\widetilde{\mathbf{X}}_{t-1}^\top = (1, \mathbf{X}_{t-1}^\top, \mathbf{X}_{t-2}^\top, \dots, \mathbf{X}_{t-p}^\top) \in \mathbb{R}^{(dp+1)}$ and

$$\boldsymbol{\rho} = \text{Vec}([\mathbf{a}_0, \mathbf{A}_1, \dots, \mathbf{A}_p]), \quad \boldsymbol{\pi} = \text{Vec}([\mathbf{b}_0, \mathbf{B}_1, \dots, \mathbf{B}_p]),$$

where $\boldsymbol{\rho}, \boldsymbol{\pi} \in \mathbb{R}^{(d(dp+1))}$, and the details of the Vec operator are in appendix B.3. The resulting model is then:

$$\mathbf{X}_t = (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \boldsymbol{\rho} + \mathbf{1}_{t>\tau} (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \boldsymbol{\pi} + \boldsymbol{\varepsilon}_t. \quad (4.31)$$

When modelling a change at time $t = \tau$, we note that the change may occur in some but not all of our coefficients. Similarly it may occur in some of the series of $\mathbf{X}$ but not all. In model (4.31), we have allowed all coefficients to change, which we call a full structural change. If we know that only a subset of coefficients change, we can use a partial structural change model, given by

$$\mathbf{X}_t = (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \boldsymbol{\rho} + \mathbf{1}_{t>\tau} (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \mathbf{S}^\top \mathbf{S} \boldsymbol{\pi} + \boldsymbol{\varepsilon}_t, \quad (4.32)$$

where $\mathbf{S}$ is a selection matrix with full row rank that contains only 0s and 1s. We have that $\mathbf{S}^\top \mathbf{S}$ is idempotent with non-zero elements only on the diagonal, and that the rank of $\mathbf{S}$ is equal to the number of coefficients that are allowed to change. To account for selection of coefficients or individual time series independently, we can specify that:

$$\mathbf{S} = \mathbf{S}_{coef} \otimes \mathbf{S}_{series},$$

where $\mathbf{S}_{series} \in \mathbb{R}^{q_s \times d}$ identifies the q_s time series subject to change, and $\mathbf{S}_{coef} \in \mathbb{R}^{q_c \times (dp+1)}$ identifies the q_c lag coefficients subject to change, with $\text{rank}(\mathbf{S}) := q_t = q_c q_s$, which represents the total number of degrees of freedom of the change. Under this model, we can formulate our hypothesis test for a change point in terms of $\mathbf{S} \boldsymbol{\pi}$ as follows.

$$\mathcal{H}_0 : \mathbf{S} \boldsymbol{\pi} = \mathbf{0}, \quad \mathcal{H}_1 : \mathbf{S} \boldsymbol{\pi} \neq \mathbf{0}, \quad (4.33)$$

so that our model under $\mathcal{H}_0$ is:

$$\mathbf{X}_t = (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \boldsymbol{\rho} + \boldsymbol{\varepsilon}_t, \quad (4.34)$$

as opposed to the alternative hypothesis, which is represented by (4.32) with $\mathbf{S} \boldsymbol{\pi} \neq \mathbf{0}$.

4.3.1 Testing for a Change

We test the hypotheses (4.33), assuming that Σ is constant. We begin by rewriting (4.32) linearly so that we can apply least squares results. For a given change point at index k , we have

$$\mathbf{X}_t = \mathbf{Z}_t^\top(k) \boldsymbol{\vartheta} + \boldsymbol{\varepsilon}_t, \quad (4.35)$$

where

$$\mathbf{Z}_t(k) = \begin{pmatrix} \widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d \\ \mathbf{1}_{t>k} (\widetilde{\mathbf{X}}_{t-1}^\top \otimes \mathbf{I}_d) \mathbf{S}^\top \end{pmatrix}, \quad \boldsymbol{\vartheta} = \begin{pmatrix} \boldsymbol{\rho} \\ \mathbf{S}\boldsymbol{\pi} \end{pmatrix},$$

so that the parameter we are testing is $\boldsymbol{\vartheta}$, as this contains $\mathbf{S}\boldsymbol{\pi}$. While previous sections relied on the likelihood ratio test, to test for a change point in the VAR setting, we use a Wald test. The main motivation for the use of this new test statistic is computational complexity, since the likelihood ratio test statistic requires the computation of a new MLE, $\widehat{\boldsymbol{\vartheta}}$, at each value of k , which involves computationally expensive matrix inversions at each step. On the other hand, the Wald statistic requires a single non-iterative evaluation of the model under $\mathcal{H}_1$, which we derive. Moreover, [4] shows that for an unknown change point, this is asymptotically equivalent to using the likelihood ratio test. Specifically, it holds that $\sup_k \mathcal{W}_k = \sup_k \text{LR}_k + o_p(1)$, where $\mathcal{W}_k$ represents the Wald statistic, defined below, and $o_p(1)$ denotes a term converging to 0 in probability.

Definition 4.8 (Wald Statistic [65]). *The Wald statistic of $\mathcal{H}_0 : \mathbf{r}(\boldsymbol{\vartheta}) = \mathbf{q}$ against $\mathcal{H}_1 : \mathbf{r}(\boldsymbol{\vartheta}) \neq \mathbf{q}$ where $\mathbf{r}$ is the restriction on $\boldsymbol{\vartheta}$ is:*

$$\mathcal{W}_k = [\mathbf{r}(\widehat{\boldsymbol{\vartheta}}) - \mathbf{q}]^\top [\widehat{\text{Var}}(\mathbf{r}(\widehat{\boldsymbol{\vartheta}}) - \mathbf{q})]^{-1} [\mathbf{r}(\widehat{\boldsymbol{\vartheta}}) - \mathbf{q}].$$

When testing for a change as in (4.33), we are only interested in testing whether $\mathbf{S}\boldsymbol{\pi} = \mathbf{0}$, not $\boldsymbol{\vartheta} = \mathbf{0}$. To reflect this in our test statistic, we define the matrix $\mathbf{R}$ as $\mathbf{R} = (\mathbf{0}, \mathbf{I}_{q_t})$, so that $\mathbf{R}\boldsymbol{\vartheta} = \mathbf{S}\boldsymbol{\pi}$. Applying this to the Wald statistic, we have $\mathbf{r}(\boldsymbol{\vartheta}) = \mathbf{R}\boldsymbol{\vartheta}$ and $\mathbf{q} = \mathbf{0}$, resulting in

$$\mathcal{W}_k = (\mathbf{R}\widehat{\boldsymbol{\vartheta}})^\top [\widehat{\text{Var}}(\mathbf{R}\widehat{\boldsymbol{\vartheta}})]^{-1} (\mathbf{R}\widehat{\boldsymbol{\vartheta}}). \quad (4.36)$$

To calculate our estimate for $\boldsymbol{\vartheta}$, we introduce the generalised least squares (GLS) estimator.

Definition 4.9 (Generalised Least Squares). *For a general linear regression system $\mathbf{y} = \mathbf{Z}\boldsymbol{\vartheta} + \boldsymbol{\varepsilon}$ with covariance matrix $\text{Var}(\boldsymbol{\varepsilon}) = \Sigma$, a generalisation of the ordinary least squares estimator is*

$$\widehat{\boldsymbol{\vartheta}}_{GLS} = (\mathbf{Z}^\top \Sigma^{-1} \mathbf{Z})^{-1} \mathbf{Z}^\top \Sigma^{-1} \mathbf{y}. \quad (4.37)$$

Under the assumption of normality, this is equivalent to the maximum likelihood estimate [2].

If the lagged observations are the same in each time series, then (4.37) collapses to the standard OLS case [71]. On the other hand, under the model (4.32) where we assume a partial structural break, the GLS estimate is “at least asymptotically more efficient” [71] than applying standard OLS to each series individually. In other words, the correlation within Σ can be used to borrow strength across dimensions such that the likelihood ratio test is more powerful than testing the univariate hypothesis for each series independently. Applying (4.37) to (4.35) for a given index k , we have:

$$\widehat{\boldsymbol{\vartheta}}(k) = \left[\sum_{t=p+1}^n \mathbf{Z}_t(k) \widehat{\Sigma}_k^{-1} \mathbf{Z}_t(k)^\top \right]^{-1} \sum_{t=p+1}^n \mathbf{Z}_t(k) \widehat{\Sigma}_k^{-1} \mathbf{X}_t, \quad (4.38)$$

which has variance given by the following, conditional on $\{\mathbf{Z}_t\}_{t=p+1}^n$, with the proof in [46].

$$\text{Var}(\hat{\boldsymbol{\vartheta}}) = \left[\sum_{t=p+1}^n \mathbf{Z}_t(k) \widehat{\boldsymbol{\Sigma}}_k^{-1} \mathbf{Z}_t(k)^\top \right]^{-1}.$$

We can substitute this into our Wald test statistic (4.36), yielding

$$\mathcal{W}_k = n [\mathbf{R} \hat{\boldsymbol{\vartheta}}(k)]^\top \left[\mathbf{R} \left(n^{-1} \sum_{t=p+1}^n \mathbf{Z}_t \widehat{\boldsymbol{\Sigma}}_k^{-1} \mathbf{Z}_t^\top \right)^{-1} \mathbf{R}^\top \right]^{-1} \mathbf{R} \hat{\boldsymbol{\vartheta}}(k),$$

where we stabilise the variance by dividing by n to allow us to take limits as $n \rightarrow \infty$. The Wald statistic and the GLLR test statistic both converge to the same Brownian bridge limit by [4], so applying an analogue to theorem 4.5 as in [5] with the usual continuous mapping $t = nu$ and $0 < u_1 < u_2 < 1$, we obtain that under $\mathcal{H}_0$

$$\mathcal{W}_{[nu]} \rightarrow_d \frac{\|\mathbf{W}(u) - u\mathbf{W}(1)\|^2}{u(1-u)},$$

where $\mathbf{W}$ is vector independent standard Brownian motion, with $\dim(\mathbf{W}) = \text{rank}(\mathbf{S})$. Therefore, as in (4.27), our estimator for the change point is $\hat{\tau} = n\hat{u}$, where

$$\hat{u} = \arg \sup_{u \in [u_1, u_2]} \frac{\|\mathbf{W}(u) - u\mathbf{W}(1)\|^2}{u(1-u)}. \quad (4.39)$$

[5] then proceeds to derive confidence intervals for the change point, with a notable result that the width of the asymptotic confidence interval does not decrease as n increases. However, it is inversely related to the number of series sharing a common break date. Intuitively, more data after the change point does not provide information about when the change occurred, whereas precision is gained by analysing multiple series simultaneously rather than individually, as discussed in section 2.4.1.

A limitation of the above model is that $\mathbf{S}$ is assumed known a priori. In other words, we assume that we know which coefficients and series we expect to undergo a change point. Misspecification of $\mathbf{S}$ could lead to a loss of power if non-changing dimensions are included, since the model is using extra degrees of freedom, or a loss of the borrowed strength if those undergoing a change are wrongly omitted. [52] provides a generalisation of the method in this chapter, treating $\mathbf{S}$ as unknown, and also handling the case where $\boldsymbol{\Sigma}$ undergoes a change alongside or independently of the VAR coefficients. Remarkably, they find that the test statistic under the null converges to the same Brownian bridge derived in this chapter.

A further limitation is that in the high-dimensional case, where $d > n$, $\boldsymbol{\Sigma}$ is singular and thus cannot be inverted to calculate the GLS estimate. [54] assumes sparsity, using a LASSO approach to estimate change points in the high-dimensional case. However, the regularised model is more complex to draw inferences from, as the asymptotic Brownian bridge result and thresholds no longer hold. An alternative approach for handling high-dimensional data is to use factor models, where it is assumed that the high-dimensional data can be projected onto a small number of latent factors, and VAR methods can be applied to these factors [8].

Chapter 5

Application to Real Data

We consider log returns between 1 January 2018 and 31 December 2022 ($n=1259$ trading days) of the S&P 500 index and also of 13 stocks from six sectors. The data is sourced from WRDS [68]. This period encompasses the beginning of the COVID-19 pandemic in early 2020, which caused a severe disruption to financial markets and so we expect to detect a change point in early 2020. We apply multiple change point detection techniques to detect changes in variance, and also VAR techniques to identify changes in cross-stock dynamics.

5.1 Change in Variance of S&P 500 Data

5.1.1 Assumptions

In this subsection, we assume that the data follows the signal-plus-noise model (2.1), though the assumption of normality is violated with a kurtosis of 10.4, compared to the theoretical normal value of 3.0 [13]. Due to market efficiency the daily mean is negligible [12] and so we test for a change in variance. As we suspect a change in variance, we standardise by the global MAD estimate, $\hat{\sigma}_{\text{MAD}}$ which is more robust to extreme values and not inflated by the change point, unlike the global variance estimate, $\hat{\sigma}_{\text{global}}$. The formula is given by $\hat{\sigma}_{\text{MAD}} = 1.4826 \times \text{Median}(|X_i - \text{Median}(X)|)$ [53]. As illustrated in figure 5.1, $\hat{\sigma}_{\text{global}}$ is inflated by the change point resulting in $\pm 3\hat{\sigma}_{\text{global}}$ (red dashed) producing bounds much wider than the typical return magnitude, while $\pm 3\hat{\sigma}_{\text{MAD}}$ (blue solid) fits the majority of the data more accurately.

We utilise the thresholds derived in section 2.3 denoting $c_s = K \ln(n)$ for the conservative threshold, and $c_l = K \ln \ln n$ for the tighter threshold. Using PELT (via the *changepoint* package [38]) with the threshold c_s for varying values of K produced figure 5.2a, which flattens between $K = 10$ and $K = 32$, stabilising at 2 change points. To ensure stability, we avoid boundary values and use $K = 15$ throughout, so that our thresholds are $c_l = 15 \ln \ln(1259) \approx 29.48$ and

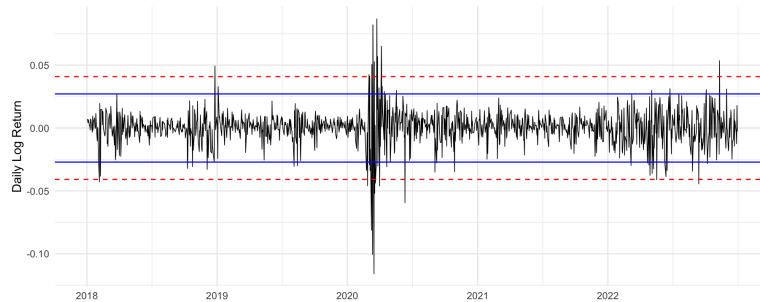


Figure 5.1: Log returns of S&P 500 between 01/01/2018 and 31/12/2022 (black) , with $\pm 3 \times \text{MAD}$ estimator (blue solid) and $\pm 3 \times \text{Global Variance}$ estimator (red dashed).

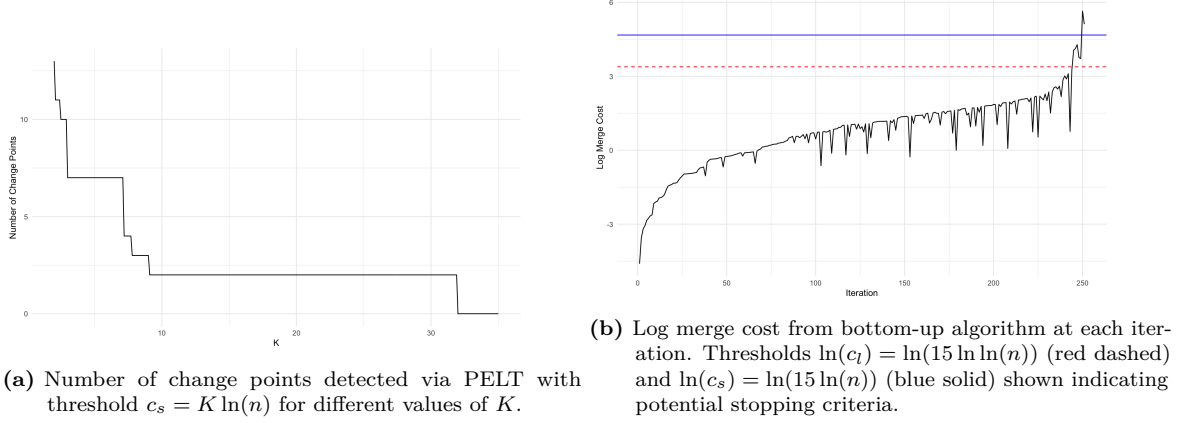


Figure 5.2: Tuning of number of change points for S&P 500 data.

$c_s = 15 \ln(1259) \approx 107.1$. Applying the bottom-up algorithm, we see from figure 5.2b that the thresholds closely match the plateau of the statistic, indicating a reasonable choice of threshold and consistency across methods.

Since the return data is heavy-tailed, we focus on the results produced under c_s , as this identifies only the most significant change points. Results under c_l are provided in appendix C for completeness, detecting additional breaks which may be of interest for further investigation.

5.1.2 Results

The results of applying PELT, binary segmentation, bottom-up and sliding window algorithms with threshold c_s are presented in figure 5.3. Note that all algorithms detected a change point within two days of 21 February 2020, suggesting this to be the most significant change. This coincides with ‘the last trading day before the market jump that contemporaneous newspaper accounts attribute to COVID-19’ [6], and so aligns with real-world events. PELT, bottom-up and binary segmentation all detected a second change within one day of 22 April 2020, which follows immediately after ‘the fall of WTI crude oil futures by more than 300%’ on 20 April [41]. We note that the binary segmentation algorithm produced identical results to PELT, highlighting its strength in accurately detecting global changes.

We note that the sliding window algorithm is implemented with $\eta = 90$, which in principle prevents detection of the 22 April change point, however this change point is also missed under smaller values of η , indicating that it is a genuine detection limitation as opposed to a result of high η . A value of $\eta = 90$ was chosen to allow the variance estimate to stabilise and reduce false positives. A change point was detected on 7 October 2019, which appears to be a false positive as it is not corroborated by any other method. This highlights the sensitivity to parameters h and η , which were tuned to produce two change points to match the number identified by PELT under c_s . Figure 5.4a shows the number of detected changes with varying values of h and η , and figure 5.4b shows number and location of change points for different values of h , with fixed $\eta = 90$, noting that the early 2020 change point is detected consistently.

5.2 Multivariate Analysis

We now consider 13 stocks as listed in table 5.1 over the same time period. We apply the INSPECT algorithm to the absolute residuals to detect sparse changes in variance, and then use VAR methods to detect a dense structural change in lag coefficients.

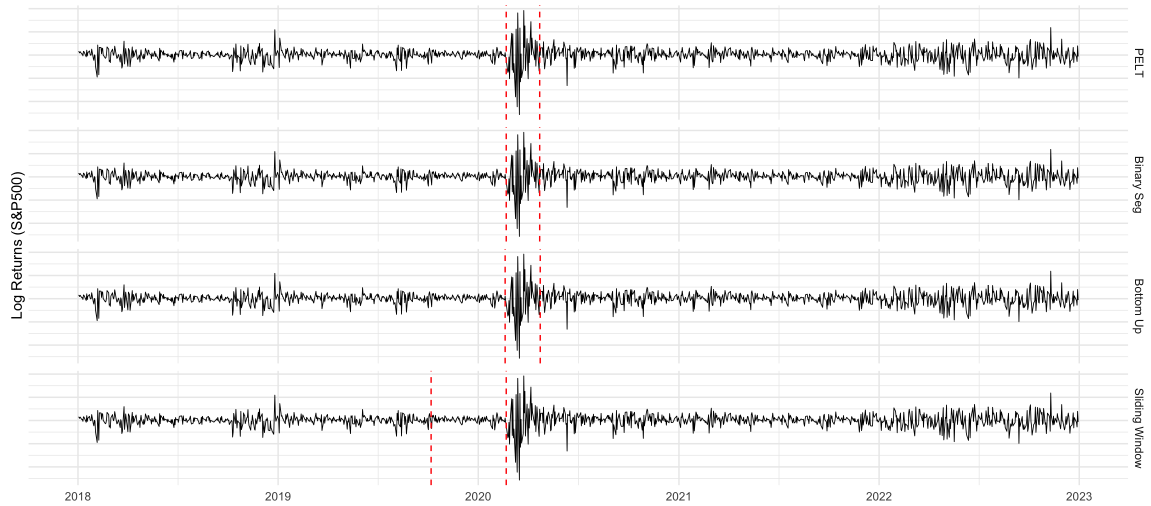
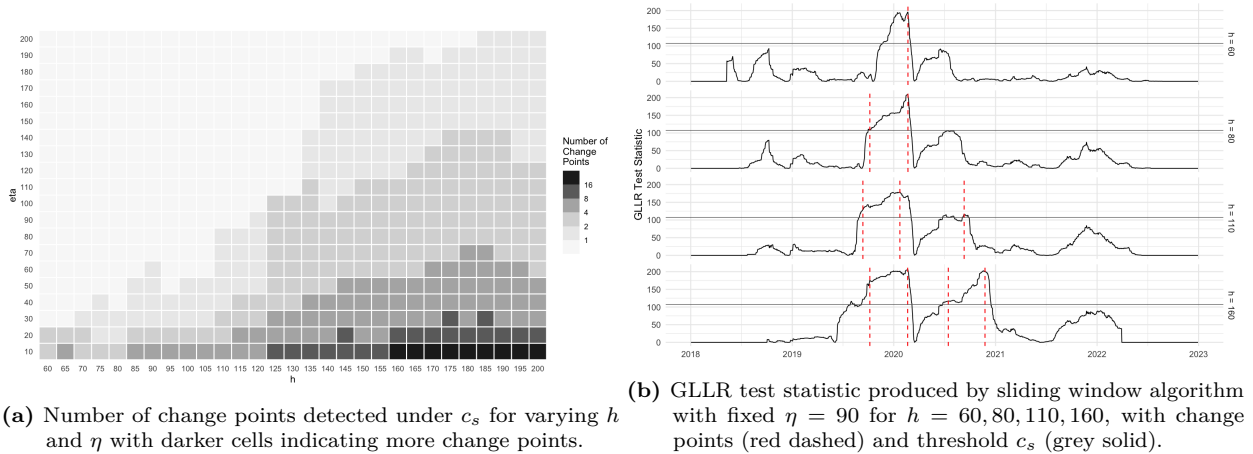


Figure 5.3: Log returns for the S&P 500 between 01/01/2018 and 31/12/2022 (black) with detected change points (red dashed) under threshold $c_s = 15 \ln(n)$ for different change point detection algorithms. The detected change points are: PELT — 21/02/2020, 22/04/2020; Binary Segmentation — 21/02/2020, 22/04/2020; Bottom-Up — 19/02/2020, 23/04/2020; Sliding Window — 21/02/2020, 07/10/2019.



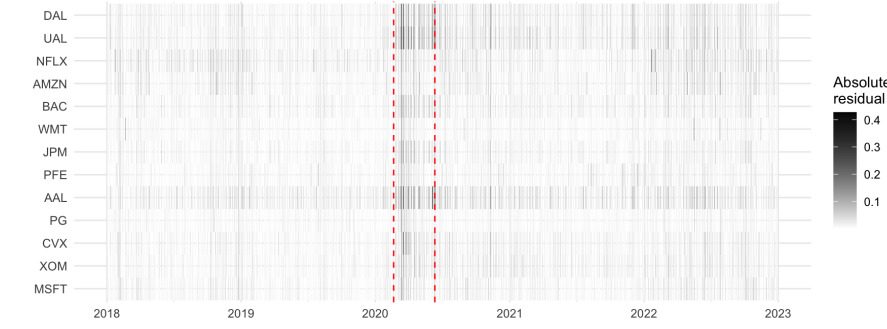
(a) Number of change points detected under c_s for varying h and η with darker cells indicating more change points.

(b) GLLR test statistic produced by sliding window algorithm with fixed $\eta = 90$ for $h = 60, 80, 110, 160$, with change points (red dashed) and threshold c_s (grey solid).

Figure 5.4: Sliding window parameter tuning for S&P 500 data under c_s .

Sector	Companies
Technology	Microsoft (MSFT), Netflix (NFLX), Amazon (AMZN)
Airlines	Delta Airlines (DAL), United Airlines (UAL), American Airlines (AAL)
Energy	ExxonMobil (XOM), Chevron (CVX)
Financials	JP Morgan (JPM), Bank of America (BAC)
Consumer Staples	Walmart (WMT), Proctor & Gamble (PG)
Healthcare	Pfizer (PFE)

Table 5.1: 13 stocks and their sector groupings according to GICS with the exception of technology, which combines Microsoft (Information Technology), Amazon (Consumer Discretionary) and Netflix (Communication Services) based on their shared characteristics in the context of this analysis.



(a) Absolute residuals for each of the stocks in 5.1 with a darker line indicating a greater value. Change points detected by INSPECT on 20/02/2020 and 11/06/2020 (red dashed), with 1000 random intervals for WBS and threshold $c = 10(\ln(13 \ln(n))) \approx 48.4$.

Stock	Weighting
UAL	0.608
DAL	0.446
XOM	0.370
AAL	0.303
CVX	0.234
JPM	0.231
BAC	0.213
SumSq	0.951

(b) 7 largest stock contributions to optimal projection vector for 20/02/2020 and their weightings, with the magnitude of this sparse vector denoted ‘Sum Sq’.

Figure 5.5: Results of applying INSPECT algorithm to log returns of stocks in table 5.1.

5.2.1 INSPECT

We use the *InspectChangepoint* package [66] on the absolute value of the residuals from each series, as these should be approximately i.i.d. [62]. Utilising wild binary segmentation within the ‘inspect’ function and a threshold $c = 10 \ln(13 \ln(n))$ resulted in detection of two change points, shown in 5.5a. We note that the first change point (20 February 2020) aligns with the findings of the univariate S&P 500 data, and the second aligns with the ‘worst one day sell-off’ of U.S. stocks since March 2020 amid growing concerns of a second wave of coronavirus [35]. The elements of the sparse projection vector and their corresponding weights for the February change are shown in 5.5b. This highlights the power of the INSPECT algorithm in identifying which series underwent a change, with the airline, energy and financials breaking simultaneously and accounting for 95% of the projection vector, whereas other sectors such as technology and consumer staples had negligible weights. This is again consistent with the context of the pandemic, as travel restrictions and economic uncertainty affect these changing stocks significantly more than those with negligible weights.

5.2.2 VAR

We use VAR to test for a change in the lag coefficients. Using the *VARselect* function from the package *vars* [51], the AIC and FPE criteria report using $p = 3$ as the optimal number of lags, whereas BIC and HQIC report $p = 1$. We choose $p = 1$ for parsimony, following the BIC and HQIC. To use the model (4.28), we require stationary data, confirmed by the augmented Dickey Fuller test [17, 61]. However, performing a Ljung-Box test [45] indicates that autocorrelation remains in the residuals. As we use the sup-Wald statistic to determine the estimated change point $\hat{\tau}$, the strict conditions for Brownian bridge thresholds are not required, and the magnitude of the change is likely sufficient for the change to be consistently detected despite violating autocovariance assumptions.

We test for a dense structural break, as it is likely that all selected stocks were affected by COVID in some manner. Therefore, under the alternative given by (4.32) we have $\mathbf{S} = \mathbf{I}$. We run a linear model for each stock and estimate the covariance matrix, $\hat{\Sigma}$, using standard OLS residuals, as each series has the same regressors. We scan over the trimmed grid $u \in [0.2, 0.8]$, corresponding to $t \in [251, 1007]$ of our series. A plot of the Wald statistic is given in (5.6) with maximum at $u = 0.447$, corresponding to 26 March 2020, aligning with the onset of the

5.3 Conclusions

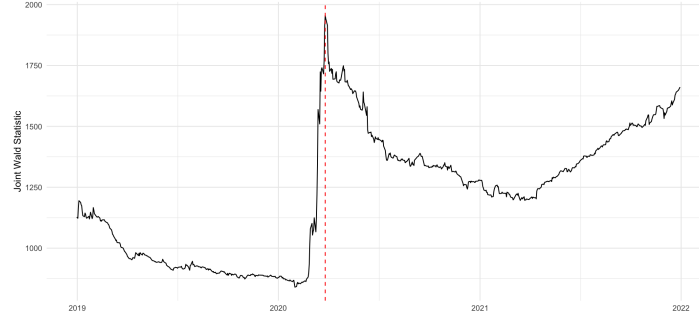


Figure 5.6: Joint Wald statistic (black) for the stocks from table 5.1 between 2018 and 2022 with the maximum joint Wald Statistic corresponding to 26/03/2020 (red dashed).

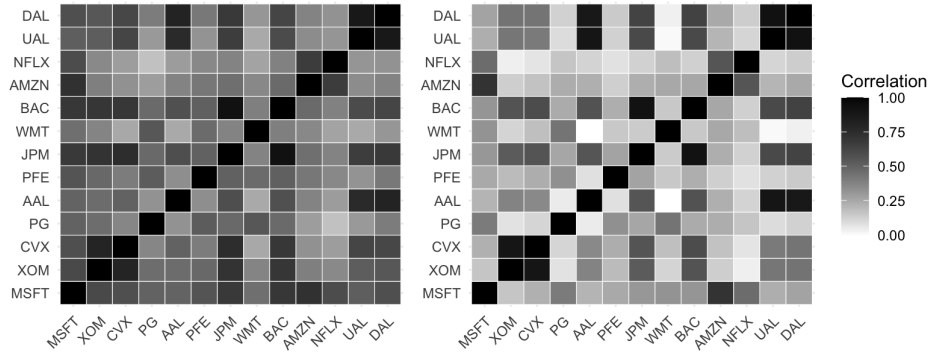


Figure 5.7: Correlation between the stocks in table 5.1 before (left) and after (right) the detected change point on 26/03/2020, where a darker square indicates more correlation.

COVID-19 pandemic.

Computing the estimated residual covariance matrix before and after the identified change point and plotting the residual correlation for each regime, we obtain figure 5.7. The change in lag parameters coincides with a change in correlation. Stocks in the same sector remain correlated (for example CVX and XOM), but between-sector correlations become considerably weaker. This is consistent with the heterogeneous impact of the pandemic across sectors, so that sectors that previously moved together diverged. This finding aligns with the INSPECT weightings, which indicated some sectors (airlines, energy, financials) were more severely impacted than others (technology, consumer staples).

5.3 Conclusions

Across both univariate and multivariate methods, a change in variance in February 2020 is consistently detected, coinciding with the onset of the COVID-19 pandemic. The INSPECT projection vector identifies airlines, energy, and financials as the primary drivers of the shared volatility break, accounting for 95% of the signal, while technology and consumer staples stocks contribute negligibly, consistent with the context of travel restrictions, economic uncertainty and the impact of the Saudi Arabia and Russia oil price war [47]. The VAR sup-Wald test confirms a corresponding change in lag coefficients on 26 March 2020, with post-break correlations indicating a weakening of between-sector relationships. The robustness of the February 2020 break across five detection methods with different assumptions and parameterisations provides strong evidence that this represents a genuine market-wide structural change.

Chapter 6

Conclusion

Throughout this report, we have used a likelihood ratio framework to develop and contrast change point detection techniques. We highlighted the importance of the choice of threshold and derived two theoretical bounds. These theoretical derivations primarily indicate the order of the threshold relative to n , and the constant factor is typically tuned empirically. Therefore, there is no fixed threshold at which a change is declared, and assumptions and context play a role in determining whether a change is significant enough to declare a change point. This is compounded by the sensitivity to hyperparameters across models, particularly for the sliding window model, whose behaviour is heavily dependent on window size. Similarly, the Bayesian approach requires specification of prior beliefs about the data, and high-dimensional contexts demand an alternative framework altogether. From this, it is clear that although we can obtain consistency results and asymptotic distributions, there is no definitive rule on when one declares a change point.

Despite this, the methods in the application section consistently detected change points corresponding to identifiable real-world changes, indicating their robustness, particularly in the context of a change as significant as the COVID-19 pandemic.

Several research directions remain open as potential extensions to this work. A deeper consideration of the Bayesian paradigm would allow us to quantify uncertainty in our estimates and incorporate prior information about the change point that is relevant to the context. Adams and MacKay [1] apply the Bayesian framework to the online change point detection problem, using new observations to repeatedly update beliefs about τ with incoming data. This has applications in data streaming, such as network traffic, high-frequency financial data, and medical sensors. The Bayesian approach would also address the threshold-selection difficulty acknowledged above, since posterior probabilities provide an interpretable basis for declaring change points without empirical tuning.

Another interesting extension is the high-dimensional context, particularly the relaxation of sparsity assumptions. Detection of a dense change in high-dimensional data is challenging due to accumulation of noise, and the difficulty depends on the sparsity of the change. Liu, Gao and Samworth [44] characterise this dependence through an exact minimax rate, identifying a phase transition between sparse and dense regimes, as well as extending the analysis to account for spatial and temporal dependence.

In both the Bayesian and high-dimensional contexts, exploring nonparametric models is of interest as these avoid the distributional assumptions made throughout this report. The Gaussian assumption is often violated in practice, particularly for financial data, as seen in the returns analysed in chapter 5, making nonparametric alternatives a significant area for further research.

Appendices

Appendix A

Supplementary Material for Chapter 2

A.1 Boole's Inequality and the Bonferroni Correction

Supplementary material for section 2.3.

Lemma A.1 (Boole's Inequality). *For a countable set of events $A_1, A_2, \dots$ we have*

$$\mathbb{P}\left[\bigcup_{i=1}^{\infty} A_i\right] \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Lemma A.2 (Bonferroni Correction [43]). *Let $\mathcal{H}_{0,1}, \dots, \mathcal{H}_{0,m}$ be a family of m null hypotheses, with corresponding test statistics $T_1, \dots, T_m$. Suppose that we apply a critical threshold c such that any null hypothesis $\mathcal{H}_{0,i}$ is rejected if $T_i > c$. Then to control the FWER at level α , the marginal probability of a Type I error for each test must satisfy:*

$$\mathbb{P}(T_i > c | \mathcal{H}_{0,i}) \leq \frac{\alpha}{m}.$$

A.2 BIC Derivation

Supplementary material for section 2.3, based on [58].

Theorem A.3. *Let $\mathbf{X} = (X_1, \dots, X_n)$ be an observed time series. Let $\ell(\boldsymbol{\theta}; \mathbf{X})$ be the log-likelihood function given $\boldsymbol{\theta} \in \Omega \subseteq \mathbb{R}^r$. Let $p(\boldsymbol{\theta})$ represent the prior distribution on these parameters. Assume that the likelihood function and prior density are twice differentiable. Then as $n \rightarrow \infty$, an approximation to the log-marginal likelihood function $\ell_m(X)$, scaled by -2 , is given by the Bayes Information Criterion:*

$$-2\ell_m(X) \approx -2\ell(\hat{\boldsymbol{\theta}}) + q \ln(n).$$

Proof. Let $g(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta})/n$. We have:

$$\begin{aligned} \mathcal{L}_M(\mathbf{X}) &= \int_{\Omega} \mathcal{L}(\boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta} \\ &= \int_{\Omega} \exp(n \cdot g(\boldsymbol{\theta})) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}. \end{aligned} \tag{A.1}$$

This integral is now suitable for a Laplace approximation which states that:

$$\int_{\Omega} \exp[n \cdot g(\boldsymbol{\theta})] f(\boldsymbol{\theta}) d\boldsymbol{\theta} = \left(\sqrt{\frac{2\pi}{n}} \right)^q \exp[n \cdot g(\boldsymbol{\theta}_0)] \left(f(\boldsymbol{\theta}_0) |\mathbf{J}(\boldsymbol{\theta}_0)|^{-1/2} + O(1/n) \right), \tag{A.2}$$

A.3 Score Function

where $\boldsymbol{\theta}_0 = \arg \max_{\boldsymbol{\theta} \in \Omega} (g(\boldsymbol{\theta}))$ and $\mathbf{J}(\boldsymbol{\theta}_0)$ is the Hessian matrix evaluated at $\boldsymbol{\theta}_0$. Since we defined $g(\boldsymbol{\theta}) = \frac{1}{n} \ell(\boldsymbol{\theta})$, we have that $\boldsymbol{\theta}_0 = \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \ell(\boldsymbol{\theta}) = \hat{\boldsymbol{\theta}}$. Therefore, we can apply (A.2) to (A.1) with $f(\boldsymbol{\theta}) = \pi(\boldsymbol{\theta})$ to get:

$$\begin{aligned} \mathcal{L}_M(\mathbf{X}) &\approx \left(\sqrt{\frac{2\pi}{n}} \right)^q \exp \left[n \cdot g(\hat{\boldsymbol{\theta}}) \right] \left(\pi(\hat{\boldsymbol{\theta}}) |\mathbf{J}(\hat{\boldsymbol{\theta}})|^{-1/2} \right) \\ &\iff \ell_m(\mathbf{X}) \approx q \ln \left(\sqrt{\frac{2\pi}{n}} \right) + \ell(\hat{\boldsymbol{\theta}}) + \ln(\pi(\hat{\boldsymbol{\theta}})) - \frac{1}{2} \ln |\mathbf{J}(\hat{\boldsymbol{\theta}})|, \\ &\iff -2\ell_m(\mathbf{X}) \approx -q \ln(2\pi) + q \ln(n) - 2\ell(\hat{\boldsymbol{\theta}}) - 2 \ln(\pi(\hat{\boldsymbol{\theta}})) + \ln |\mathbf{J}(\hat{\boldsymbol{\theta}})|, \end{aligned} \quad (\text{A.3})$$

after taking logarithms and multiplying by -2 . Then the order of the terms is as follows:

$$-2\ell_m(\mathbf{X}) \approx \underbrace{-q \ln(2\pi)}_{O(1)} + \underbrace{q \ln(n)}_{O(\ln n)} - \underbrace{2\ell(\hat{\boldsymbol{\theta}})}_{O_p(n)} - \underbrace{2 \ln(\pi(\hat{\boldsymbol{\theta}}))}_{O_p(1)} + \underbrace{\ln |\mathbf{J}(\hat{\boldsymbol{\theta}})|}_{O_p(1)},$$

where $O_p(a_n)$ denotes a sequence that is a_n -bounded in probability. For strong consistency, and to validate the Laplace approximation, we let $n \rightarrow \infty$. This allows us to ignore $O(1)$ terms, resulting in the BIC:

$$-2\ell_m(\mathbf{X}) \approx -2\ell(\hat{\boldsymbol{\theta}}) + q \ln(n).$$

□

A.3 Score Function

This section provides the proof of lemma 2.20, deriving the connection between the score process $\mathbf{Z}_n$ and the GLLR test statistic. Recall the partial sum score process defined in (2.29):

$$\mathbf{Z}_n(u, \boldsymbol{\theta}) = \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nu \rfloor} \mathbf{s}(X_t, \boldsymbol{\theta})$$

We first note that

$$\begin{aligned} \mathbf{Z}_n(0, \hat{\boldsymbol{\theta}}) &= 0, \\ \mathbf{Z}_n(1, \hat{\boldsymbol{\theta}}) &= \frac{1}{\sqrt{n}} \sum_{t=1}^n \mathbf{s}(X_t, \hat{\boldsymbol{\theta}}) = 0, \end{aligned}$$

with the latter holding by definition of the MLE:

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}) \implies \frac{\partial}{\partial \boldsymbol{\theta}} \ell(\hat{\boldsymbol{\theta}}) = \mathbf{0} \implies \sum_{t=1}^n \mathbf{s}(X_t, \hat{\boldsymbol{\theta}}) = \mathbf{0}. \quad (\text{A.4})$$

If we fix $u = 1$ and perform a first order Taylor expansion of $\mathbf{Z}_n(1, \hat{\boldsymbol{\theta}})$ around an arbitrary point $\boldsymbol{\theta}$, we see that

$$\begin{aligned} \mathbf{Z}_n(1, \hat{\boldsymbol{\theta}}) &\approx \mathbf{Z}_n(1, \boldsymbol{\theta}) + \left[\frac{\partial}{\partial \boldsymbol{\theta}} \mathbf{Z}_n(1, \boldsymbol{\theta}) \right] (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \\ &\implies 0 \approx \mathbf{Z}_n(1, \boldsymbol{\theta}) + \left[\frac{1}{\sqrt{n}} \frac{\partial}{\partial \boldsymbol{\theta}} \sum_{t \leq n} \mathbf{s}(X_t, \boldsymbol{\theta}) \right] (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \end{aligned}$$

Using (2.28) and the fact that $-n\mathbf{J}(\boldsymbol{\theta}) \approx \mathbf{H}(\boldsymbol{\theta})$ asymptotically, we can rearrange to obtain

$$\begin{aligned} \mathbf{Z}_n(1, \boldsymbol{\theta}) &\approx \sqrt{n}\mathbf{J}(\boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}), \\ \implies (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) &\approx \frac{1}{\sqrt{n}}\mathbf{J}^{-1}(\boldsymbol{\theta})\mathbf{Z}_n(1, \boldsymbol{\theta}). \end{aligned} \quad (\text{A.5})$$

Now we have established the score function, we want to consider its link to the generalised log-likelihood ratio test statistic. By performing a second order Taylor expansion of $\ell(\boldsymbol{\theta})$ around the MLE, $\hat{\boldsymbol{\theta}}$, we see:

$$\begin{aligned} \ell(\boldsymbol{\theta}) &\approx \ell(\hat{\boldsymbol{\theta}}) + (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \frac{\partial}{\partial \boldsymbol{\theta}} \ell(\hat{\boldsymbol{\theta}}) + \frac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \mathbf{H}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \\ &= \ell(\hat{\boldsymbol{\theta}}) + \frac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \mathbf{H}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \end{aligned}$$

Again with the second equivalence holding due to (A.4). If we consider the GLLR test statistic from (2.12), we can apply this approximation with $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_0$ and $\hat{\boldsymbol{\theta}}$ the MLE of the respective segment. Considering the first segment of length k , the Hessian is approximated by $-k\mathbf{J}(\hat{\boldsymbol{\theta}}_1)$:

$$\begin{aligned} \ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:k}) &\approx \ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) - \left[\ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) + \frac{1}{2}(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_1)^\top (-k\mathbf{J}(\hat{\boldsymbol{\theta}}_1))(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_1) \right] \\ &= \frac{1}{2}(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_1)^\top (k\mathbf{J}(\hat{\boldsymbol{\theta}}_1))(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_1) \end{aligned} \quad (\text{A.6})$$

Then applying (A.5), we have:

$$(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_1) \approx -\frac{\sqrt{n}}{k}\mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n\left(\frac{k}{n}, \boldsymbol{\theta}_0\right),$$

and substituting this into (A.6), we get:

$$\begin{aligned} \ell(\hat{\boldsymbol{\theta}}_1; \mathbf{X}_{1:k}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{1:k}) &\approx \frac{1}{2} \left[-\frac{\sqrt{n}}{k}\mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n\left(\frac{k}{n}, \hat{\boldsymbol{\theta}}_0\right) \right]^\top (k\mathbf{J}(\hat{\boldsymbol{\theta}}_1)) \left[-\frac{\sqrt{n}}{k}\mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n\left(\frac{k}{n}, \boldsymbol{\theta}_0\right) \right] \\ &= \frac{n}{2k}\mathbf{Z}_n\left(\frac{k}{n}, \hat{\boldsymbol{\theta}}_0\right)^\top \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n\left(\frac{k}{n}, \hat{\boldsymbol{\theta}}_0\right) \end{aligned}$$

Similarly for the second segment:

$$\begin{aligned} \ell(\hat{\boldsymbol{\theta}}_2; \mathbf{X}_{k+1:n}) - \ell(\hat{\boldsymbol{\theta}}_0; \mathbf{X}_{k+1:n}) &\approx \frac{1}{2}(\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_2)^\top [(n-k)\mathbf{J}(\hat{\boldsymbol{\theta}}_2)](\hat{\boldsymbol{\theta}}_0 - \hat{\boldsymbol{\theta}}_2) \\ &= \frac{n}{2(n-k)}\mathbf{Z}_n\left(\frac{k}{n}, \hat{\boldsymbol{\theta}}_0\right)^\top \mathbf{J}^{-1}\mathbf{Z}_n\left(\frac{k}{n}, \hat{\boldsymbol{\theta}}_0\right) \end{aligned}$$

Substituting these into (2.12), noting the cancellation of the factor of 2, and writing $k' = \frac{k}{n}$ we see

$$\begin{aligned} \text{LR}_k(\mathbf{X}) &= \frac{1}{k'}\mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0)^\top \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0) + \frac{1}{1-k'}\mathbf{Z}_n(k', \boldsymbol{\theta}_0)^\top \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0) \\ &= \frac{1}{k'(1-k')}\mathbf{Z}_n(k', \boldsymbol{\theta}_0)^\top \mathbf{J}^{-1}(\hat{\boldsymbol{\theta}}_0)\mathbf{Z}_n(k', \hat{\boldsymbol{\theta}}_0) \end{aligned} \quad (\text{A.7})$$

A.4 Slutsky's Theorem

From [46, App. C]

Theorem A.4 (Slutsky's Theorem). *If $g : \mathbb{R}^k \rightarrow \mathbb{R}^m$ is a continuous function, then:*

$$\begin{aligned} x_T \rightarrow_p x &\implies g(x_T) \rightarrow_p g(x) \\ x_T \rightarrow_d x &\implies g(x_T) \rightarrow_d g(x) \end{aligned}$$

A.5 Multivariate Likelihood at the MLE

Supplementary Material for section 2.4.1.

$$\begin{aligned} \sum_{i=1}^n (\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top \widehat{\boldsymbol{\Sigma}}^{-1} (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) &= \sum_{t=1}^n \text{tr} \left((\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top \widehat{\boldsymbol{\Sigma}}^{-1} (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) \right) \\ &= \sum_{t=1}^n \text{tr} \left(\widehat{\boldsymbol{\Sigma}}^{-1} (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) (\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top \right), \end{aligned}$$

using the fact that a scalar is equal to its trace, followed the cyclic property of the trace. By linearity of the trace, we then have

$$\begin{aligned} \sum_{i=1}^n (\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top \widehat{\boldsymbol{\Sigma}}^{-1} (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) &= \text{tr} \widehat{\boldsymbol{\Sigma}}^{-1} \left(\sum_{t=1}^n (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) (\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top \right) \\ &= n \text{tr}(\widehat{\boldsymbol{\Sigma}}^{-1} \widehat{\boldsymbol{\Sigma}}) \\ &= nd, \end{aligned}$$

since the sample covariance MLE is $\widehat{\boldsymbol{\Sigma}} = n^{-1} \sum_{t=1}^n (\mathbf{X}_t - \hat{\boldsymbol{\mu}}) (\mathbf{X}_t - \hat{\boldsymbol{\mu}})^\top$. Therefore the exponential term in the likelihood in () simplifies to $\exp(nd/2)$, which is constant with respect to $\widehat{\boldsymbol{\Sigma}}$.

A.6 Normal-Normal Conjugacy

Supplementary material for section 2.5.

Theorem A.5 (Conjugate Normal Sampling with Known Variance). *Suppose we have data $\mathbf{X} = (X_1, \dots, X_n)$ such that*

$$X_t \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2) \quad \text{for } t = 1, 2, \dots, n,$$

where we assume σ^2 is known. Suppose our prior for the parameter μ is:

$$\mu \sim \mathcal{N}(m_0, v_0),$$

for m_0, v_0 known. Then the posterior distribution for μ given the data $\mathbf{X}$ is

$$\mu | \mathbf{X} \sim \mathcal{N}(m_1, v_1),$$

where

$$\begin{aligned} m_1 &= \frac{v_1}{v_0} m_0 + \frac{nv_1}{\sigma^2} \bar{x}, \\ v_1 &= \frac{1}{1/v_0 + n/\sigma^2} = \frac{v_0 \sigma^2 / n}{v_0 + \sigma^2 / n}. \end{aligned}$$

Then by using Chib's marginal likelihood identity (rearranging Bayes):

$$\mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau) = \frac{\ell(\mathbf{X}_{1:\tau}|\tau, \mu_1)p(\mu_1)}{p(\mu_1|\mathbf{X}_{1:\tau})}$$

Choosing $\mu_1 = m_1$, and using independence of the data we see:

$$\begin{aligned} \mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau) &= \frac{\prod_{t=1}^{\tau} f(X_t|\tau, \mu_1 = m_1)p(\mu_1 = m_1)}{p(\mu_1|\mathbf{X}_{1:\tau})} \\ &= \sqrt{2\pi v_1} \prod_{t=1}^{\tau} \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(X_t - m_1)^2\right) \right] \frac{1}{\sqrt{2\pi v_0}} \exp\left(-\frac{1}{2v_0}(m_1 - m_0)^2\right) \\ &= (2\pi)^{-\tau/2} \sqrt{\frac{v_1}{v_0}} \exp\left[-\frac{1}{2}\left(\sum_{t=1}^{\tau} (X_t - m_1)^2 + \frac{1}{v_0}(m_1 - m_0)^2\right)\right], \end{aligned} \quad (\text{A.8})$$

where the term $(X_t - m_1)^2$ represents the residuals of the data against the posterior mean (m_1), the term $(m_1 - m_0)^2/v_0$ checks how well the posterior mean fits the initial prior beliefs, and multiplying by $\sqrt{v_1}$ weights the likelihood by the posterior variance. Applying the same logic to the post-change point data, we have:

$$\mathcal{L}_M(\mathbf{X}_{\tau+1:n}|\tau) = (2p)^{-(n-\tau)/2} \sqrt{\frac{v_2}{v_0}} \exp\left[-\frac{1}{2}\left(\sum_{t=\tau+1}^n (X_t - m_2)^2 + \frac{1}{v_0}(m_2 - m_0)^2\right)\right] \quad (\text{A.9})$$

We are also required to specify a prior for τ , representing prior beliefs about where the change point may be. If we assume that all values of $t = 1, \dots, n$ are equally likely to be our change point τ , this results in the discrete uniform prior:

$$p(\tau) = \frac{1}{n}.$$

Since this is independent of τ we can write:

$$p(\tau|\mathbf{X}) \propto \mathcal{L}_M(\mathbf{X}_{1:\tau}|\tau)\mathcal{L}_M(\mathbf{X}_{\tau+1:n}|\tau) \quad (\text{A.10})$$

Combining (A.8) and (A.9) according to (A.10) gives:

$$\begin{aligned} p(\tau|\mathbf{X}) &\propto \frac{(2p)^{n/2}}{n} \frac{\sqrt{v_1 v_2}}{v_0} \exp\left[-\frac{1}{2}\left(\sum_{t=1}^{\tau} (X_t - m_1)^2 + \sum_{t=\tau+1}^n (X_t - m_2)^2 + \frac{(m_1 - m_0)^2 + (m_2 - m_0)^2}{v_0}\right)\right], \\ &\propto \frac{\sqrt{v_1 v_2}}{v_0} \exp\left[-\frac{1}{2}\left(\sum_{t=1}^{\tau} (X_t - m_1)^2 + \sum_{t=\tau+1}^n (X_t - m_2)^2 + \frac{(m_1 - m_0)^2 + (m_2 - m_0)^2}{v_0}\right)\right]. \end{aligned}$$

A.7 Animations of the Algorithms in Chapter 3

Animation of the sliding window algorithm - <https://youtu.be/FQ2yJUb3vJU>

Animation of the bottom-up algorithm - <https://youtu.be/eCNi8ouTpiI>

Appendix B

Supplementary Material for Chapter 4

B.1 Causality Implies Weak Stationarity

Supplementary Material for Chapter 4 We prove this using the conditions given in definition 2.2.

Proof. For the first condition, we see that:

$$\mathbb{E}[X_t] = \mathbb{E} \left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \right] = \sum_{j=0}^{\infty} \psi_j \mathbb{E}[\varepsilon_{t-j}] = 0 \forall t, \quad (\text{B.1})$$

and so the expectation, μ , is independent of the index. For the second condition:

$$\text{Var}[X_t] = \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \right)^2 \right] - \left(\mathbb{E} \left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \right] \right)^2,$$

where the second term is equal to zero by (B.1). This leaves:

$$\begin{aligned} \text{Var}[X_t] &= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \right)^2 \right] = \mathbb{E} \left[\left(\sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i} \right) \left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \right) \right] \\ &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j \mathbb{E}[\varepsilon_{t-i} \varepsilon_{t-j}], \end{aligned}$$

where we can use that ε_t are independent, to get

$$\mathbb{E}[\varepsilon_{t-i} \varepsilon_{t-j}] = \begin{cases} \mathbb{E}[\varepsilon_{t-i}^2] & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (\text{B.2})$$

Substituting this in we see

$$\text{Var}[X_t] = \sum_{i=0}^{\infty} \psi_i^2 \mathbb{E}[\varepsilon_{t-i}^2] = \sigma^2 \sum_{i=0}^{\infty} \psi_i^2,$$

which again is independent of t .

$$\text{Cov}[X_t, X_{t+k}] = \mathbb{E}[X_t X_{t+k}] - \mathbb{E}[X_t] \mathbb{E}[X_{t+k}] = \mathbb{E}[X_t X_{t+k}],$$

using (B.1). Then substituting in we see

$$\begin{aligned} \text{Cov}[X_t, X_{t+k}] &= \mathbb{E} \left[\sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i} \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j+k} \right] \\ &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j \mathbb{E}[\varepsilon_{t-i} \varepsilon_{t-j+k}], \end{aligned}$$

and by similar logic to (B.2), we have

$$\mathbb{E}[\varepsilon_{t-i}\varepsilon_{t-j+k}] = \begin{cases} \sigma^2 & \text{if } i = j - k, \\ 0 & \text{if } i \neq j - k. \end{cases}$$

This results in

$$\text{Cov}[X_t, X_{t+k}] = \sum_{i=0}^{\infty} \psi_i \psi_{i+k} \sigma^2,$$

which is independent of t and so all conditions for weak stationarity hold. $\square$

B.2 Properties of AR(p) Processes

B.2.1 Mean

We now derive the mean stated in (4.3).

We have $\mathbb{E}[X_t] = \mu$. Taking expectations we see:

$$\begin{aligned} \mathbb{E}[X_t] &= \mathbb{E}[\phi_0] + \mathbb{E}[\phi_1 X_{t-1}] + \dots + \mathbb{E}[\phi_p X_{t-p}] + \mathbb{E}[\varepsilon_t] \\ &= \phi_0 + \phi_1 \mathbb{E}[X_{t-1}] + \dots + \phi_p \mathbb{E}[X_{t-p}]. \end{aligned}$$

Stationarity implies that $\mathbb{E}[X_t] = \mu \forall t$, therefore

$$\begin{aligned} \mu &= \phi_0 + \phi_1 \mu + \dots + \phi_p \mu \\ &= \phi_0 + \mu \sum_{i=1}^p \phi_i \end{aligned}$$

Therefore,

$$\mu = \frac{\phi_0}{1 - \sum_{i=1}^p \phi_i}.$$

B.2.2 Variance

We now derive the variance stated in (4.4) for the case $\mathbb{E}[X_t] = 0$, with the non-zero case following by use of section B.2.3.

$$\begin{aligned} \mathbb{E}[X_t X_t] &= \mathbb{E}[\phi_0 X_t] + \mathbb{E}[\phi_1 X_{t-1} X_t] + \dots + \mathbb{E}[\phi_p X_{t-p} X_t] + \mathbb{E}[X_t \varepsilon_t] \\ &= \phi_0 \mathbb{E}[X_t] + \phi_1 \mathbb{E}[X_{t-1} X_t] + \dots + \phi_p \mathbb{E}[X_{t-p} X_t] + \mathbb{E}[X_t \varepsilon_t] \\ &= \phi_1 \gamma_1 + \dots + \phi_p \gamma_p + \mathbb{E}[X_t \varepsilon_t] \\ &= \phi_1 \gamma_1 + \dots + \phi_p \gamma_p + \mathbb{E} \left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \varepsilon_t \right] \\ &= \phi_1 \gamma_1 + \dots + \phi_p \gamma_p + \sigma^2 \end{aligned}$$

since $\mathbb{E}[\varepsilon_{t-j} \varepsilon_t] = 0$ for $j \neq 0$, and $\mathbb{E}[\varepsilon_t \varepsilon_t] = \sigma^2$.

B.2.3 Equivalence to Non-Zero Intercept

We now show that any stationary AR(p) with non-zero intercept can be re-expressed as a stationary AR(p) with $\phi_0 = 0$.

Theorem B.1. *Let $X_t = \phi_0 + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \varepsilon_t$ and $\widetilde{X}_t = \phi_1 \widetilde{X}_{t-1} + \dots + \phi_p \widetilde{X}_{t-p} + \varepsilon_t$ be two AR(p) processes satisfying causality and therefore stationary processes. Let $\mathbb{E}[X_t] = \mu$. Then X_t can be rewritten as stationary AR(p) process:*

$$\widetilde{X}_t = X_t - \mu.$$

i.e., equivalent up to translation.

Proof. Substituting $\widetilde{X}_t = X_t - \mu$ in (4.1), we see

$$\begin{aligned} \widetilde{X}_t + \mu &= \phi_0 + \sum_{i=1}^p \phi_i (\widetilde{X}_{t-i} + \mu) + \varepsilon_t \\ &= \phi_0 + \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \mu \sum_{i=1}^p \phi_i + \varepsilon_t \end{aligned}$$

So that

$$\begin{aligned} \widetilde{X}_t &= \phi_0 - \mu + \mu \sum_{i=1}^p \phi_i + \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \varepsilon_t \\ &= \phi_0 + \mu \left(\sum_{i=1}^p \phi_i - 1 \right) + \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \varepsilon_t \\ &= \phi_0 + \frac{\phi_0}{1 - \sum_{i=1}^p \phi_i} \left(\sum_{i=1}^p \phi_i - 1 \right) + \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \varepsilon_t \\ &= \phi_0 - \phi_0 + \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \varepsilon_t \\ &= \sum_{i=1}^p \phi_i \widetilde{X}_{t-i} + \varepsilon_t. \end{aligned}$$

□

B.3 Vec Operator and Kronecker Product

Taken from new introduction to multi 2005 [46, App. A]

Definition B.2 (Vec Operator). *Let $A = (\mathbf{a}_1, \dots, \mathbf{a}_n)$ be an $(m \times n)$ matrix with m columns, $\mathbf{a}_i$. The Vec operator transforms A into an $(mn \times 1)$ vector by stacking the columns:*

$$\text{Vec}(A) = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_n \end{bmatrix}$$

Definition B.3 (Kronecker Product). *Let $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ be $(m \times n)$ and $(p \times q)$ matrices respectively. The Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is given by:*

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \dots & a_{1n}\mathbf{B} \\ \vdots & & \vdots \\ a_{m1}\mathbf{B} & \dots & a_{mn}\mathbf{B} \end{bmatrix}$$

We use the following properties:

- (i) $\text{Vec}(\mathbf{ABC}) = (\mathbf{I} \otimes \mathbf{AB})\text{Vec}(\mathbf{C})$
- (ii) $\text{Vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{Vec}(\mathbf{B})$
- (iii) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC} \otimes \mathbf{BD})$
- (iv) $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$
- (v) $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \cdot \text{rank}(\mathbf{B})$

Appendix C

Supplementary Material for Chapter 5

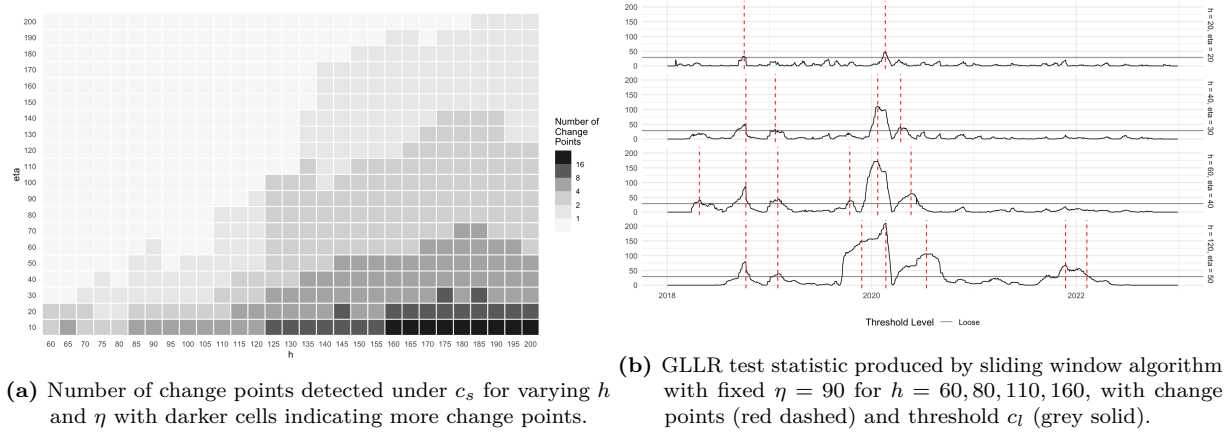


Figure C.1: Sliding window parameter tuning for S&P 500 data under c_l .

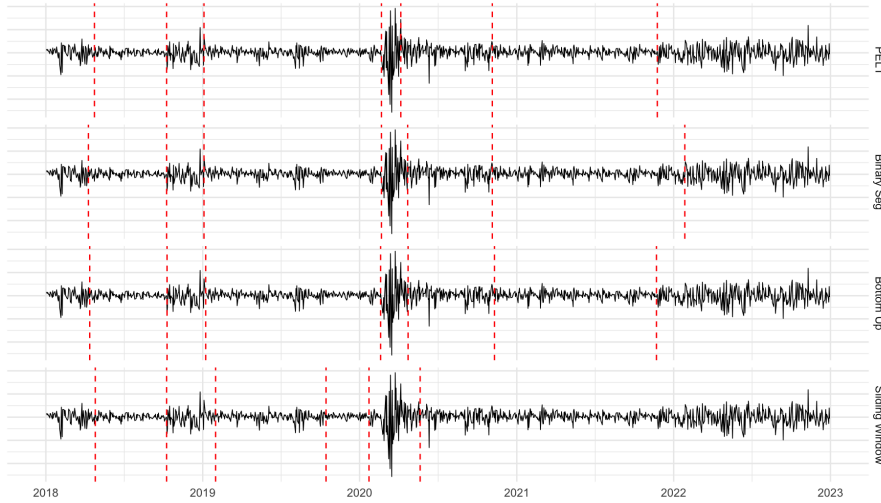


Figure C.2: Log returns for the S&P 500 between 01/01/2018 and 31/12/2022 (black) with detected change points (red dashed) under the tighter threshold $c_l = 15 \ln \ln(n) \approx 29.48$ for different change point detection algorithms, indicating relatively close agreement between PELT, binary segmentation, and bottom-up, with sliding window only detecting changes before 2021.

Bibliography

- [1] Ryan Prescott Adams and David MacKay. Bayesian online changepoint detection. *arXiv (Cornell University)*, 01 2007.
- [2] Alexander C Aitken. Iv.—on least squares and linear combination of observations. *Proceedings of the Royal Society of Edinburgh*, 55:42–48, 1936.
- [3] Elena Andreou and Eric Ghysels. Detecting multiple breaks in financial market volatility dynamics. *Journal of Applied Econometrics*, 17:579–600, 2002.
- [4] Donald W. K. Andrews. Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61:821, 07 1993.
- [5] Jushan Bai, Robin L. Lumsdaine, and James H. Stock. Testing for and dating common breaks in multivariate time series. *The Review of Economic Studies*, 65:395–432, 1998.
- [6] Scott R. Baker, Nicholas Bloom, Steven J. Davis, Kyle Kosr, Marco Sammon, and Tasaneeya Viratyosin. The unprecedented stock market reaction to covid-19. *The Review of Asset Pricing Studies*, 10:742–758, 07 2020.
- [7] Daniel Barry and J. A. Hartigan. A bayesian analysis for change point problems. *Journal of the American Statistical Association*, 88:309, 03 1993.
- [8] Ben S. Bernanke, Jean Boivin, and Piotr E Eliazs. Measuring the effects of monetary policy: A factor-augmented vector autoregressive (favar) approach. *The Quarterly Journal of Economics*, 120:387–422, 02 2005.
- [9] Patrick Billingsley. *Convergence of Probability Measures*. John Wiley & Sons, 06 1999.
- [10] Richard C. Bradley. Basic properties of strong mixing conditions. a survey and some open questions. *Probability Surveys*, 2:107–144, 2005.
- [11] Peter J Brockwell and Richard A Davis. *Introduction to Time Series and Forecasting*. Springer, 2016.
- [12] John Y. Campbell, Andrew W. Lo, A. Craig MacKinlay, and Robert F. Whitelaw. The econometrics of financial markets. *Macroeconomic Dynamics*, 2, 12 1998.
- [13] George Casella and Roger L Berger. *Statistical Inference*. Brooks/Cole Cengage Learning, 2017.
- [14] Jie Chen and Ajay K. Gupta. Testing and locating variance changepoints with application to stock prices. *Journal of the American Statistical Association*, 92:739–739, 06 1997.
- [15] D A Darling and P. Erdős. A limit theorem for the maximum of normalized sums of independent random variables. *Duke Mathematical Journal*, 23, 03 1956.
- [16] Richard A Davis, Dawei Huang, and YiChing Yao. Testing for a change in the parameter values and order of an autoregressive model. *The Annals of Statistics*, 23:282–304, 1995.

BIBLIOGRAPHY

- [17] David A. Dickey and Wayne A. Fuller. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74:427–431, 06 1979.
- [18] Idris A Eckley, Paul Fearnhead, and Rebecca Killick. Analysis of changepoint models. *Cambridge University Press eBooks*, pages 205–224, 08 2011.
- [19] Birte Eichinger and Claudia Kirch. A mosum procedure for the estimation of multiple random change points. *Bernoulli*, 24, 02 2018.
- [20] M Fazel, H Hindi, and S Boyd. Rank minimization and applications in system theory, 2004.
- [21] Paul Fearnhead. Exact and efficient bayesian inference for multiple changepoint problems. *Statistics and Computing*, 16:203–213, 06 2006.
- [22] Paul Fearnhead and Piotr Fryzlewicz. Detecting a single change-point. "Draft chapter from the forthcoming book "Change-Point Detection and Data Segmentation", 2022.
- [23] Klaus Frick, Axel Munk, and Hannes Sieling. Multiscale change point inference. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 76:495–580, 05 2014.
- [24] Piotr Fryzlewicz. Wild binary segmentation for multiple changepoint detection. *The Annals of Statistics*, 42:2243–2281, 2014.
- [25] Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. Chapman and Hall/CRC, 11 2013.
- [26] Gene Howard Golub and Charles F Van. *Matrix Computations*. The Johns Hopkins University Press, Cop, 2013.
- [27] Peter J Green. Reversible jump markov chain monte carlo computation and bayesian model determination. *Biometrika*, 82:711–711, 12 1995.
- [28] James D. Hamilton. *Time Series Analysis*. Princeton University Press, 1994.
- [29] E. J. Hannan and B. G. Quinn. The determination of the order of an autoregression. *Journal of the Royal Statistical Society: Series B (Methodological)*, 41:190–195, 01 1979.
- [30] Douglas M. Hawkins and K. D. Zamba. A change-point model for a shift in variance. *Journal of Quality Technology*, 37:21–31, 01 2005.
- [31] Nils Lid Hjort and Alexander Koning. Tests for constancy of model parameters over time. *Journal of Nonparametric Statistics*, 14:113–132, 01 2002.
- [32] Lajos Horváth and Emanuel Parzen. Limit theorems for fisherscore change processes. *Lecture Notes Monograph Series*, 23:157–169, 1994.
- [33] Marie Husková and Ales Slabý. Permutation tests for multiple changes. *Kybernetika*, 37:605–622, 2001.
- [34] B. Jackson, J.D. Scargle, D. Barnes, S. Arabhi, A. Alt, P. Gioumousis, E. Gwin, P. San, L. Tan, and Tun Tao Tsai. An algorithm for optimal partitioning of data on an interval. *IEEE Signal Processing Letters*, 12:105–108, 02 2005.

- [35] Madeleine Johnson. Why did the stock market plunge on june 11?, 2020.
- [36] E. Keogh, S. Chu, D. Hart, and M. Pazzani. An online algorithm for segmenting time series. *Proceedings 2001 IEEE International Conference on Data Mining*, 2001.
- [37] R. Killick, P. Fearnhead, and I. A. Eckley. Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107:1590–1598, 10 2012.
- [38] Rebecca Killick, Kaylea Haynes, Harjit Hullait, Paul Fearnhead, Robin Long, and Jamie Lee. *changepoint: Methods for Changepoint Detection*, 2024. R package version 2.3.
- [39] J Kuelbs and Walter Philipp. Almost sure invariance principles for partial sums of mixing b -valued random variables. *Annals of probability*, 8, 12 1980.
- [40] Bernard Laurent and P Massart. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics*, 28, 10 2000.
- [41] Thai-Ha Le, Anh Tu Le, and Ha-Chi Le. The historic oil price fluctuation during the covid-19 pandemic: What are the causes? *Research in International Business and Finance*, 58:101489, 12 2021.
- [42] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88:365–411, 02 2004.
- [43] Erich L. Lehmann and Joseph P Romano. *Testing Statistical Hypotheses*. Springer, 2010.
- [44] Haoyang Liu, Chao Gao, and Richard J. Samworth. Minimax rates in sparse, high-dimensional change point detection. *The Annals of Statistics*, 49, 04 2021.
- [45] G. M. Ljung and G. E. P. Box. On a measure of lack of fit in time series models. *Biometrika*, 65:297, 08 1978.
- [46] Helmut Lütkepohl. *New Introduction to Multiple Time Series Analysis*. Springer Berlin Heidelberg, 2005.
- [47] Richie Ruchuan Ma, Tao Xiong, and Yukun Bao. The russia–saudi arabia oil price war during the covid-19 pandemic. *Energy Economics*, 102:105517, 08 2021.
- [48] Michael Messer, Marietta Kirchner, Julia Schiemann, Jochen Roeper, Ralph Neininger, and Gaby Schneider. A multiple filter test for the detection of rate changes in renewal processes with varying variance. *The Annals of Applied Statistics*, 8, 12 2014.
- [49] Abdelkader Mokkadem. Mixing properties of arma processes. *Stochastic Processes and their Applications*, 29:309–315, 03 2002.
- [50] E. S. Page. A test for a change in a parameter occurring at an unknown point. *Biometrika*, 42:523–527, 1955.
- [51] Bernhard Pfaff and Matthieu Stigler. *vars: VAR Modelling*, 2024. R package version 1.6-1.
- [52] Zhongjun Qu and Pierre Perron. Estimating and testing structural changes in multivariate regressions. *Econometrica*, 75:459–502, 03 2007.

BIBLIOGRAPHY

- [53] Peter J. Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88:1273–1273, 12 1993.
- [54] Abolfazl Safikhani and Ali Shojaie. Joint structural break detection and parameter estimation in high-dimensional nonstationary var models. *Journal of the American Statistical Association*, 117:251–264, 07 2020.
- [55] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6:461–464, 03 1978.
- [56] Allen J. Scott and M. Knott. A cluster analysis method for grouping means in the analysis of variance. *Biometrics*, 30:507, 09 1974.
- [57] Adrian F. M. Smith. A bayesian approach to inference about a change-point in a sequence of random variables. *Biometrika*, 62:407–416, 08 1975.
- [58] Joram Soch. Derivation of the bayesian information criterion, 01 2020.
- [59] S. M. Srivastava and J. K. Worsley. Likelihood ratio tests for a change in the multivariate normal mean. *Journal of the American Statistical Association*, 81:199–204, 1986.
- [60] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58:267–288, 01 1996.
- [61] Adrian Trapletti and Kurt Hornik. *tseries: Time Series Analysis and Computational Finance*, 2024. R package version 0.10-58.
- [62] Ruey S. Tsay. *Analysis of Financial Time Series*. John Wiley & Sons, Inc., 08 2010.
- [63] A. W. van der Vaart. *M-and Z-Estimators*. Cambridge University Press, 10 1998.
- [64] E. S. Venkatraman and A. B. Olshen. A faster circular binary segmentation algorithm for the analysis of array cgh data. *Bioinformatics*, 23:657–663, 01 2007.
- [65] Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54:426–482, 1943.
- [66] Tengyao Wang, Bertille Follain, and Richard Samworth. *InspectChangepoint: High-Dimensional Changepoint Estimation via Sparse Projection*, 2022. R package version 1.2.
- [67] Tengyao Wang and Richard J. Samworth. High dimensional change point estimation via sparse projection. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80:57–83, 08 2017.
- [68] Wharton Research Data Services. WRDS. <https://wrds.wharton.upenn.edu>, 2024.
- [69] Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9:60–62, 03 1938.
- [70] Yi-Ching Yao. Estimating the number of change-points via schwarz’ criterion. *Statistics & Probability Letters*, 6:181–189, 02 1988.
- [71] Arnold Zellner. An efficient method of estimating seemingly unrelated regressions and tests for aggregation bias. *Journal of the American Statistical Association*, 57:348, 06 1962.